SCRIPT-BASED MODEL USER MANUAL · 12 NOVEMBER 2026

H2RES User Manual

For H2RES_12.11.2026: main.py, policies, native data files, workbook variations, electricity DSM and product based industry.

62 source-linked equations · 378 unique settings explained · Complete workflow and dependency guide · 32 illustrated source/data views · Reference bibliography · Desktop GUI excluded

H2RES user guide: 12 November 2026

Model inspected: `D:\POSAO\H2RES_12.11.2026`. **Scope:** running and configuring the optimization model through its Python scripts and native data folder. The desktop GUI is excluded. For mathematical details, use the model manual; for exact file and sheet names, use the data reference.

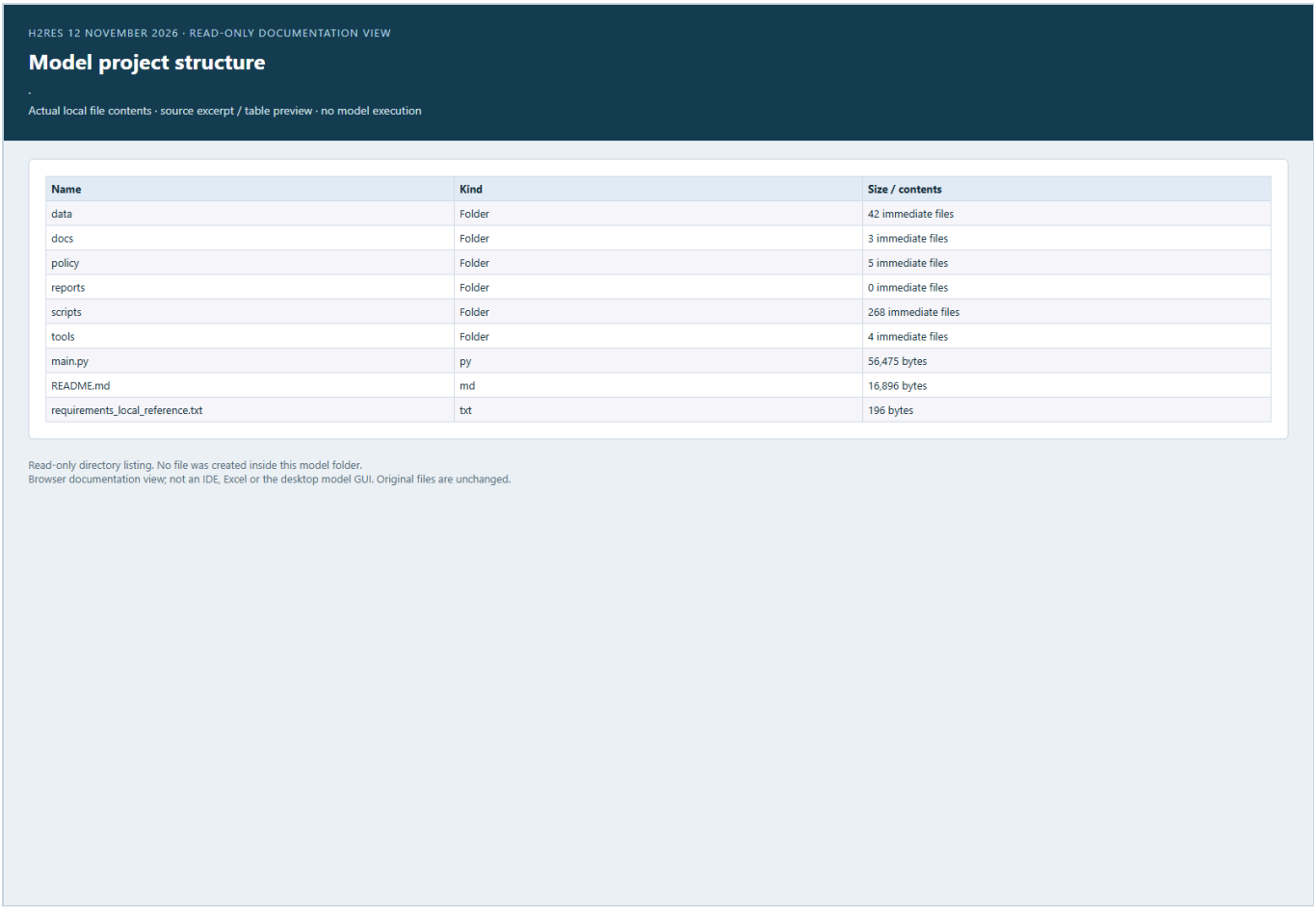
For complete configuration instructions, continue to the main configuration guide. It covers all module and time-mode groups, table-level activation, dependencies, recipe variations and interpretation. The complete option reference explains every recorded setting; the equation chapter provides numbered physical/economic relationships and their source locations.

Screenshots below are browser captures of explicitly labelled, read-only documentation views of the actual local source, directory listings and worksheet cells. They are not screenshots of an IDE or Excel, and they do not depict optimization results. Worksheet previews show stored formulas without recalculating them. Full filenames, source hashes and screenshot hashes are recorded in the capture manifest. Nothing was changed in the model to make these images.

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1. Where to start

Open the complete `H2RES_12.11.2026` folder in your editor. The entry point is `main.py`; it loads the policy files, selects inputs, reads data, constructs optimization problems and exports results. Running it starts optimization. Do not run a `Build_model*.py` or cost/solver script independently: these expect variables created by earlier stages in the shared execution namespace.



Model project structure

The folders most frequently used are:

Location	What to open it for
`main.py`	Scenario name, feature switches, time mode, input selection, execution
`policy/`	Transition, economic, transport and numerical assumptions
`data/`	Native CSV and Excel inputs for the selected configuration
`scripts/`	Readers, variables, constraints, objective, solver and exporters
`docs/`	Supplied module descriptions; this guide complements those notes
`reports/`	Existing diagnostics and integration evidence; inspect their stated scope
`results/<scenario>/`	Results created when the selected run executes; location depends on mode

Root-level `diagnose_*`, rebuild and acceptance scripts are maintenance tools. Their presence does not make them the normal launch command. The supplied `RUN_INSTRUCTIONS.txt` includes historical rebuild-harness instructions; use `main.py` for the workflow described here.

2. Install and launch the model environment

The model environment is separate from the data-preparation application's environment. The local reference specifies NumPy 2.4.4, pandas 3.0.2, openpyxl 3.1.5, matplotlib 3.10.8, Plotly 6.3.0 and gurobipy 12.0.3. It is a local version reference, not a complete cross-platform environment lock. Gurobi also needs a suitable license for the problem size.

Example PowerShell commands, entered in the model folder:

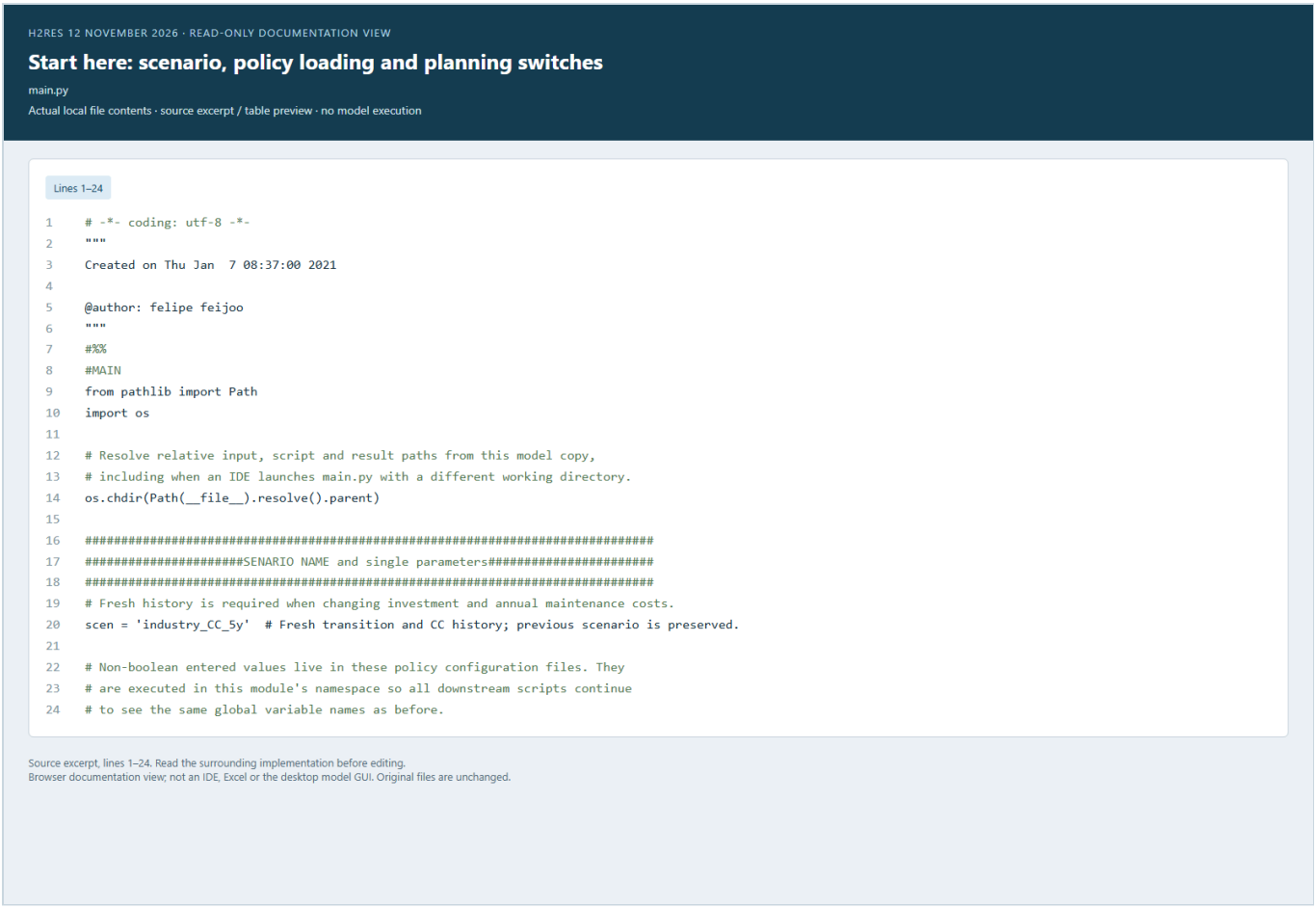
```
Set-Location 'D:\POSAO\H2RES_12.11.2026'
py -3.12 -m venv .venv
& '.\.venv\Scripts\python.exe' -m pip install -r requirements_local_reference.txt
& '.\.venv\Scripts\python.exe' -c "import numpy, pandas, openpyxl, matplotlib, plotly, gurobipy; print('Imports OK')"
```

Review the scenario and inputs described below before the final launch:

```
& '.\.venv\Scripts\python.exe' -u main.py
```

If a configured model environment already exists, select that interpreter rather than creating another one. In an editor, run `main.py` with that same interpreter. The script resolves its working directory to its own model folder, including when an editor starts it from elsewhere. Do not import `main.py` as a quick package test: its top-level execution can start a solve.

3. Edit the scenario and main switches



Scenario name and entry script

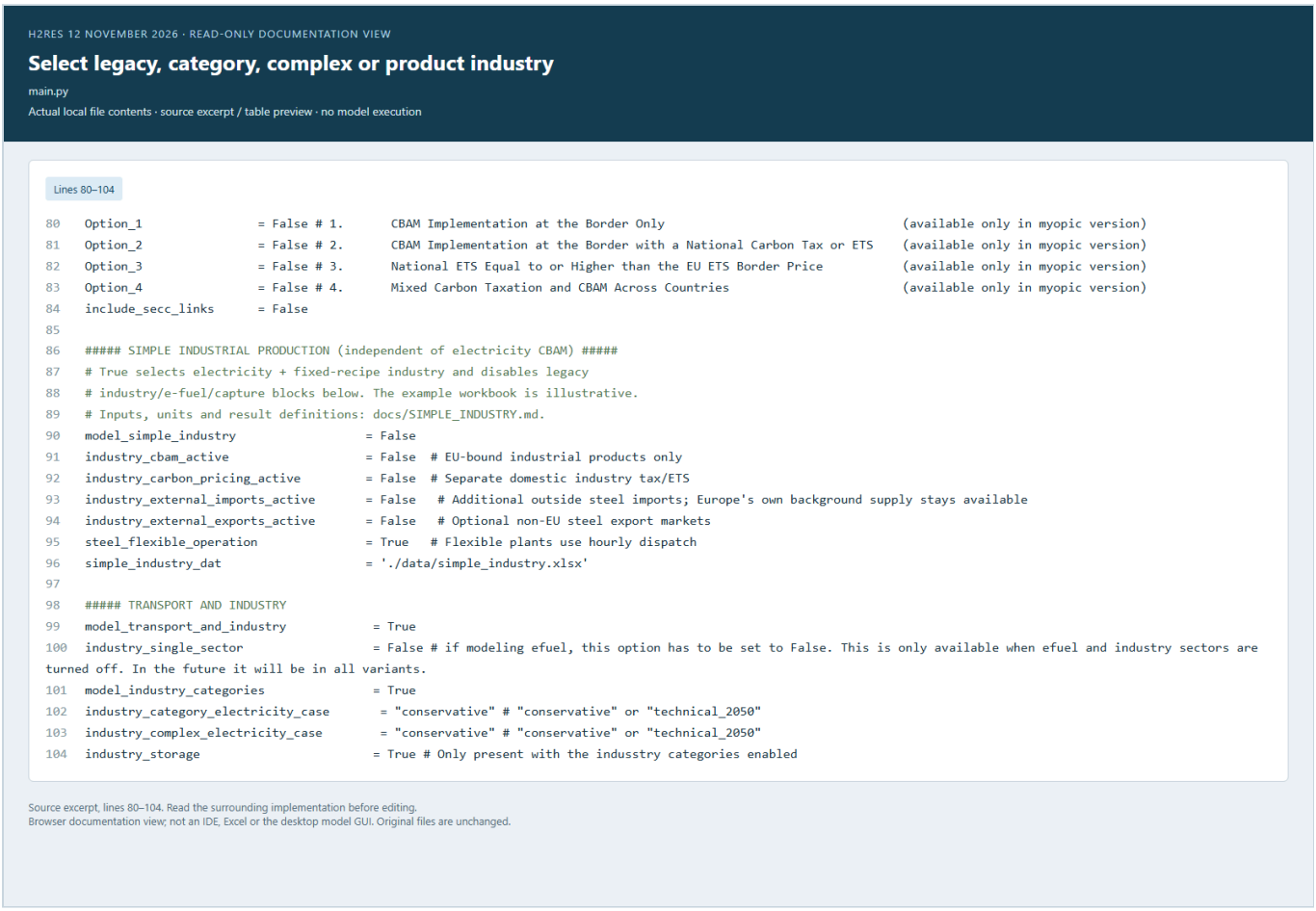
Set `scen` to a new name for a materially different study, for example `HR_categories_baseline_2026_11_12`. The supplied name is `industry_CC_5y`. A name is a results/history identifier; its text does not configure the actual year grid or policy. Those come from switches and input tables.

The delivered selection is:

Control	Inspected value	Consequence
`Myopic`	`True`	Sequential planning with historical handoff
`partial_perfect_foresight`	`False`	Partial-foresight sampling is not selected
`model_heating`	`True`	Heating blocks are enabled
`model_transport_and_industry`	`True`	Legacy transport/industry blocks are enabled
`model_industry_categories`	`True`	Industry is represented by categories
`model_complex_industry`	`False`	Complex process replacement is not selected
`industry_category_electricity_case`	`conservative`	Main category technology workbook is used
`model_efuel`	`True`	E-fuel technology and policy families are selected
`model_simple_industry`	`False`	Separate product-industry workbook is not read

Control	Inspected value	Consequence
`DSM_day` , `DSM_week`	Both `False`	Basic electricity load is not shifted by DSM
`disable_transport_infrastructure`	`False`	This debug switch does not suppress infrastructure
`myopic_resume_from_results`	`True`	Compatible completed history may be reused

Exact capitalization matters. The configuration inventory records all inspected assignments in model_configuration.csv. This table is a snapshot, not a recommendation to leave every setting unchanged.



Industry selection switches

Regular, category and complex industry require a coherent selection. Categories and complex industry are mutually exclusive. Category thermal storage requires categories. Simple product industry configures a different run and disables overlapping legacy blocks. Standalone H2 transmission and terminals are disabled in the delivered e-fuel selection; do not infer activity just because the relevant file is present.

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4. Policy files you will frequently edit

`main.py` executes the policy modules before deriving most run selections. Use these files for their corresponding settings; a workbook does not replace a Python policy switch.

H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

Physical and transition assumptions

policy/model_policy.py

Actual local file contents · source excerpt / table preview · no model execution

Lines 1–26

```
"""Non-boolean model and policy inputs loaded by main.py.

Keep scenario switches and input-file paths in main.py.  This file contains
the entered values that would otherwise make the main entry point difficult to
review.  It is executed in main.py's global namespace, so the variable names
remain fully backward compatible with the existing scripts.
"""

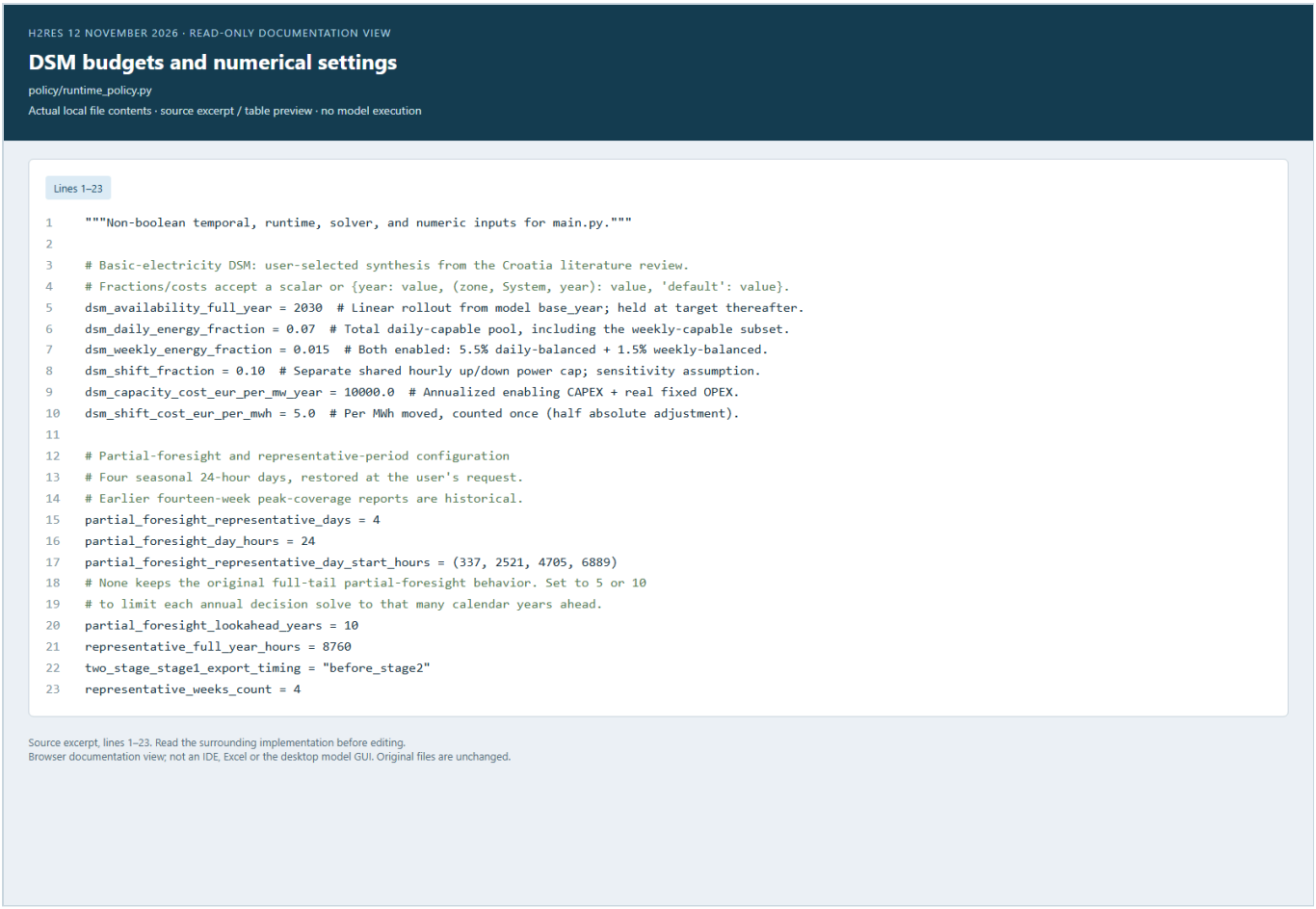
# Annual generation build limits: the validated OPTIMAL 2045 case doubled
# these four workbook limits. Total installed-capacity ceilings stay separate.
generation_annual_build_limit_multipliers = {
    "HR_Wind": 2.0,
    "HR_Wind_1": 2.0,
    "HR_PV": 2.0,
    "HR_PV_1": 2.0,
}

# Land-use and heating assumptions
dh_chp_heat_max_annual_change = 0.25  # Both directions; fixed fraction of each plant's base-year heat output
dh_chp_heat_reduction_limit_years = 10  # Limit decreases through base_year + 10; increases remain limited in all years
dh_chp_heat_history_tolerance_mwh = 1.0  # Numerical tolerance for completed history only; model limits stay exact
individual_gas_heat_max_annual_change = 0.25  # Both directions; fraction of each individual gas boiler's base-year heat output
individual_gas_heat_reduction_limit_years = 5  # Limit decreases through base_year + 5; increases remain limited in all years
fossil_heating_history_tolerance_mwh = 1.0  # Numerical tolerance when checking already completed annual heat history
fossil_boiler_phaseout_years = {"coal": 2030, "gas": 2035, "oil": 2035}  # Linear annual heat caps from base-year output; no CHP
dh_chp_base_year_heat_targets_twh = {
```

Source excerpt, lines 1–26. Read the surrounding implementation before editing.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

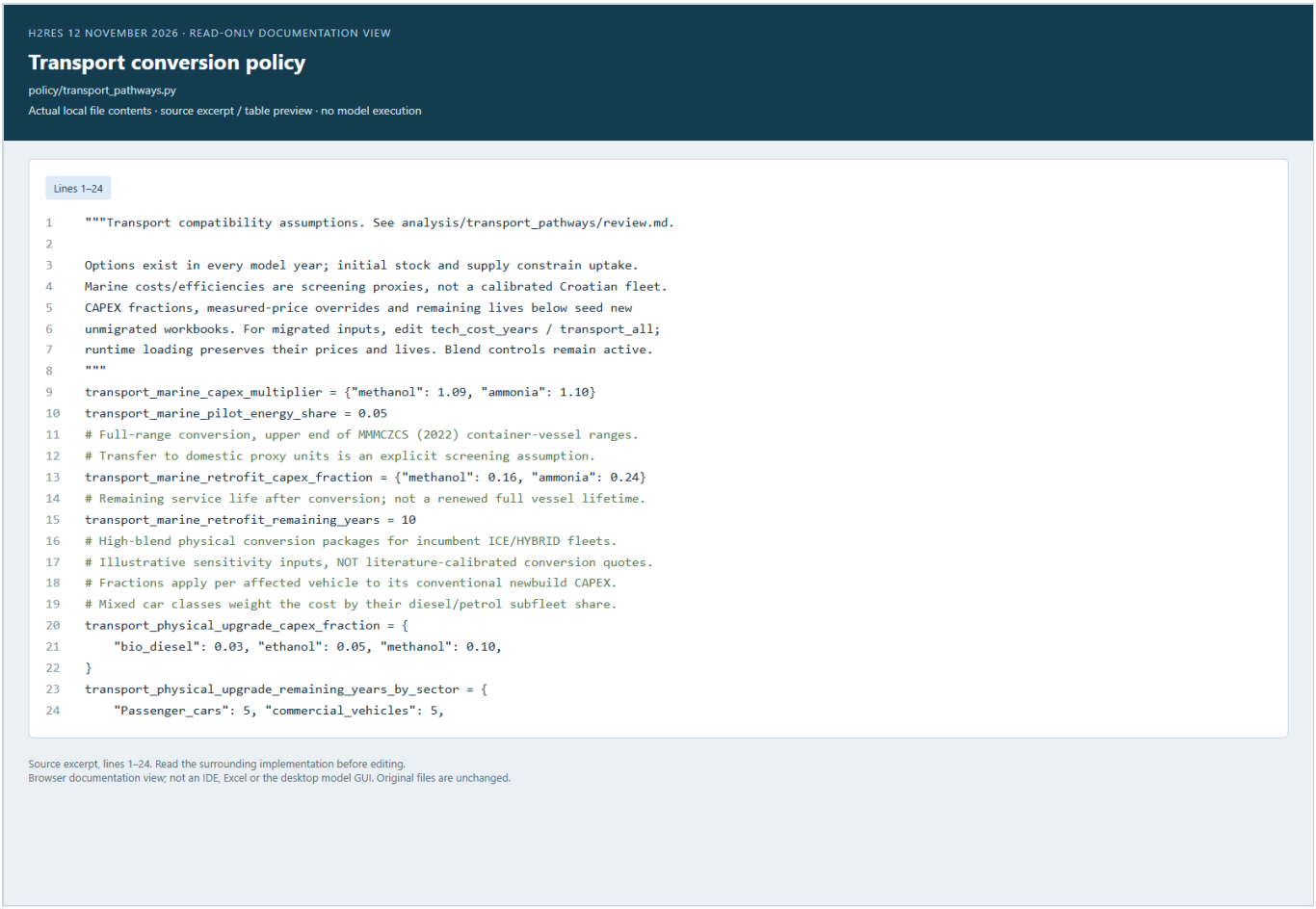
Model policy assumptions

`policy/model_policy.py` contains physical/economic and transition assumptions. In this snapshot these include the industry's 2045 fossil-combustion transition completion year, heating turnover restrictions, annual investment multipliers, discounting and active systems for the product-industry example. The discount rate, discount base year and capital finance rate must remain consistent with the way costs are reported. Fossil combustion retirement is not a claim of zero industrial process emissions.



Runtime policy and DSM parameters

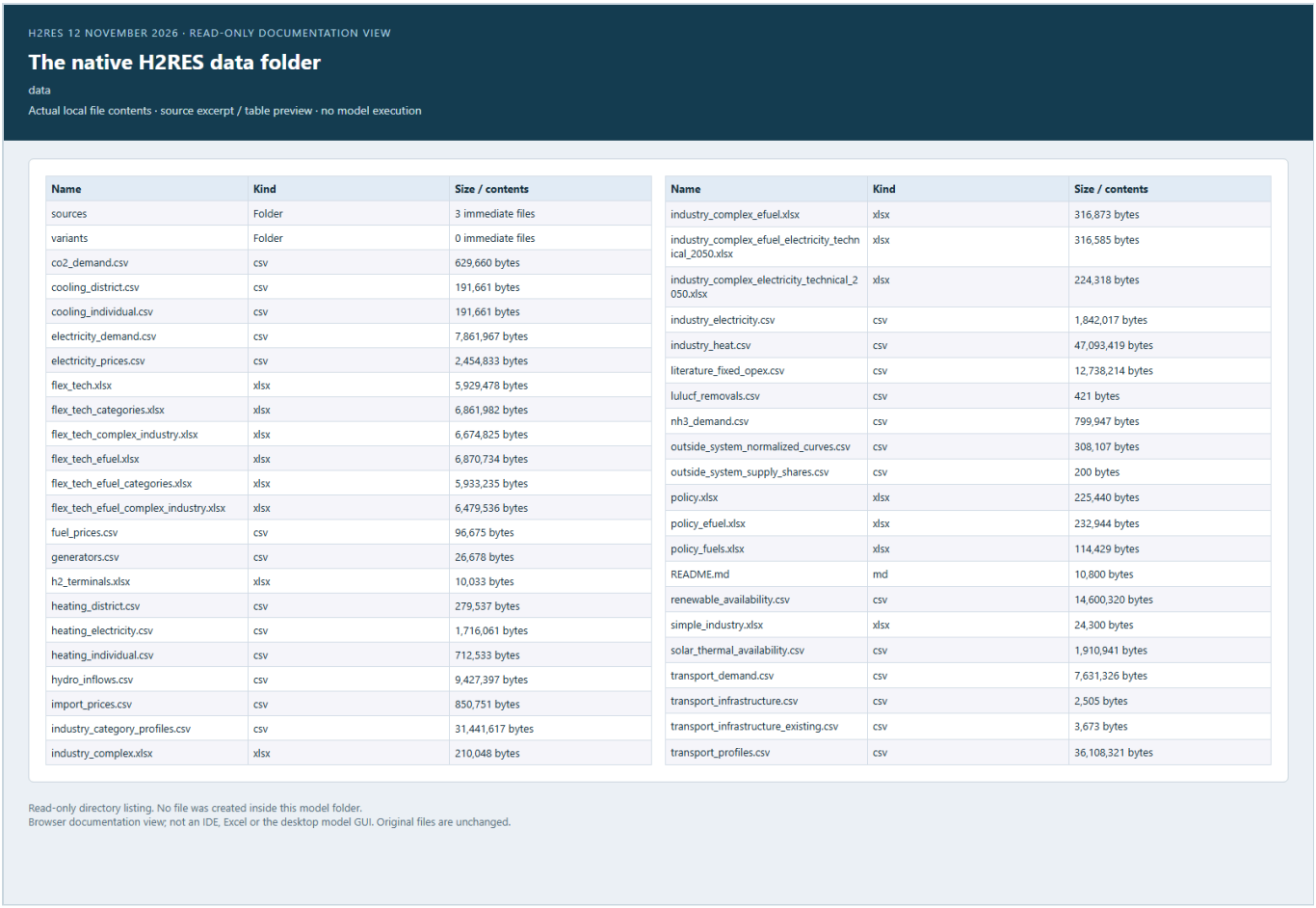
`policy/runtime_policy.py` contains solver methods, tolerances, numerical scaling, time limits, recovery settings and DSM parameters. In this snapshot stage-1 primary Method is 2, rolling Method is 1, and several time limits are zero, meaning unlimited in the applicable wrappers. A long solve is therefore not necessarily stuck. Changing tolerances can change acceptance; record numerical settings with the result.



Transport pathway policy

`policy/transport_pathways.py` contains transport conversion and compatibility assumptions. Keep these consistent with `transport_all`, `transport_ICE_prim` where used, the annual transport targets and the hourly service profiles.

5. The data folder and exact workbook selection



H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

The native H2RES data folder

data

Actual local file contents · source excerpt / table preview · no model execution

Name	Kind	Size / contents
sources	Folder	3 immediate files
variants	Folder	0 immediate files
co2_demand.csv	csv	629,660 bytes
cooling_district.csv	csv	191,661 bytes
cooling_individual.csv	csv	191,661 bytes
electricity_demand.csv	csv	7,861,967 bytes
electricity_prices.csv	csv	2,454,833 bytes
flex_tech.xlsx	xlsx	5,929,478 bytes
flex_tech_categories.xlsx	xlsx	6,861,982 bytes
flex_tech_complex_industry.xlsx	xlsx	6,674,825 bytes
flex_tech_efuel.xlsx	xlsx	6,870,734 bytes
flex_tech_efuel_categories.xlsx	xlsx	5,933,235 bytes
flex_tech_efuel_complex_industry.xlsx	xlsx	6,479,536 bytes
fuel_prices.csv	csv	96,675 bytes
generators.csv	csv	26,678 bytes
h2_terminals.xlsx	xlsx	10,033 bytes
heating_district.csv	csv	279,537 bytes
heating_electricity.csv	csv	1,716,061 bytes
heating_individual.csv	csv	712,533 bytes
hydro_inflows.csv	csv	9,427,397 bytes
import_prices.csv	csv	850,751 bytes
industry_category_profiles.csv	csv	31,441,617 bytes
industry_complex.xlsx	xlsx	210,048 bytes

Name	Kind	Size / contents
industry_complex_efuel.xlsx	xlsx	316,873 bytes
industry_complex_efuel_electricity_technical_2050.xlsx	xlsx	316,585 bytes
industry_complex_electricity_technical_2050.xlsx	xlsx	224,318 bytes
industry_electricity.csv	csv	1,842,017 bytes
industry_heat.csv	csv	47,093,419 bytes
literature_fixed_opex.csv	csv	12,738,214 bytes
lulucf_removals.csv	csv	421 bytes
nh3_demand.csv	csv	799,947 bytes
outside_system_normalized_curves.csv	csv	308,107 bytes
outside_system_supply_shares.csv	csv	200 bytes
policy.xlsx	xlsx	225,440 bytes
policy_efuel.xlsx	xlsx	232,944 bytes
policy_fuels.xlsx	xlsx	114,429 bytes
README.md	md	10,800 bytes
renewable_availability.csv	csv	14,600,320 bytes
simple_industry.xlsx	xlsx	24,300 bytes
solar_thermal_availability.csv	csv	1,910,941 bytes
transport_demand.csv	csv	7,631,326 bytes
transport_infrastructure.csv	csv	2,505 bytes
transport_infrastructure_existing.csv	csv	3,673 bytes
transport_profiles.csv	csv	36,108,321 bytes

Read-only directory listing. No file was created inside this model folder.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

Native model data directory

Keep CSVs and workbooks together under `data/`, including the nested `variants/technical_2050/` folder. The model reads specific relative paths. Renaming a workbook to look more convenient does not change the file-selection code. The inspected input inventory contains 46 files, including nested source records and variants; workbook inventories cover 774 worksheets across the supplied workbooks.

H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

How the model selects its input workbooks

main.py

Actual local file contents · source excerpt / table preview · no model execution

Lines 316–342

```
316  flex_tech_directory = './data'
317  if model_efuel:
318      flex_tech_name += '_efuel'
319  if model_complex_non_metallic_minerals:
320      flex_tech_name += '_complex_industry'
321  if model_industry_categories:
322      flex_tech_name += '_categories'
323  # Select the electricity-limit scenario for e-fuel industry categories.
324  if model_efuel and model_industry_categories and not model_complex_non_metallic_minerals:
325      if industry_category_electricity_case not in ("conservative", "technical_2050"):
326          raise ValueError("industry_category_electricity_case must be conservative or technical_2050")
327      if industry_category_electricity_case == "technical_2050":
328          flex_tech_directory += '/variants/technical_2050'
329  flex_tech_dat = flex_tech_directory + '/' + flex_tech_name + '.xlsx'
330
331  heat_dem_industry_dat = (
332      './data/industry_category_profiles.csv'
333      if model_industry_categories
334      else './data/industry_heat.csv'
335  )
336  if model_efuel:
337      policy_dat_fuels = './data/policy_fuels.xlsx'
338      if exogeneous_NH3_demand:
339          NH3_demand_dat = './data/nh3_demand.csv'
340  else:
341      demand_co2 = './data/co2_demand.csv'
342
```

Source excerpt, lines 316–342. Read the surrounding implementation before editing.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

Input file-selection code

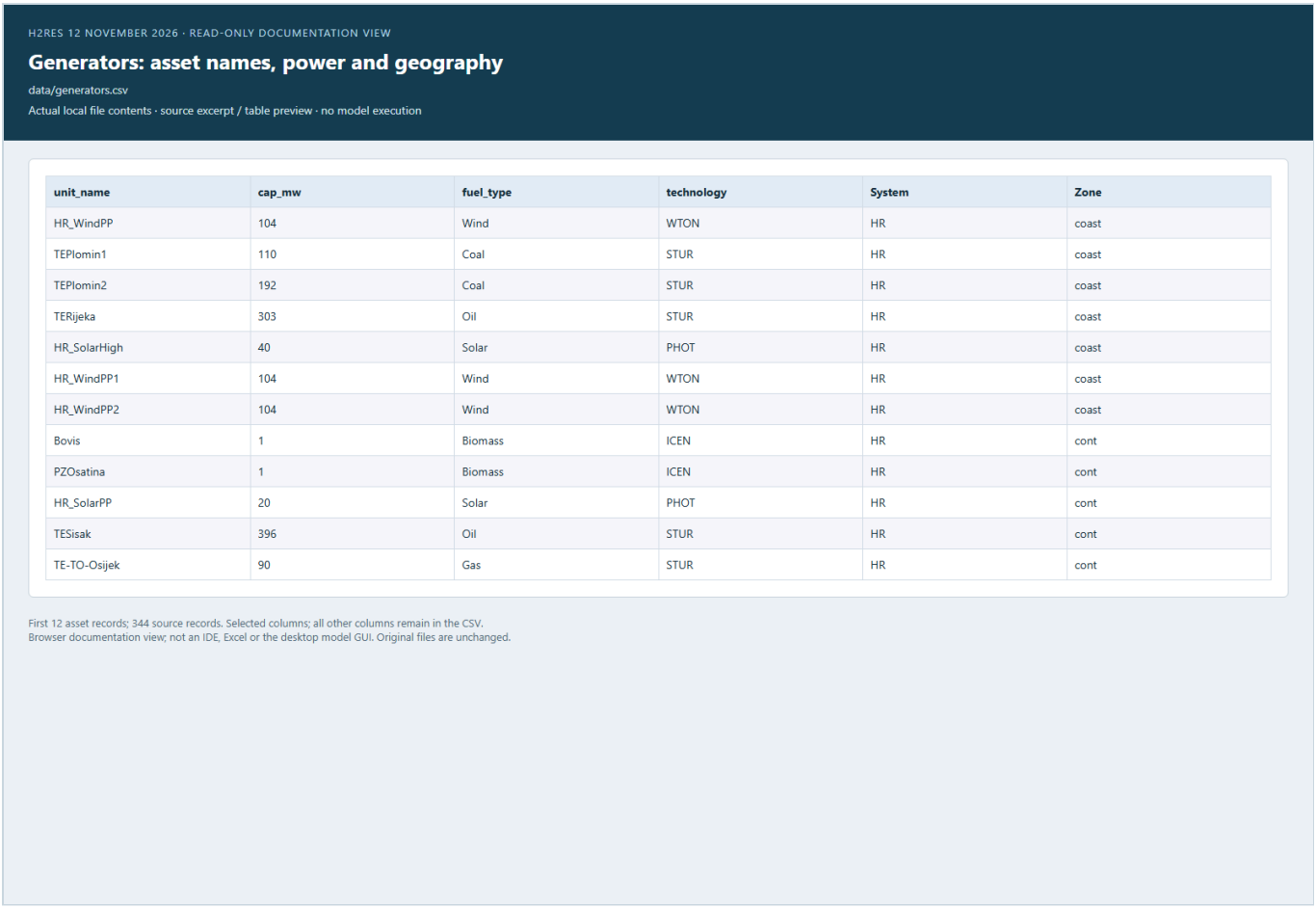
The normal technology selection is:

Industry representation	`model_efuel=False`	`model_efuel=True`
Regular: categories and complex both false	`flex_tech.xlsx`	`flex_tech_efuel.xlsx`
Categories	`flex_tech_categories.xlsx`	`flex_tech_efuel_categories.xlsx`
Complex	`flex_tech_complex_industry.xlsx`	`flex_tech_efuel_complex_industry.xlsx`

For e-fuel categories with `industry_category_electricity_case='technical_2050'`, the technology path is `data/variants/technical_2050/flex_tech_efuel_categories.xlsx`. The main conservative path has the same basename directly under `data/`. Complex electricity assumptions select `industry_complex*_electricity_technical_2050.xlsx`; they do not rename the associated flex-tech workbook.

`model_efuel=True` selects `policy_efuel.xlsx`, otherwise `policy.xlsx`. E-fuel carrier price assumptions additionally use `policy_fuels.xlsx`. Category demand timing uses `industry_category_profiles.csv`; ordinary industry timing uses `industry_heat.csv`. Follow the selected filenames printed by the reader.

6. Generators and their related tables



H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

Generators: asset names, power and geography

data/generators.csv

Actual local file contents · source excerpt / table preview · no model execution

unit_name	cap_mw	fuel_type	technology	System	Zone
HR_WindPP	104	Wind	WTON	HR	coast
TEPlomin1	110	Coal	STUR	HR	coast
TEPlomin2	192	Coal	STUR	HR	coast
TERijeka	303	Oil	STUR	HR	coast
HR_SolarHigh	40	Solar	PHOT	HR	coast
HR_WindPP1	104	Wind	WTON	HR	coast
HR_WindPP2	104	Wind	WTON	HR	coast
Bovis	1	Biomass	ICEN	HR	cont
PZOsatina	1	Biomass	ICEN	HR	cont
HR_SolarPP	20	Solar	PHOT	HR	cont
TESisak	396	Oil	STUR	HR	cont
TE-TO-Osijek	90	Gas	STUR	HR	cont

First 12 asset records; 344 source records. Selected columns; all other columns remain in the CSV.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

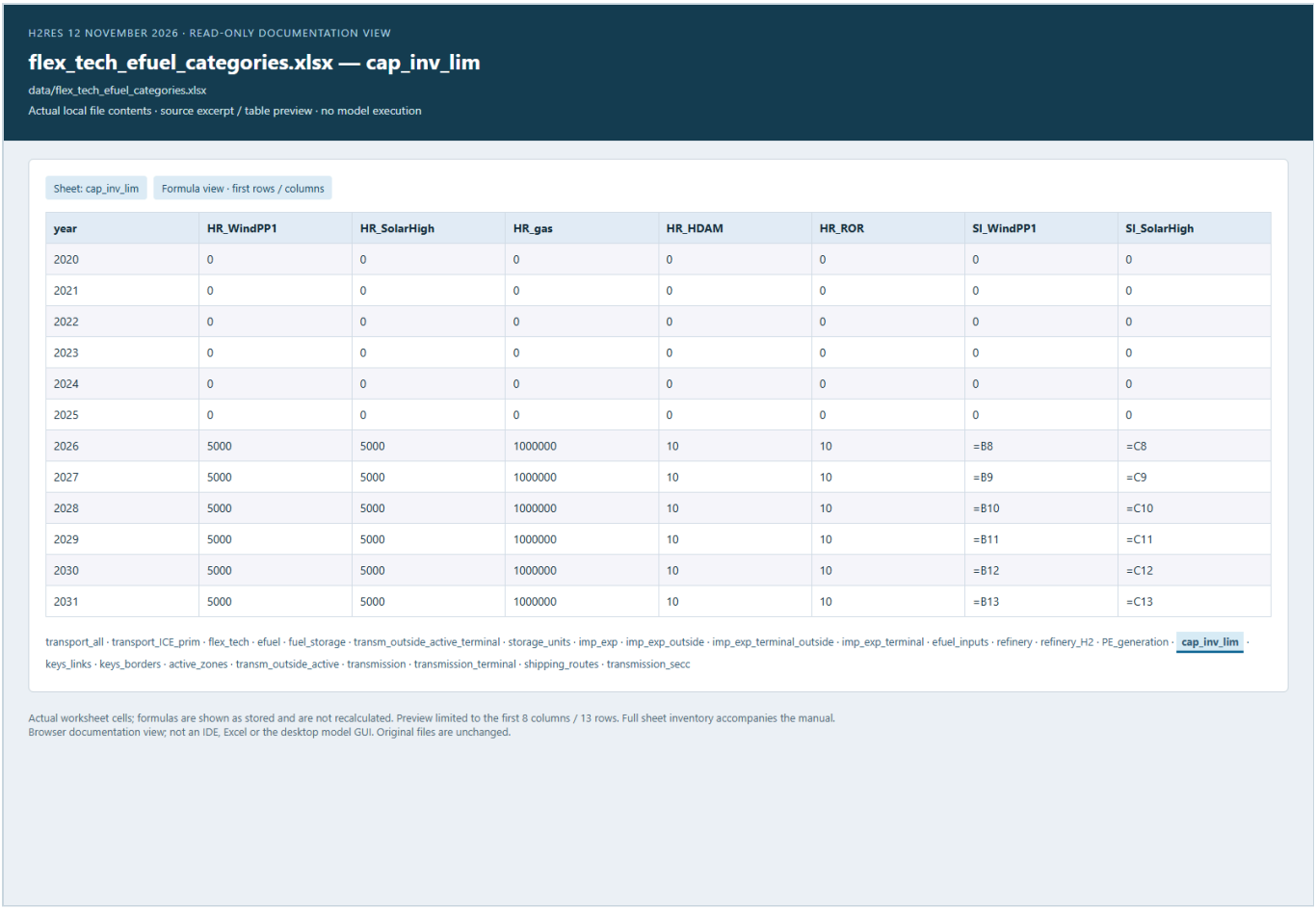
Generator asset preview

`data/generators.csv` is the native generator inventory. Each asset has an exact `unit_name`, installed `cap_mw`, fuel/technology and `System` / `Zone`. Other native fields describe efficiency, costs, ramping, CHP heat coupling, lifetime/retirement and hydro storage characteristics. MW power and GWh storage energy are different quantities.

A generator must be represented consistently wherever its identifier is used. When adding or renaming a unit, review the following relationships:

Related input	What must agree
`active_zones`	The asset's zone and System belong to the intended model geography
`renewable_availability.csv`	Required VRE/run-of-river curve columns use exact generator IDs
`hydro_inflows.csv`	Relevant reservoir/pumped-hydro columns use the required exact IDs
`cap_inv_lim`	Applicable unit, geography and annual investment coverage are present
`tech_cost_years`	Applicable generator/technology capital-cost trajectory is unambiguous
`fixedOPEX_cost_years`	Applicable fixed-cost trajectory and units agree
Policy calibration tables	Generation and hydro targets match actual modeled assets and fuels

These are table-specific relationships: not every workbook is stored as a simple `(year,unit_name,zone,System)` table. Some native sheets contain multiple stacked/wide blocks. Follow the input reader and preserve its keys rather than adding a superficially similar column. A zone with no existing generator can still have demand and receive electricity through modeled links; do not invent a power plant merely to populate every zone.



H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

flex_tech_efuel_categories.xlsx — cap_inv_lim

data/flex_tech_efuel_categories.xlsx

Actual local file contents · source excerpt / table preview · no model execution

Sheet: cap_inv_lim | Formula view · first rows / columns

year	HR_WindPP1	HR_SolarHigh	HR_gas	HR_HDAM	HR_ROR	SI_WindPP1	SI_SolarHigh
2020	0	0	0	0	0	0	0
2021	0	0	0	0	0	0	0
2022	0	0	0	0	0	0	0
2023	0	0	0	0	0	0	0
2024	0	0	0	0	0	0	0
2025	0	0	0	0	0	0	0
2026	5000	5000	1000000	10	10	=B8	=C8
2027	5000	5000	1000000	10	10	=B9	=C9
2028	5000	5000	1000000	10	10	=B10	=C10
2029	5000	5000	1000000	10	10	=B11	=C11
2030	5000	5000	1000000	10	10	=B12	=C12
2031	5000	5000	1000000	10	10	=B13	=C13

transport_all · transport_ICE_prim · flex_tech · efuel · fuel_storage · transm_outside_active_terminal · storage_units · imp_exp · imp_exp_outside · imp_exp_terminal_outside · imp_exp_terminal · efuel_inputs · refinery · refinery_H2 · PE_generation · **cap_inv_lim** · keys_links · keys_borders · active_zones · transm_outside_active · transmission · transmission_terminal · shipping_routes · transmission_secc

Actual worksheet cells; formulas are shown as stored and are not recalculated. Preview limited to the first 8 columns / 13 rows. Full sheet inventory accompanies the manual.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

Investment limit worksheet

`cap_inv_lim` controls additions in the units expected for each asset family. Zero prevents additions for the applicable entry; it does not remove existing capacity. Sparse decision years require correct aggregation of underlying annual limits. A blank is not universally equivalent to zero: the reader's scalar fallback and unrestricted-limit conventions matter.

H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

flex_tech_efuel_categories.xlsx — tech_cost_years

data/flex_tech_efuel_categories.xlsx

Actual local file contents · source excerpt / table preview · no model execution

Sheet: tech_cost_years | Formula view · first rows / columns

year	Air_to_water_HP	Ground_to_water_HP	Water_to_water_HP	Electric_heater	Air_to_water_HP_DH	Ground_to_water_HP_DH	Water_to_water_HP_DH
2020	0.892	1.168	0.48	0.626	0.892	1.168	0.48
2021	0.8832	1.1564	0.48	0.6234	0.8832	1.1564	0.48
2022	0.8744	1.1448	0.48	0.6208	0.8744	1.1448	0.48
2023	0.8656	1.1332	0.48	0.6182	0.8656	1.1332	0.48
2024	0.8568	1.1216	0.48	0.6156	0.8568	1.1216	0.48
2025	1.83837610111069	2.3285933140012	0.48	0.289597855227882	1.1	1.11	1.19
2026	1.7887399463807	2.26909739152373	0.46	0.288164273625858	1.0945	1.0984	1.1781
2027	1.73910379165071	2.20960146904625	0.44	0.286730692023834	1.089	1.0868	1.1662
2028	1.68946763692072	2.15010554656878	0.42	0.28529711042181	1.0835	1.0752	1.1543
2029	1.63983148219073	2.0906096240913	0.4	0.283863528819785	1.078	1.0636	1.1424
2030	1.59019532746074	2.03111370161383	0.38	0.282429947217761	1.0725	1.052	1.1305
2031	1.63363079279969	2.08920522487077	0.38	0.281031848547889	1.067	1.0466	1.1186

transport_all · transport_ICE_prim · flex_tech · efuel · fuel_storage · transm_outside_active_terminal · storage_units · imp_exp · imp_exp_outside · imp_exp_terminal_outside · imp_exp_terminal · efuel_inputs · refinery · refinery_H2 · PE_generation · cap_inv_lim · keys_links · keys_borders · active_zones · transm_outside_active · transmission · transmission_terminal · shipping_routes · transmission_secc

Actual worksheet cells; formulas are shown as stored and are not recalculated. Preview limited to the first 8 columns / 13 rows. Full sheet inventory accompanies the manual.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

Technology cost worksheet

`tech_cost_years` supplies cost trajectories. Confirm the physical unit and price basis before transferring a row between generation, energy storage, heating, vehicles or capture. A unit may not have two conflicting applicable trajectories. Check costs and investment limits together whenever the GENCO inventory changes.

H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

flex_tech_efuel_categories.xlsx — active_zones

data/flex_tech_efuel_categories.xlsx

Actual local file contents · source excerpt / table preview · no model execution

Sheet: active_zones | Formula view · first rows / columns

zone	System	active	implemented into the model	NUTS		System	active
DE1	DE	N	Y	1		DE	=Table3[[#This Row], [active]]
RS1	RS	N	Y	1		RS	=Table3[[#This Row], [active]]
IT1	IT	N	Y	2		IT	=Table3[[#This Row], [active]]
UA1	UA	N	Y	1		UA	=Table3[[#This Row], [active]]
HU1	HU	N	Y	1		HU	=Table3[[#This Row], [active]]
FR1	FR	N	Y	1		FR	=Table3[[#This Row], [active]]
AT1	AT	N	Y	1		AT	=Table3[[#This Row], [active]]
UK1	UK	N	Y	1		UK	=Table3[[#This Row], [active]]
NO1	NO	N	Y	1		NO	=Table3[[#This Row], [active]]
SE1	SE	N	Y	1		SE	=Table3[[#This Row], [active]]
PL1	PL	N	Y	1		PL	=Table3[[#This Row], [active]]
RU1	RU	N	Y	1		RUS	=Table3[[#This Row], [active]]

transport_all · transport_ICE_prim · flex_tech · efuel · fuel_storage · transm_outside_active_terminal · storage_units · imp_exp · imp_exp_outside · imp_exp_terminal_outside · imp_exp_terminal · efuel_inputs · refinery · refinery_H2 · PE_generation · cap_inv_lim · keys_links · keys_borders · **active_zones** · transm_outside_active · transmission · transmission_terminal · shipping_routes · transmission_secc

Active geography worksheet

`active_zones` contains retained countries and zones as well as selection columns. A country appearing in this worksheet is not automatically active. Inspect `active` and the reader's scoping rules, then ensure matching demand, transmission, heating, industry and transport inputs. The preview starts at the first stored rows; inactive rows in a screenshot do not imply the model has no active zones.

7. Flex-tech variations: what changes and what to review

7.1 Regular industry without e-fuels



H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

flex_tech.xlsx — simple_industry

data/flex_tech.xlsx

Actual local file contents · source excerpt / table preview · no model execution

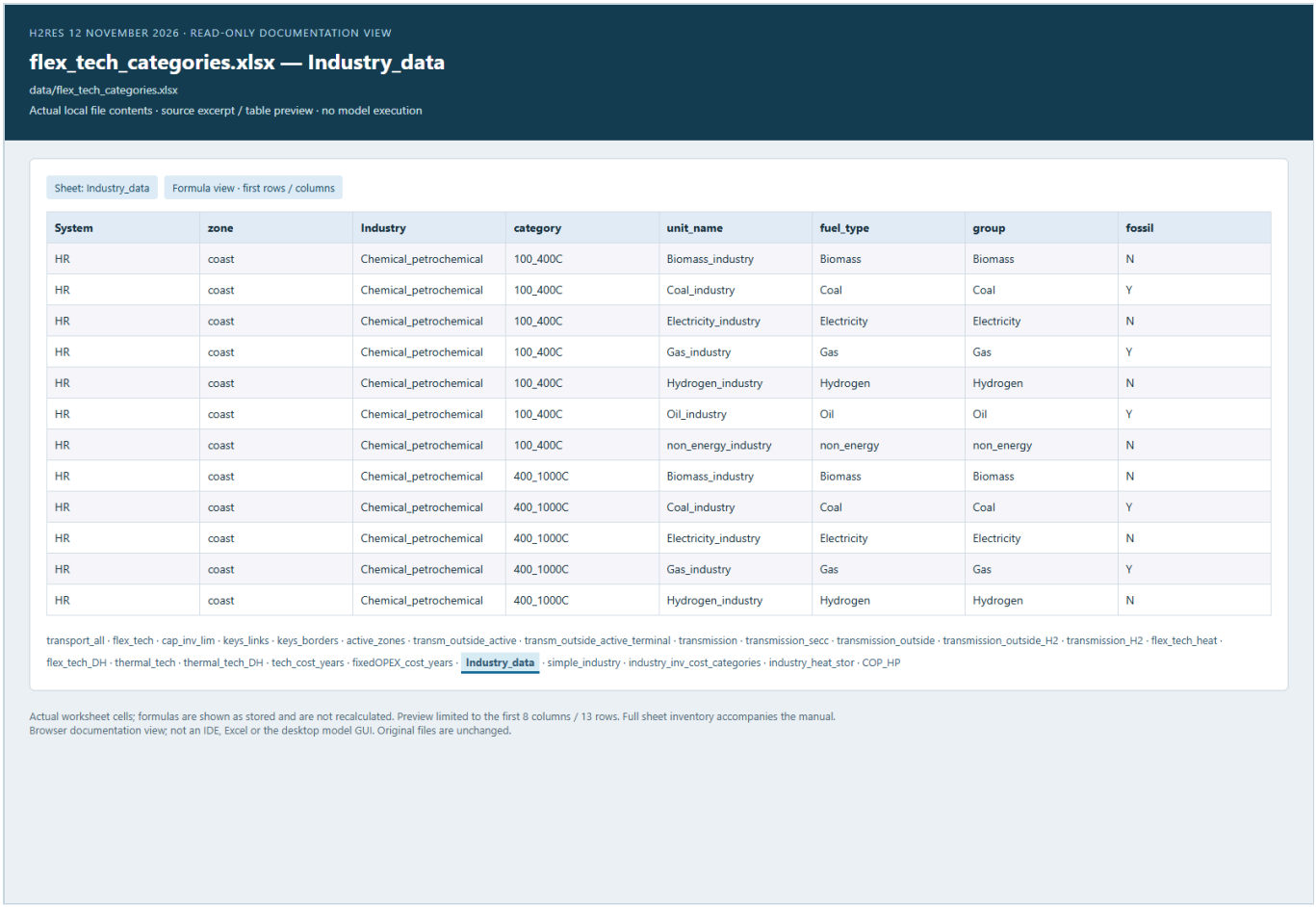
Sheet: simple_industry | Formula view · first rows / columns

System	zone	Industry	unit_name	fuel_type	efficiency	cap_mw	CO2Intensity
HR	cont	remainder	Gas_industry	Gas	0.99	=SUMIFS('_remainder_calc'!$G$2:$G$4408,'_remainder_calc'!$A$2:$A$4408,$A2,'_remainder_calc'!$B$2:$B$4408,$B2,'_remainder_calc'!$D$2:$D$4408,$D2)	0.23
HR	cont	remainder	Coal_industry	Coal	0.99	=SUMIFS('_remainder_calc'!$G$2:$G$4408,'_remainder_calc'!$A$2:$A$4408,$A3,'_remainder_calc'!$B$2:$B$4408,$B3,'_remainder_calc'!$D$2:$D$4408,$D3)	0.38
HR	cont	remainder	Biomass_industry	Biomass	0.83	=SUMIFS('_remainder_calc'!$G$2:$G$4408,'_remainder_calc'!$A$2:$A$4408,$A4,'_remainder_calc'!$B$2:$B$4408,$B4,'_remainder_calc'!$D$2:$D$4408,$D4)	0.39
HR	cont	remainder	Electricity_industry	Electricity	0.99	=SUMIFS('_remainder_calc'!$G$2:$G$4408,'_remainder_calc'!$A$2:$A$4408,$A5,'_remainder_calc'!$B$2:$B$4408,$B5,'_remainder_calc'!$D$2:$D$4408,$D5)	0
HR	cont	remainder	Oil_industry	Oil	0.99	=SUMIFS('_remainder_calc'!$G$2:$G$4408,'_remainder_calc'!$A$2:$A$4408,$A6,'_remainder_calc'!$B$2:$B$4408,$B6,'_remainder_calc'!$D$2:$D$4408,$D6)	0.4
HR	cont	remainder	Hydrogen_industry	Hydrogen	0.99	=SUMIFS('_remainder_calc'!$G$2:$G$4408,'_remainder_calc'!$A$2:$A$4408,$A7,'_remainder_calc'!$B$2:$B$4408,$B7,'_remainder_calc'!$D$2:$D$4408,$D7)	0

Regular technology workbook

`flex_tech.xlsx` provides the regular technology layout. The `simple_industry` worksheet is the pooled legacy service representation. It must reconcile with detailed-sector inclusion in `Industry_data`: an excluded detailed service belongs in the pool, while a retained detailed service must not be counted again. Preserve demand and capacity units while recomputing the remainder. A formula visible in a cell does not establish that its cached numeric value is current.

7.2 Category industry without e-fuels



H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

flex_tech_categories.xlsx — Industry_data

data/flex_tech_categories.xlsx

Actual local file contents · source excerpt / table preview · no model execution

Sheet: Industry_data | Formula view · first rows / columns

System	zone	Industry	category	unit_name	fuel_type	group	fossil
HR	coast	Chemical_petrochemical	100_400C	Biomass_industry	Biomass	Biomass	N
HR	coast	Chemical_petrochemical	100_400C	Coal_industry	Coal	Coal	Y
HR	coast	Chemical_petrochemical	100_400C	Electricity_industry	Electricity	Electricity	N
HR	coast	Chemical_petrochemical	100_400C	Gas_industry	Gas	Gas	Y
HR	coast	Chemical_petrochemical	100_400C	Hydrogen_industry	Hydrogen	Hydrogen	N
HR	coast	Chemical_petrochemical	100_400C	Oil_industry	Oil	Oil	Y
HR	coast	Chemical_petrochemical	100_400C	non_energy_industry	non_energy	non_energy	N
HR	coast	Chemical_petrochemical	400_1000C	Biomass_industry	Biomass	Biomass	N
HR	coast	Chemical_petrochemical	400_1000C	Coal_industry	Coal	Coal	Y
HR	coast	Chemical_petrochemical	400_1000C	Electricity_industry	Electricity	Electricity	N
HR	coast	Chemical_petrochemical	400_1000C	Gas_industry	Gas	Gas	Y
HR	coast	Chemical_petrochemical	400_1000C	Hydrogen_industry	Hydrogen	Hydrogen	N

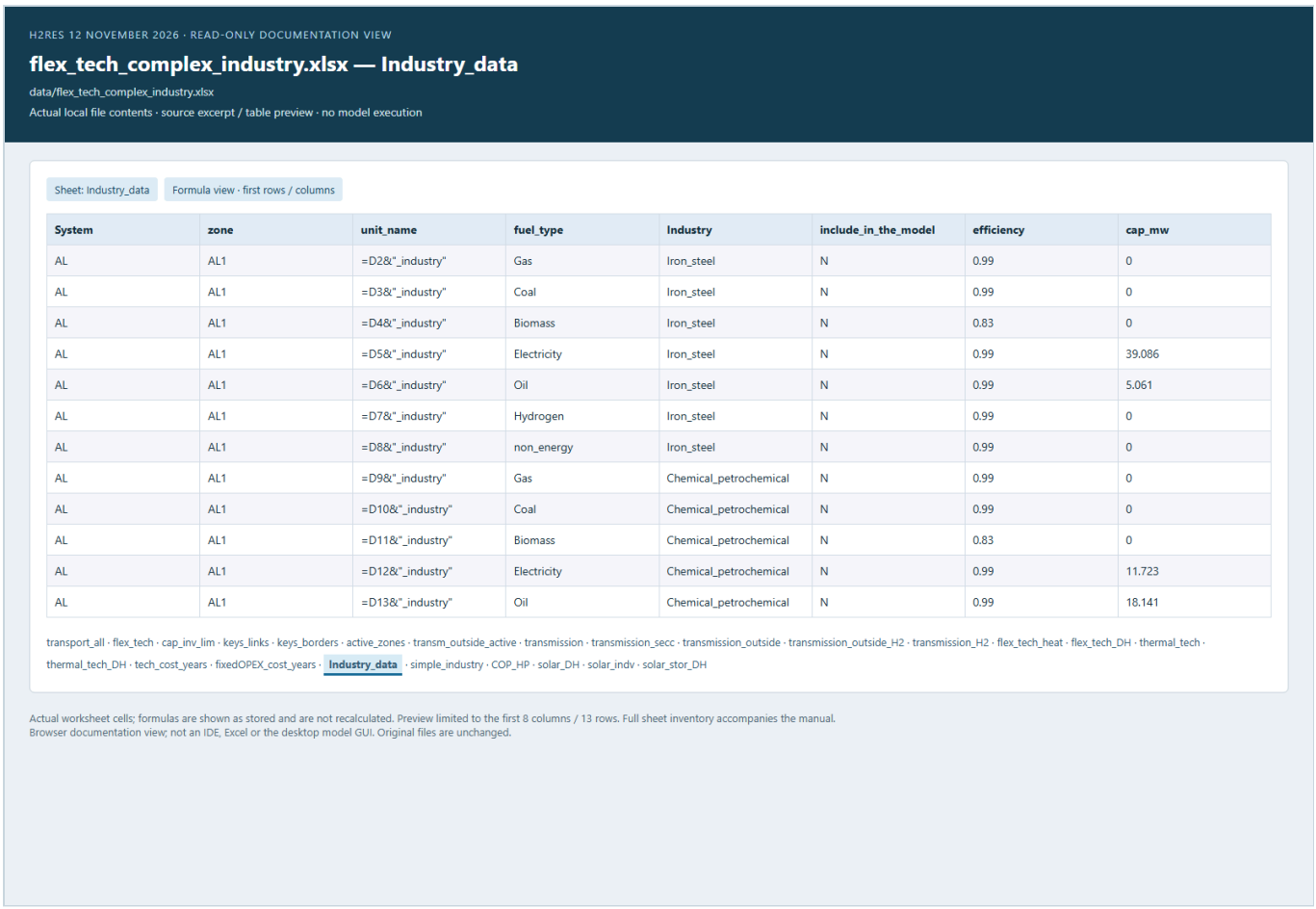
transport_all · flex_tech · cap_inv_lim · keys_links · keys_borders · active_zones · transm_outside_active · transm_outside_active_terminal · transmission · transmission_secc · transmission_outside · transmission_outside_H2 · transmission_H2 · flex_tech_heat · flex_tech_DH · thermal_tech · thermal_tech_DH · tech_cost_years · fixedOPEX_cost_years · **Industry_data** · simple_industry · industry_inv_cost_categories · industry_heat_stor · COP_HP

Actual worksheet cells; formulas are shown as stored and are not recalculated. Preview limited to the first 8 columns / 13 rows. Full sheet inventory accompanies the manual.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

Category technology workbook

`flex_tech_categories.xlsx` distinguishes industry and service/temperature category. Review `Industry_data`, `industry_inv_cost_categories`, applicable heat-pump limits and category thermal storage. The `simple_industry` remainder can legitimately be empty when all applicable services are detailed. Do not populate it merely because an empty sheet looks suspicious.

7.3 Complex industry without e-fuels



H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

flex_tech_complex_industry.xlsx — Industry_data

data/flex_tech_complex_industry.xlsx

Actual local file contents · source excerpt / table preview · no model execution

Sheet: Industry_data | Formula view · first rows / columns

System	zone	unit_name	fuel_type	Industry	include_in_the_model	efficiency	cap_mw
AL	AL1	=D2&"_industry"	Gas	Iron_steel	N	0.99	0
AL	AL1	=D3&"_industry"	Coal	Iron_steel	N	0.99	0
AL	AL1	=D4&"_industry"	Biomass	Iron_steel	N	0.83	0
AL	AL1	=D5&"_industry"	Electricity	Iron_steel	N	0.99	39.086
AL	AL1	=D6&"_industry"	Oil	Iron_steel	N	0.99	5.061
AL	AL1	=D7&"_industry"	Hydrogen	Iron_steel	N	0.99	0
AL	AL1	=D8&"_industry"	non_energy	Iron_steel	N	0.99	0
AL	AL1	=D9&"_industry"	Gas	Chemical_petrochemical	N	0.99	0
AL	AL1	=D10&"_industry"	Coal	Chemical_petrochemical	N	0.99	0
AL	AL1	=D11&"_industry"	Biomass	Chemical_petrochemical	N	0.83	0
AL	AL1	=D12&"_industry"	Electricity	Chemical_petrochemical	N	0.99	11.723
AL	AL1	=D13&"_industry"	Oil	Chemical_petrochemical	N	0.99	18.141

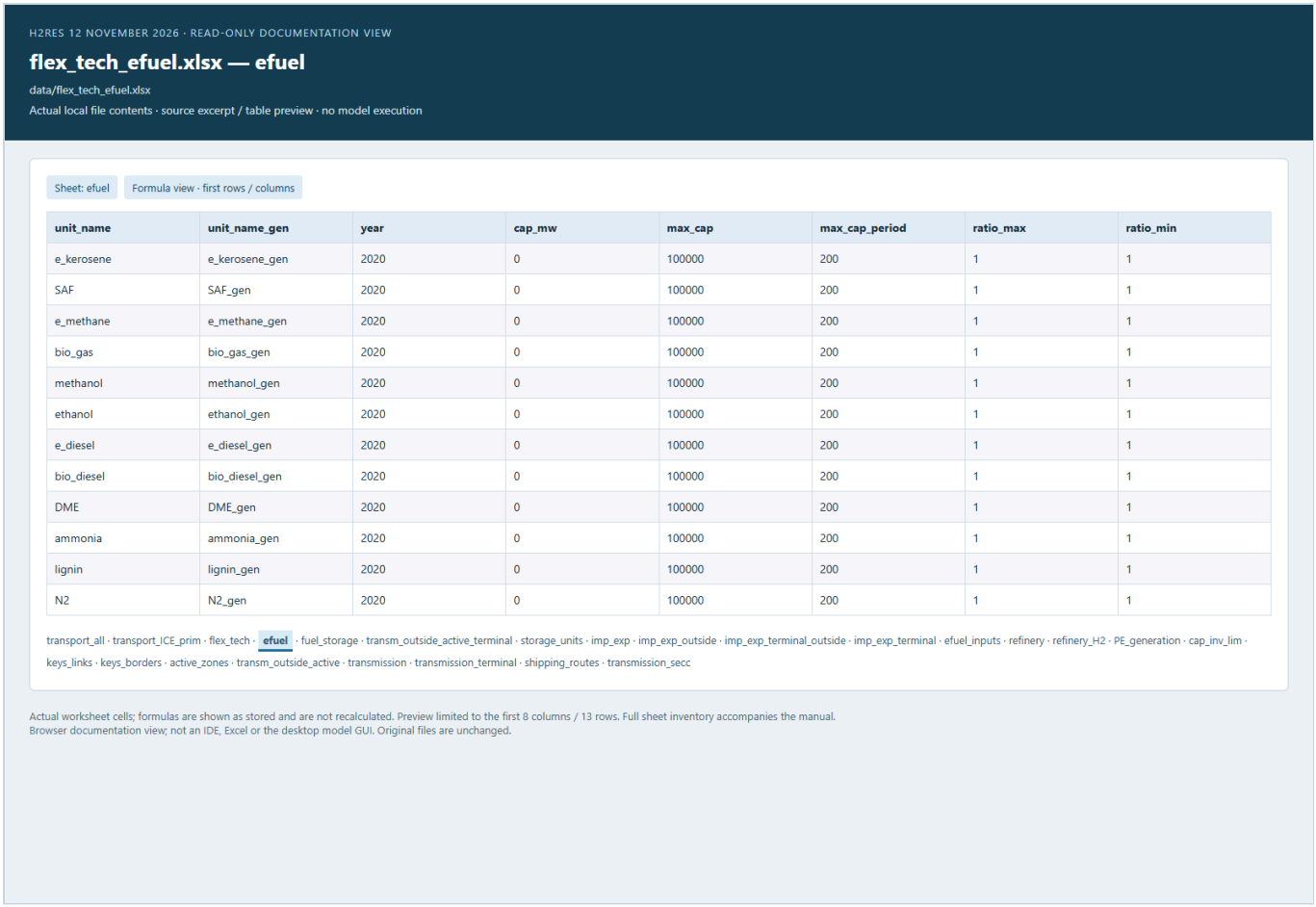
transport_all · flex_tech · cap_inv_lim · keys_links · keys_borders · active_zones · transm_outside_active · transmission · transmission_secc · transmission_outside · transmission_outside_H2 · transmission_H2 · flex_tech_heat · flex_tech_DH · thermal_tech · thermal_tech_DH · tech_cost_years · fixedOPEX_cost_years · **Industry_data** · simple_industry · COP_HP · solar_DH · solar_indv · solar_stor_DH

Actual worksheet cells; formulas are shown as stored and are not recalculated. Preview limited to the first 8 columns / 13 rows. Full sheet inventory accompanies the manual.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

Complex technology workbook

`flex_tech_complex_industry.xlsx` supplies the common technologies while `industry_complex.xlsx` supplies the process representation. Selected complex sectors replace their overlapping legacy services. Changing the process workbook alone does not activate all its sectors: `complex_industry_sectors` also scopes the model; its delivered policy default is `Non_metallic_minerals`.

7.4 Regular industry with e-fuels



H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

flex_tech_efuel.xlsx — efuel

data/flex_tech_efuel.xlsx

Actual local file contents · source excerpt / table preview · no model execution

Sheet: efuel | Formula view · first rows / columns

unit_name	unit_name_gen	year	cap_mw	max_cap	max_cap_period	ratio_max	ratio_min
e_kerosene	e_kerosene_gen	2020	0	100000	200	1	1
SAF	SAF_gen	2020	0	100000	200	1	1
e_methane	e_methane_gen	2020	0	100000	200	1	1
bio_gas	bio_gas_gen	2020	0	100000	200	1	1
methanol	methanol_gen	2020	0	100000	200	1	1
ethanol	ethanol_gen	2020	0	100000	200	1	1
e_diesel	e_diesel_gen	2020	0	100000	200	1	1
bio_diesel	bio_diesel_gen	2020	0	100000	200	1	1
DME	DME_gen	2020	0	100000	200	1	1
ammonia	ammonia_gen	2020	0	100000	200	1	1
lignin	lignin_gen	2020	0	100000	200	1	1
N2	N2_gen	2020	0	100000	200	1	1

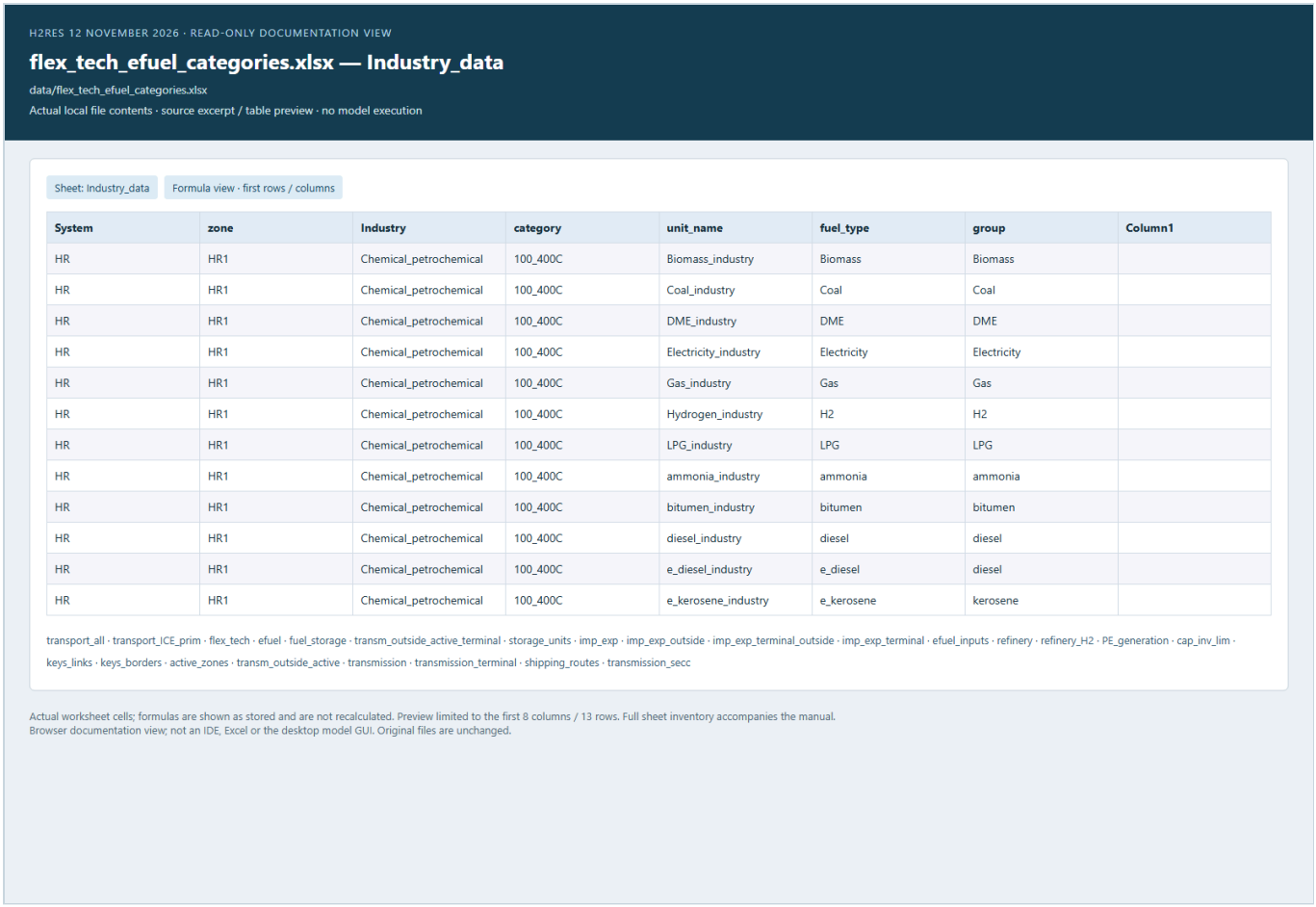
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Actual worksheet cells; formulas are shown as stored and are not recalculated. Preview limited to the first 8 columns / 13 rows. Full sheet inventory accompanies the manual.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

Regular e-fuel technology workbook

`flex_tech_efuel.xlsx` adds the e-fuel conversion, feedstock, storage, trade and applicable refinery/terminal layout. Review `efuel`, `efuel_inputs`, `fuel_storage`, `storage_units`, import/export sheets and transmission/routes together. A conversion output creates corresponding input and auxiliary demand; it is not free energy.

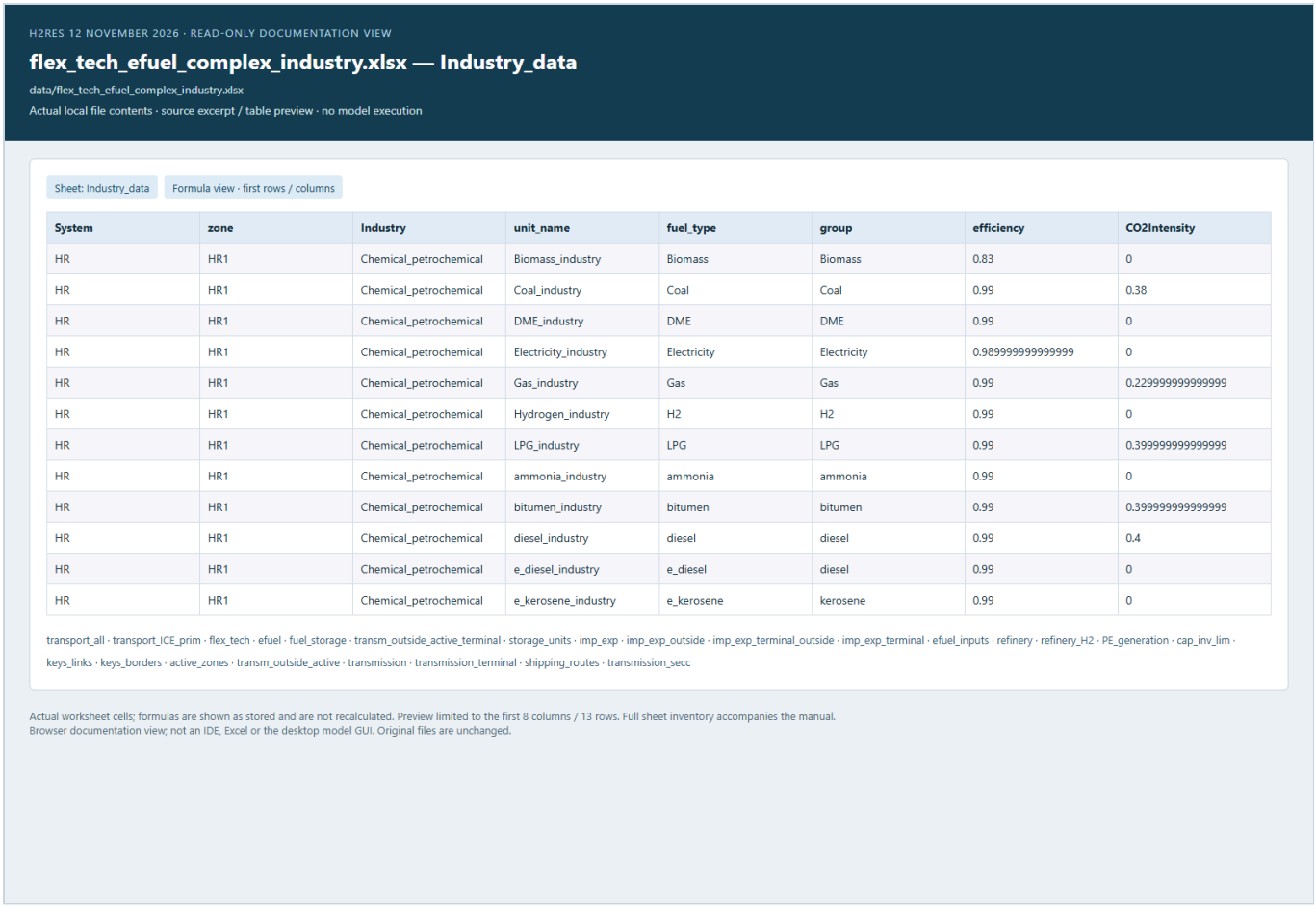
7.5 Category industry with e-fuels



E-fuel category technology workbook

`flex_tech_efuel_categories.xlsx` is the delivered conservative selection. In addition to service data, examine category costs, `industry_heat_stor`, industrial capture and process-emission sheets, compatibility rules and replacement ceilings. The `simple_industry` worksheet remains a pooled remainder, not the separate product model.

7.6 Complex industry with e-fuels



H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

flex_tech_efuel_complex_industry.xlsx — Industry_data

data/flex_tech_efuel_complex_industry.xlsx

Actual local file contents · source excerpt / table preview · no model execution

Sheet: Industry_data | Formula view · first rows / columns

System	zone	Industry	unit_name	fuel_type	group	efficiency	CO2Intensity
HR	HR1	Chemical_petrochemical	Biomass_industry	Biomass	Biomass	0.83	0
HR	HR1	Chemical_petrochemical	Coal_industry	Coal	Coal	0.99	0.38
HR	HR1	Chemical_petrochemical	DME_industry	DME	DME	0.99	0
HR	HR1	Chemical_petrochemical	Electricity_industry	Electricity	Electricity	0.989999999999999	0
HR	HR1	Chemical_petrochemical	Gas_industry	Gas	Gas	0.99	0.229999999999999
HR	HR1	Chemical_petrochemical	Hydrogen_industry	H2	H2	0.99	0
HR	HR1	Chemical_petrochemical	LPG_industry	LPG	LPG	0.99	0.399999999999999
HR	HR1	Chemical_petrochemical	ammonia_industry	ammonia	ammonia	0.99	0
HR	HR1	Chemical_petrochemical	bitumen_industry	bitumen	bitumen	0.99	0.399999999999999
HR	HR1	Chemical_petrochemical	diesel_industry	diesel	diesel	0.99	0.4
HR	HR1	Chemical_petrochemical	e_diesel_industry	e_diesel	diesel	0.99	0
HR	HR1	Chemical_petrochemical	e_kerosene_industry	e_kerosene	kerosene	0.99	0

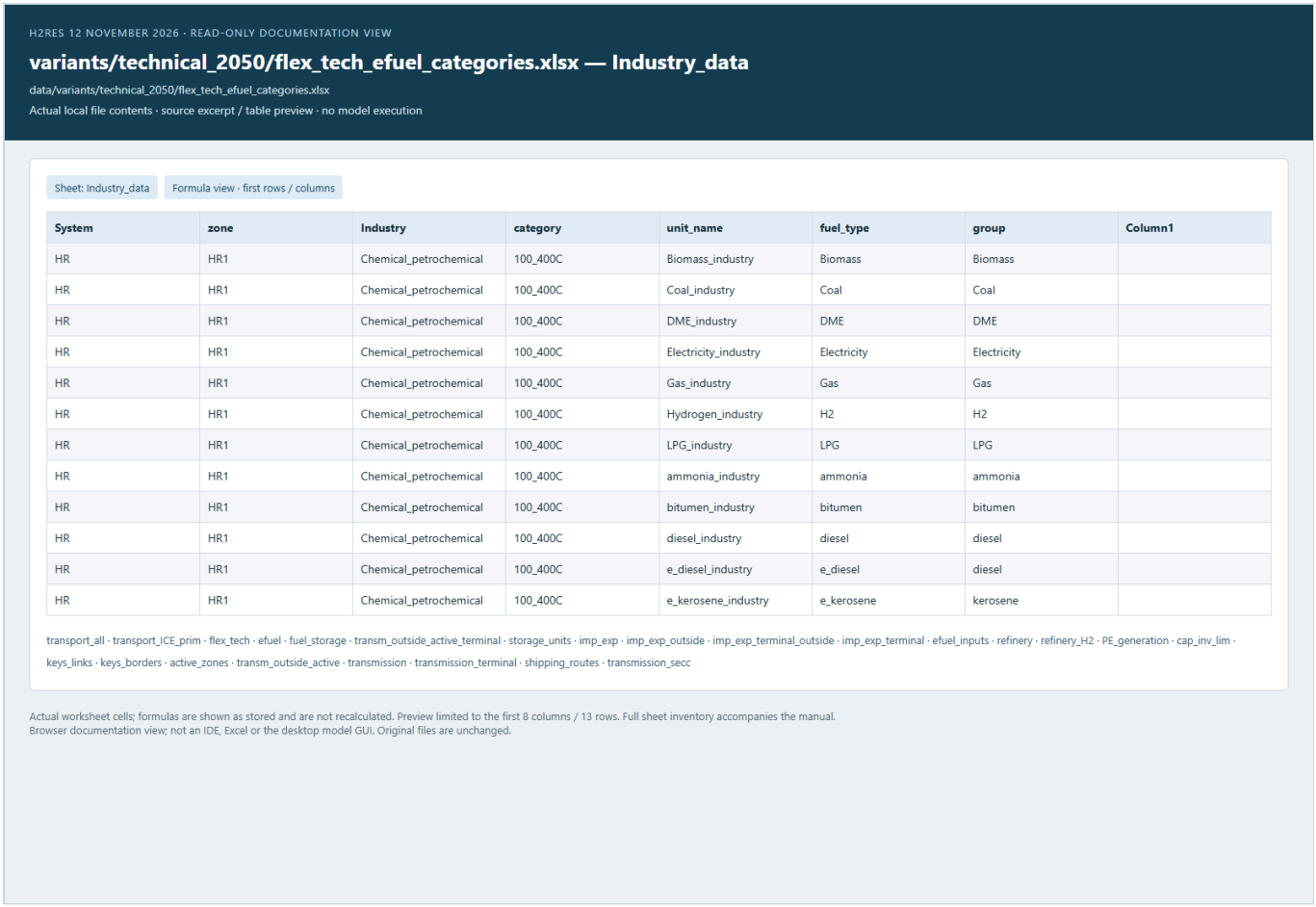
transport_all · transport_ICE_prim · flex_tech · efuel · fuel_storage · transm_outside_active_terminal · storage_units · imp_exp · imp_exp_outside · imp_exp_terminal_outside · imp_exp_terminal · efuel_inputs · refinery · refinery_H2 · PE_generation · cap_inv_lim · keys_links · keys_borders · active_zones · transm_outside_active · transmission · transmission_terminal · shipping_routes · transmission_secc

Actual worksheet cells; formulas are shown as stored and are not recalculated. Preview limited to the first 8 columns / 13 rows. Full sheet inventory accompanies the manual.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

E-fuel complex technology workbook

`flex_tech_efuel_complex_industry.xlsx` is paired with `industry_complex_efuel.xlsx`. Keep process heat/electricity, non-energy feedstocks, carrier conversion and process carbon consistent. Separate process and other-carbon balances are enabled in the current `main.py`; untagged imported CO2 cannot be assumed eligible for geological storage.

7.7 Technical-2050 category case



H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

variants/technical_2050/flex_tech_efuel_categories.xlsx — Industry_data

data/variants/technical_2050/flex_tech_efuel_categories.xlsx

Actual local file contents · source excerpt / table preview · no model execution

Sheet: Industry_data | Formula view · first rows / columns

System	zone	Industry	category	unit_name	fuel_type	group	Column1
HR	HR1	Chemical_petrochemical	100_400C	Biomass_industry	Biomass	Biomass	
HR	HR1	Chemical_petrochemical	100_400C	Coal_industry	Coal	Coal	
HR	HR1	Chemical_petrochemical	100_400C	DME_industry	DME	DME	
HR	HR1	Chemical_petrochemical	100_400C	Electricity_industry	Electricity	Electricity	
HR	HR1	Chemical_petrochemical	100_400C	Gas_industry	Gas	Gas	
HR	HR1	Chemical_petrochemical	100_400C	Hydrogen_industry	H2	H2	
HR	HR1	Chemical_petrochemical	100_400C	LPG_industry	LPG	LPG	
HR	HR1	Chemical_petrochemical	100_400C	ammonia_industry	ammonia	ammonia	
HR	HR1	Chemical_petrochemical	100_400C	bitumen_industry	bitumen	bitumen	
HR	HR1	Chemical_petrochemical	100_400C	diesel_industry	diesel	diesel	
HR	HR1	Chemical_petrochemical	100_400C	e_diesel_industry	e_diesel	diesel	
HR	HR1	Chemical_petrochemical	100_400C	e_kerosene_industry	e_kerosene	kerosene	

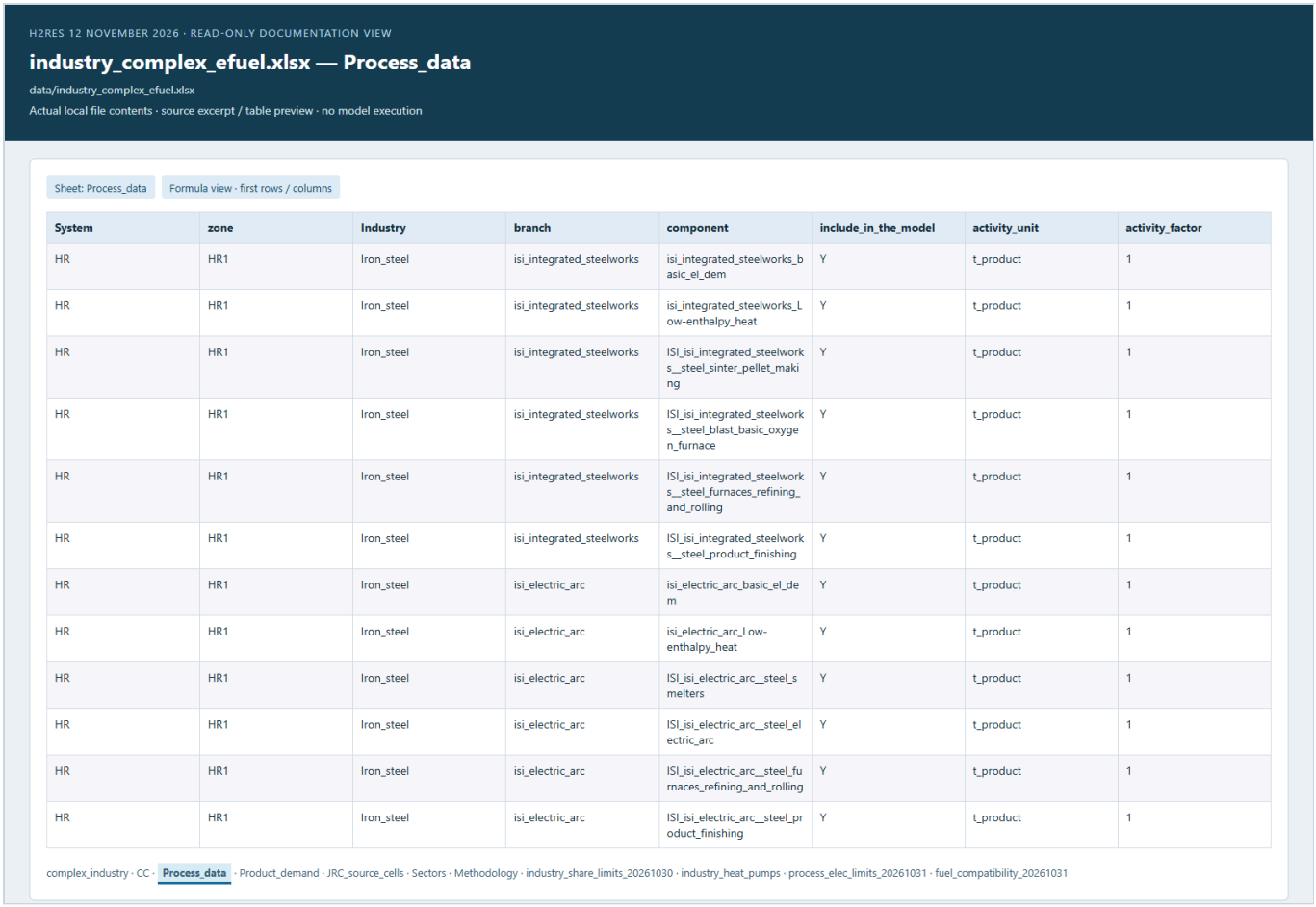
transport_all · transport_ICE_prim · flex_tech · efuel · fuel_storage · transm_outside_active_terminal · storage_units · imp_exp · imp_exp_outside · imp_exp_terminal_outside · imp_exp_terminal · efuel_inputs · refinery · refinery_H2 · PE_generation · cap_inv_lim · keys_links · keys_borders · active_zones · transm_outside_active · transmission · transmission_terminal · shipping_routes · transmission_secc

Actual worksheet cells; formulas are shown as stored and are not recalculated. Preview limited to the first 8 columns / 13 rows. Full sheet inventory accompanies the manual.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

Technical electricity variant

`variants/technical_2050/flex_tech_efuel_categories.xlsx` changes the relevant industrial electricity assumptions. Select it through `industry_category_electricity_case`; do not overwrite the conservative workbook. This is a technical scenario variant, not a forecast or an automatic renewable-energy target.

7.8 Process input workbook



H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

industry_complex_efuel.xlsx — Process_data

data/industry_complex_efuel.xlsx

Actual local file contents · source excerpt / table preview · no model execution

Sheet: Process_data | Formula view · first rows / columns

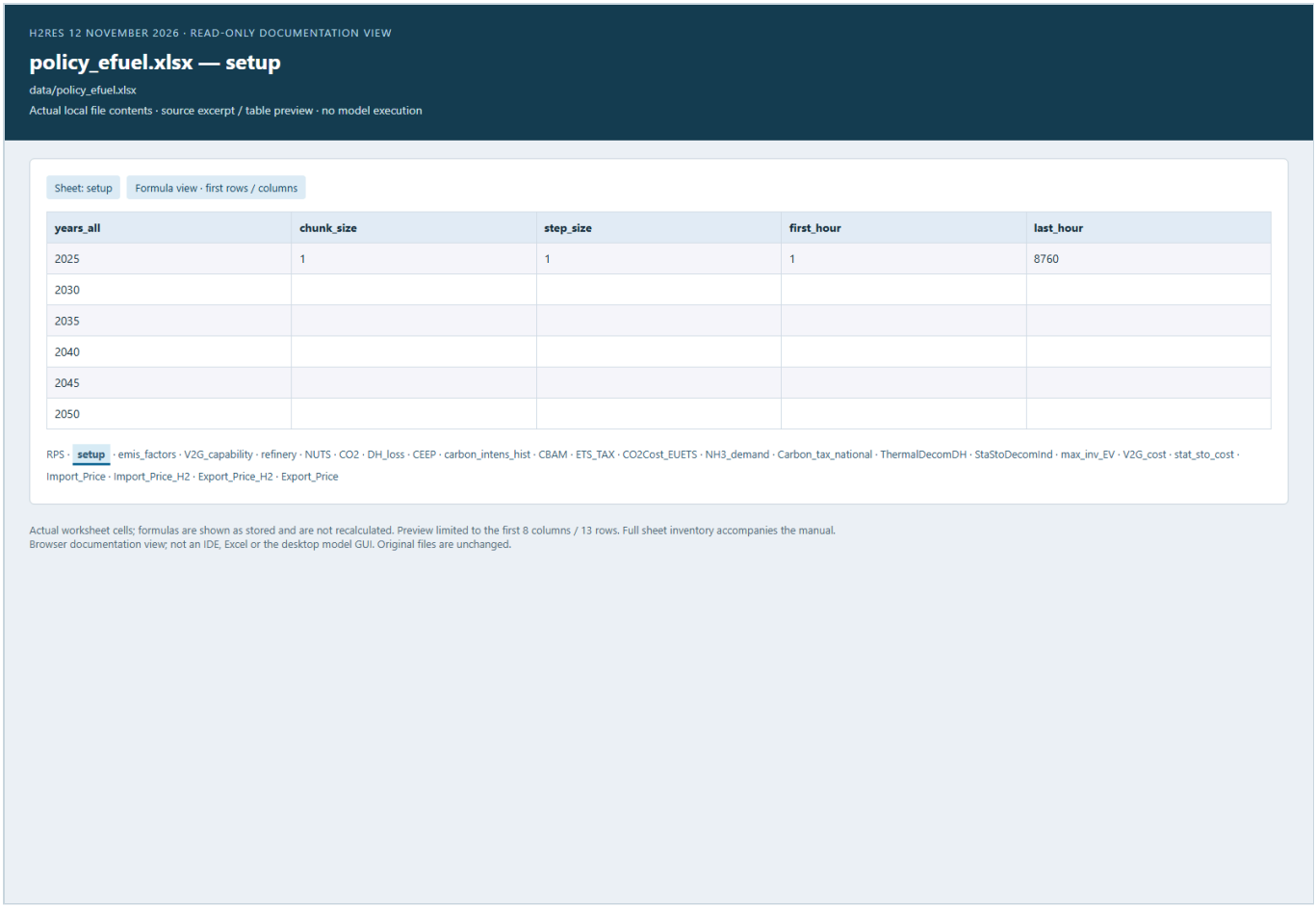
System	zone	Industry	branch	component	include_in_the_model	activity_unit	activity_factor
HR	HR1	Iron_steel	isi_integrated_steelworks	isi_integrated_steelworks_basic_el_dem	Y	t_product	1
HR	HR1	Iron_steel	isi_integrated_steelworks	isi_integrated_steelworks_Low-enthalpy_heat	Y	t_product	1
HR	HR1	Iron_steel	isi_integrated_steelworks	ISI_isi_integrated_steelworks__steel_sinter_pellet_making	Y	t_product	1
HR	HR1	Iron_steel	isi_integrated_steelworks	ISI_isi_integrated_steelworks__steel_blast_basic_oxygen_furnace	Y	t_product	1
HR	HR1	Iron_steel	isi_integrated_steelworks	ISI_isi_integrated_steelworks__steel_furnaces_refining_and_rolling	Y	t_product	1
HR	HR1	Iron_steel	isi_integrated_steelworks	ISI_isi_integrated_steelworks__steel_product_finishing	Y	t_product	1
HR	HR1	Iron_steel	isi_electric_arc	isi_electric_arc_basic_el_dem	Y	t_product	1
HR	HR1	Iron_steel	isi_electric_arc	isi_electric_arc_Low-enthalpy_heat	Y	t_product	1
HR	HR1	Iron_steel	isi_electric_arc	ISI_isi_electric_arc__steel_smelters	Y	t_product	1
HR	HR1	Iron_steel	isi_electric_arc	ISI_isi_electric_arc__steel_electric_arc	Y	t_product	1
HR	HR1	Iron_steel	isi_electric_arc	ISI_isi_electric_arc__steel_furnaces_refining_and_rolling	Y	t_product	1
HR	HR1	Iron_steel	isi_electric_arc	ISI_isi_electric_arc__steel_product_finishing	Y	t_product	1

complex_industry · CC · **Process_data** · Product_demand · JRC_source_cells · Sectors · Methodology · industry_share_limits_20261030 · industry_heat_pumps · process_elec_limits_20261031 · fuel_compatibility_20261031

Complex process data

The complex-industry companion workbooks contain `Process_data`, `Product_demand`, `complex_industry`, `CC` and methodology/compatibility sheets. Check product and component identifiers, activity units, useful service, final inputs, capture eligibility and the retained legacy remainder. The technical-electricity case must use the matching process workbook for its e-fuel selection.

8. Policy workbooks, years and demand trajectories



Policy setup and planning horizon

The inspected `policy_efuel.xlsx` `setup` sheet has decision years 2025, 2030, 2035, 2040, 2045 and 2050. Its first row of values sets `chunk_size=1`, `step_size=1`, `first_hour=1` and `last_hour=8760`. A five-year spacing between decisions is different from `chunk_size`. Follow the reader's construction of each planning window; the scenario-name suffix is not a year-grid control.

When changing the horizon, also extend applicable annual costs, limits, demand changes, carbon/RPS trajectories, transport paths, supplier/market inputs and retirement assumptions. Changing only `setup` can create missing-year data or an inconsistent scenario. The weather/source year of an hourly shape is not the same as a future decision year.

H2RES 12 NOVEMBER 2026 · READ-ONLY DOCUMENTATION VIEW

policy_efuel.xlsx — CO2

data/policy_efuel.xlsx

Actual local file contents · source excerpt / table preview · no model execution

Sheet: CO2 | Formula view · first rows / columns

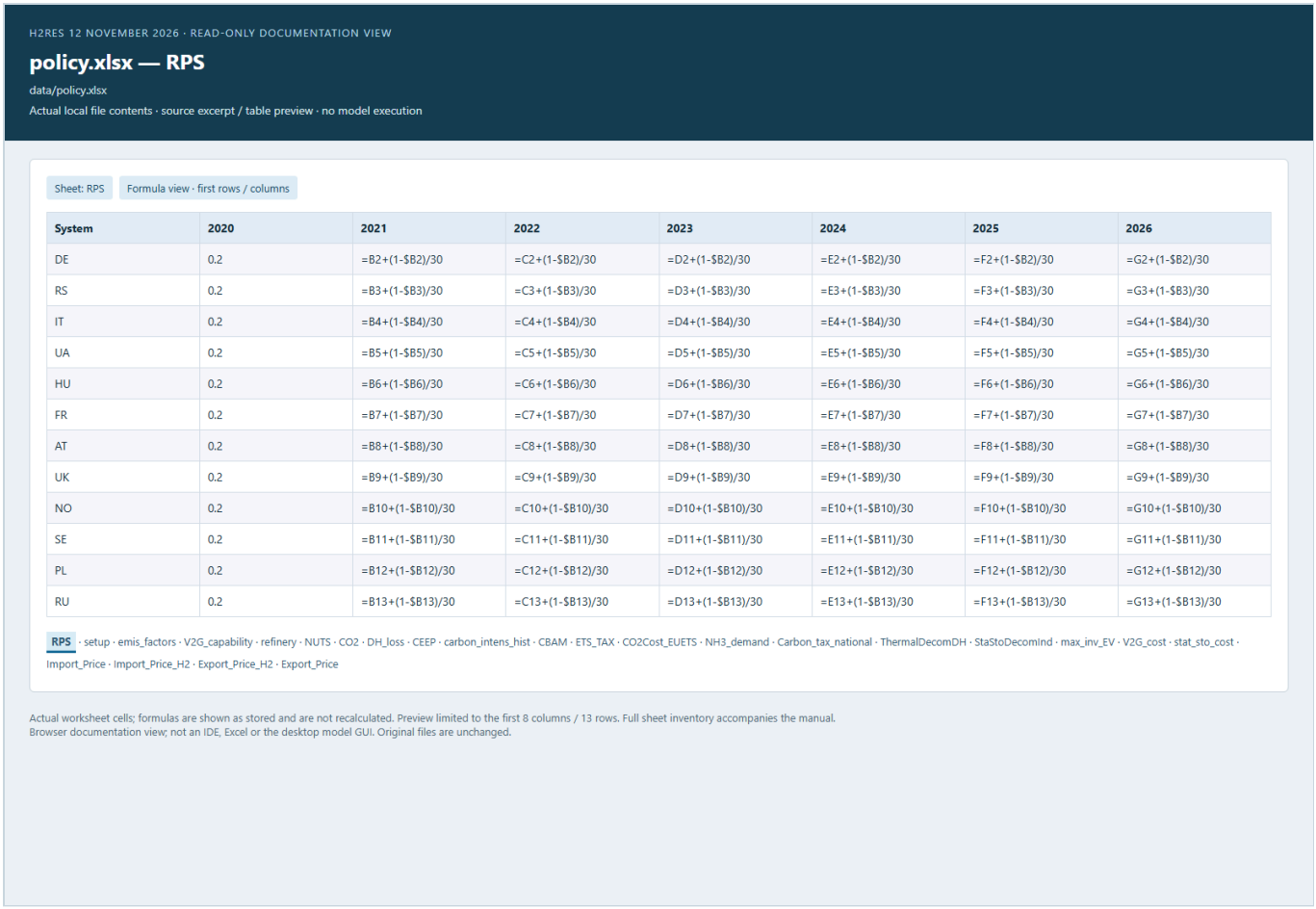
System	2020	2021	2022	2023	2024	2025	2026
DE	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827215	25462792.3308709	22966216.1794088
RS	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827215	25462792.3308709	22966216.1794088
IT	295000000	288076226.412059	278790538.874335	266595592.768421	251013478.650865	231786981.330548	209060728.754248
UA	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827215	25462792.3308709	22966216.1794088
HU	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827215	25462792.3308709	22966216.1794088
FR	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827215	25462792.3308709	22966216.1794088
AT	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827215	25462792.3308709	22966216.1794088
UK	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827215	25462792.3308709	22966216.1794088
NO	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827215	25462792.3308709	22966216.1794088
SE	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827215	25462792.3308709	22966216.1794088
PL	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827215	25462792.3308709	22966216.1794088
RU	32407013.0879993	31646406.9141595	30626334.3800371	29286667.3357598	27574905.3827214	25462792.3308709	22966216.1794088

RPS · setup · emis_factors · V2G_capability · refinery · NUTS · CO2 · DH_loss · CEEP · carbon_intens_hist · CBAM · ETS_TAX · CO2Cost_EUETS · NH3_demand · Carbon_tax_national · ThermalDecomDH · StaStoDecomInd · max_inv_EV · V2G_cost · stat_sto_cost · Import_Price · Import_Price_H2 · Export_Price_H2 · Export_Price

Actual worksheet cells; formulas are shown as stored and are not recalculated. Preview limited to the first 8 columns / 13 rows. Full sheet inventory accompanies the manual.
Browser documentation view; not an IDE, Excel or the desktop model GUI. Original files are unchanged.

Carbon policy worksheet

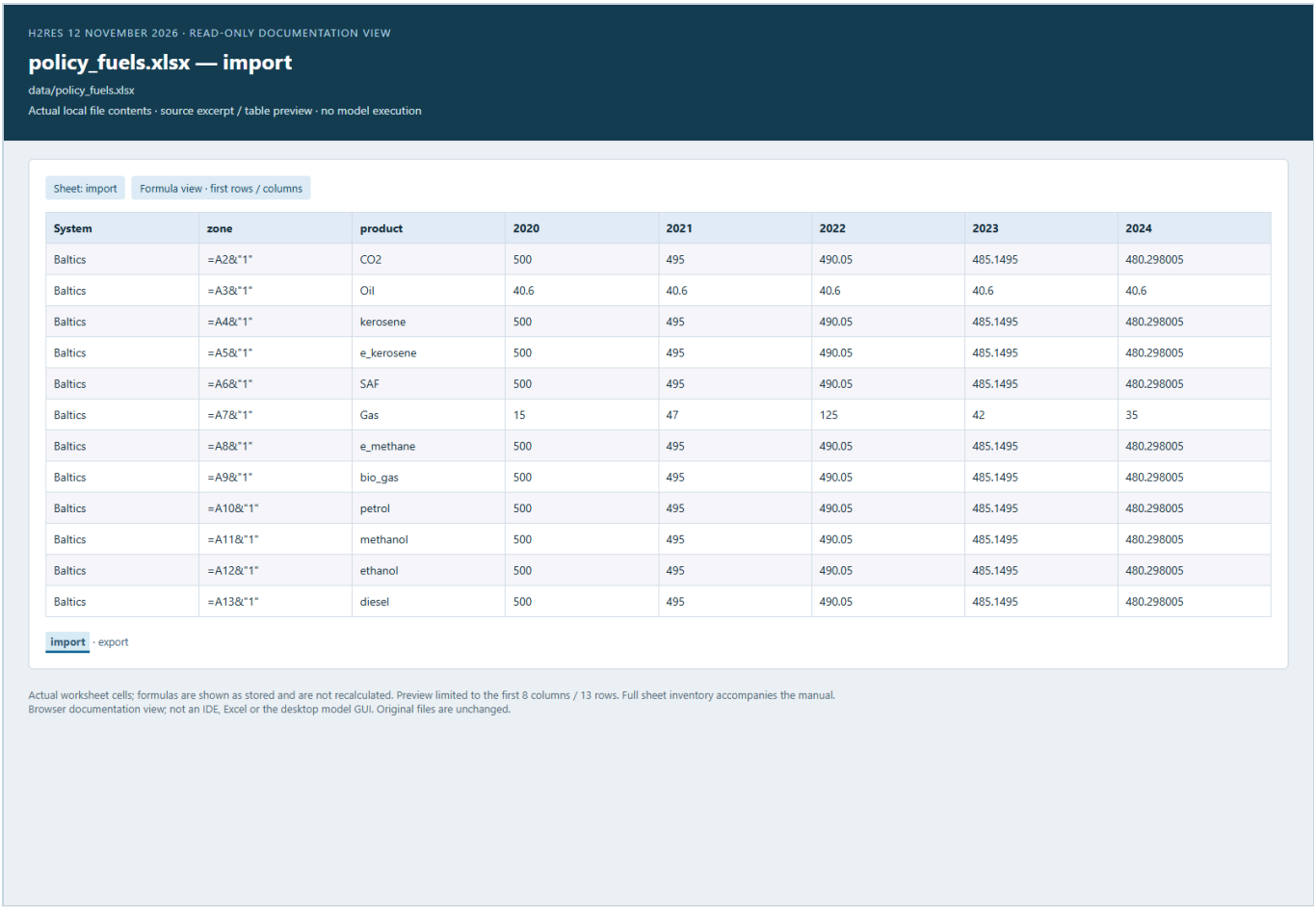
The `CO2` sheet supplies the carbon-limit trajectories selected by the model. Read the implemented units and System mapping. Carbon limits, ETS/carbon prices, capture, LULUCF removals and CBAM switches have different roles. The current configuration enables the carbon limit and disables LULUCF removals.



System	2020	2021	2022	2023	2024	2025	2026
DE	0.2	=B2+(1-$B2)/30	=C2+(1-$B2)/30	=D2+(1-$B2)/30	=E2+(1-$B2)/30	=F2+(1-$B2)/30	=G2+(1-$B2)/30
RS	0.2	=B3+(1-$B3)/30	=C3+(1-$B3)/30	=D3+(1-$B3)/30	=E3+(1-$B3)/30	=F3+(1-$B3)/30	=G3+(1-$B3)/30
IT	0.2	=B4+(1-$B4)/30	=C4+(1-$B4)/30	=D4+(1-$B4)/30	=E4+(1-$B4)/30	=F4+(1-$B4)/30	=G4+(1-$B4)/30
UA	0.2	=B5+(1-$B5)/30	=C5+(1-$B5)/30	=D5+(1-$B5)/30	=E5+(1-$B5)/30	=F5+(1-$B5)/30	=G5+(1-$B5)/30
HU	0.2	=B6+(1-$B6)/30	=C6+(1-$B6)/30	=D6+(1-$B6)/30	=E6+(1-$B6)/30	=F6+(1-$B6)/30	=G6+(1-$B6)/30
FR	0.2	=B7+(1-$B7)/30	=C7+(1-$B7)/30	=D7+(1-$B7)/30	=E7+(1-$B7)/30	=F7+(1-$B7)/30	=G7+(1-$B7)/30
AT	0.2	=B8+(1-$B8)/30	=C8+(1-$B8)/30	=D8+(1-$B8)/30	=E8+(1-$B8)/30	=F8+(1-$B8)/30	=G8+(1-$B8)/30
UK	0.2	=B9+(1-$B9)/30	=C9+(1-$B9)/30	=D9+(1-$B9)/30	=E9+(1-$B9)/30	=F9+(1-$B9)/30	=G9+(1-$B9)/30
NO	0.2	=B10+(1-$B10)/30	=C10+(1-$B10)/30	=D10+(1-$B10)/30	=E10+(1-$B10)/30	=F10+(1-$B10)/30	=G10+(1-$B10)/30
SE	0.2	=B11+(1-$B11)/30	=C11+(1-$B11)/30	=D11+(1-$B11)/30	=E11+(1-$B11)/30	=F11+(1-$B11)/30	=G11+(1-$B11)/30
PL	0.2	=B12+(1-$B12)/30	=C12+(1-$B12)/30	=D12+(1-$B12)/30	=E12+(1-$B12)/30	=F12+(1-$B12)/30	=G12+(1-$B12)/30
RU	0.2	=B13+(1-$B13)/30	=C13+(1-$B13)/30	=D13+(1-$B13)/30	=E13+(1-$B13)/30	=F13+(1-$B13)/30	=G13+(1-$B13)/30

Renewable electricity policy worksheet

`RPS` controls the implemented renewable-electricity share policy. It does not by itself impose 100% renewable final energy across heating, industry and transport. The preparation tool's whole-energy preset is an additional adapter policy with its own coverage and compatibility limits; it is not implied by setting RPS to one.



System	zone	product	2020	2021	2022	2023	2024
Baltics	=A2&"1"	CO2	500	495	490.05	485.1495	480.298005
Baltics	=A3&"1"	Oil	40.6	40.6	40.6	40.6	40.6
Baltics	=A4&"1"	kerosene	500	495	490.05	485.1495	480.298005
Baltics	=A5&"1"	e_kerosene	500	495	490.05	485.1495	480.298005
Baltics	=A6&"1"	SAF	500	495	490.05	485.1495	480.298005
Baltics	=A7&"1"	Gas	15	47	125	42	35
Baltics	=A8&"1"	e_methane	500	495	490.05	485.1495	480.298005
Baltics	=A9&"1"	bio_gas	500	495	490.05	485.1495	480.298005
Baltics	=A10&"1"	petrol	500	495	490.05	485.1495	480.298005
Baltics	=A11&"1"	methanol	500	495	490.05	485.1495	480.298005
Baltics	=A12&"1"	ethanol	500	495	490.05	485.1495	480.298005
Baltics	=A13&"1"	diesel	500	495	490.05	485.1495	480.298005

E-fuel price policy workbook

`policy_fuels.xlsx` supplies applicable e-fuel carrier price assumptions. Keep these distinct from primary fuel prices in `fuel_prices.csv` and electricity import prices. Do not count the same feedstock purchase again at every conversion stage.

Other frequently reviewed policy sheets include `dem_change`, `Industry_dem_change`, `Heat_dem_change`, `Heat_dem_change_DH`, `transport_targets`, `transport_targets_efuel`, `biomass_caps`, `generation_targets`, `hydro_caps`, `outside_trade`, `CO2Cost_EUETS`, `Carbon_tax_national`, `CBAM` and `ETS_TAX`. Presence is not activation: both the selected workbook and Python flags determine whether a policy applies.

---

9. Hourly electricity, heat and transport inputs

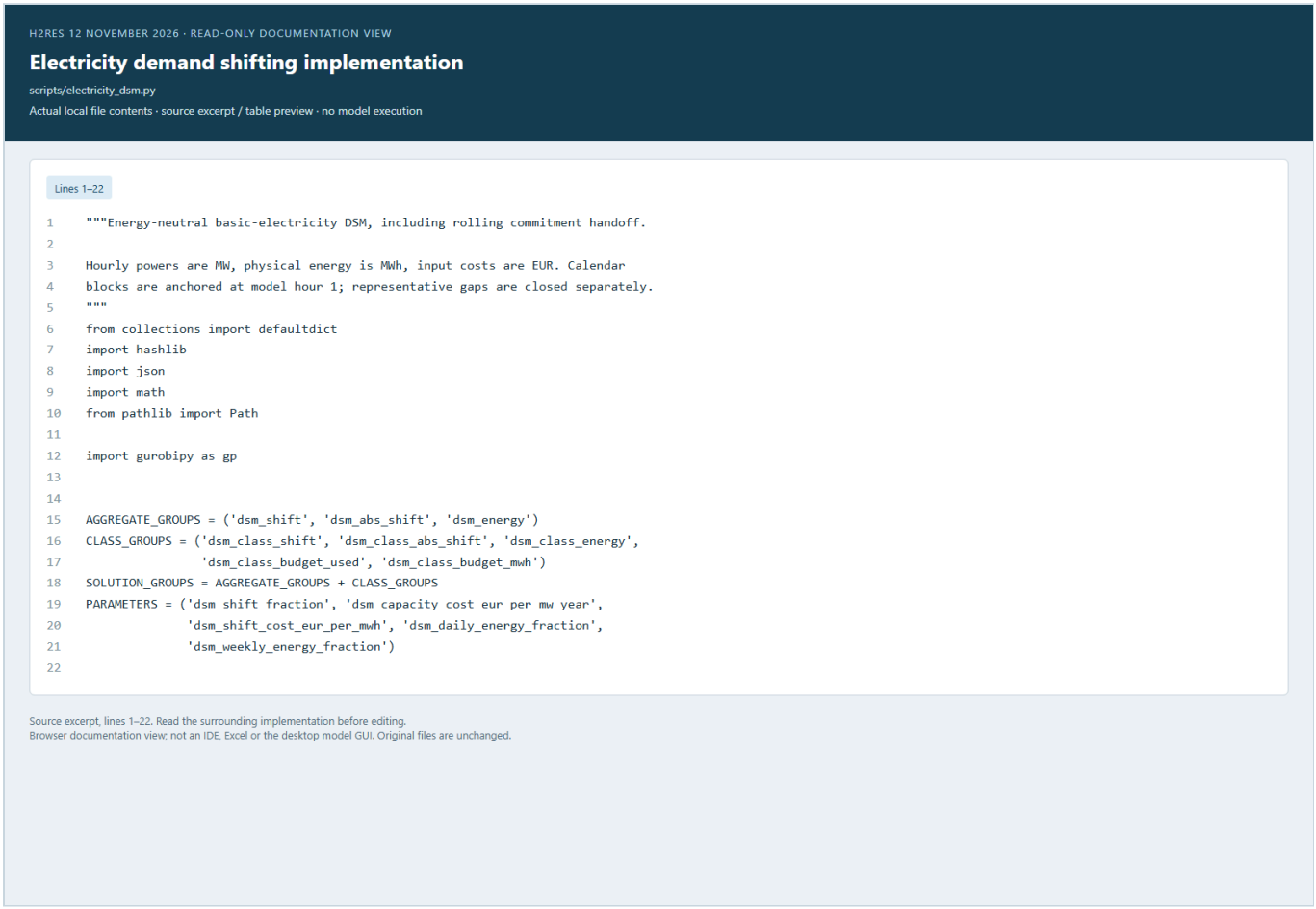
`electricity_demand.csv` is the basic electricity-demand profile by period, System and Zone. `industry_electricity.csv` and `heating_electricity.csv` participate in the model's sector-accounting boundary; follow the reader so that electricity already embedded in basic load is not charged twice. `renewable_availability.csv` and `hydro_inflows.csv` use their exact asset columns and expected normalization. Check full hourly coverage and period ordering.

Heating uses `heating_individual.csv`, `heating_district.csv`, relevant electric-heating data, the flex/thermal heating sheets, COP data and solar availability where enabled. The delivered solar individual/DH and cooling switches are off. Changing a switch does not create missing profiles or costs.

Transport uses `transport_profiles.csv`, `transport_demand.csv`, fleet/technology sheets, compatible fuels, policy paths and infrastructure CSVs. Service, final energy and vehicles are different units. V2G availability is not the same as driving demand. Navigation modes also need coherent inland/coastal allocation in a regional dataset.

---

10. New electricity demand shifting



DSM implementation source

For a DSM study, choose a fresh `scen` and enable the applicable controls in `main.py`:

```
DSM_day = True
DSM_week = True
dsm_energy_budget_active = True
dsm_availability_ramp_active = True
```

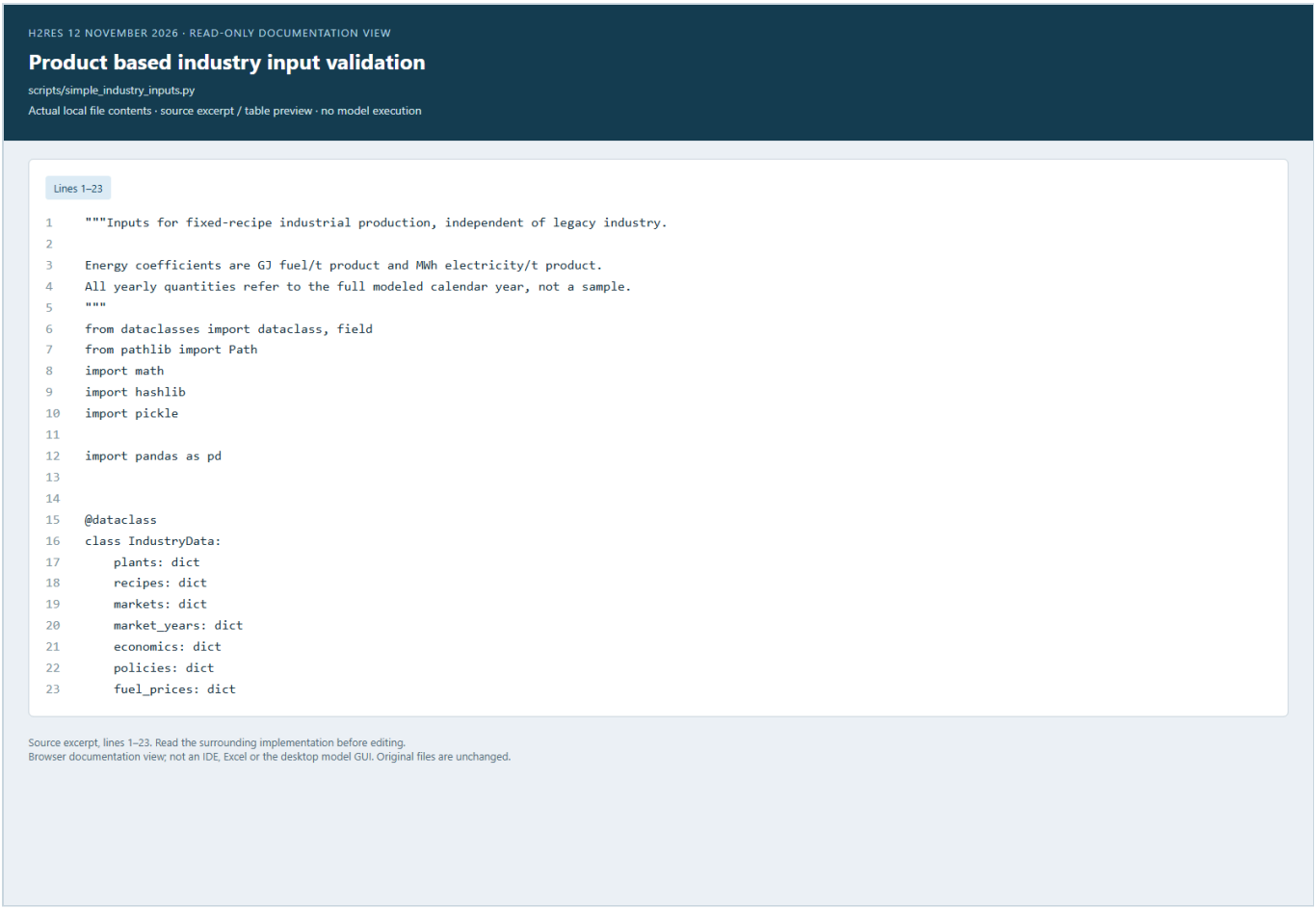
Review the parameters at the beginning of `policy/runtime_policy.py`. The supplied energy pools are 7% daily-capable including a 1.5% weekly subset; both together mean 5.5% daily-balanced and 1.5% weekly-balanced. Rollout reaches its target in 2030. The shared hourly adjustment cap is 10% of baseline load, capacity cost is EUR 10,000/MW/year and activation cost EUR 5/MWh moved. These are scenario assumptions.

DSM shifts basic electricity load while conserving energy within its daily/weekly blocks; it does not automatically shift every sector's separate modeled demand. Its hourly power and moved-energy limits are separate. For example, the symmetric 10% hourly cap can bind before the full 7% energy allowance is usable. Read model manual section 19 and the supplied DSM notes before changing sampled/rolling behavior.

Demand-response background is provided by R35, R36 and R37; these sources do not validate the local energy-pool percentages or costs.

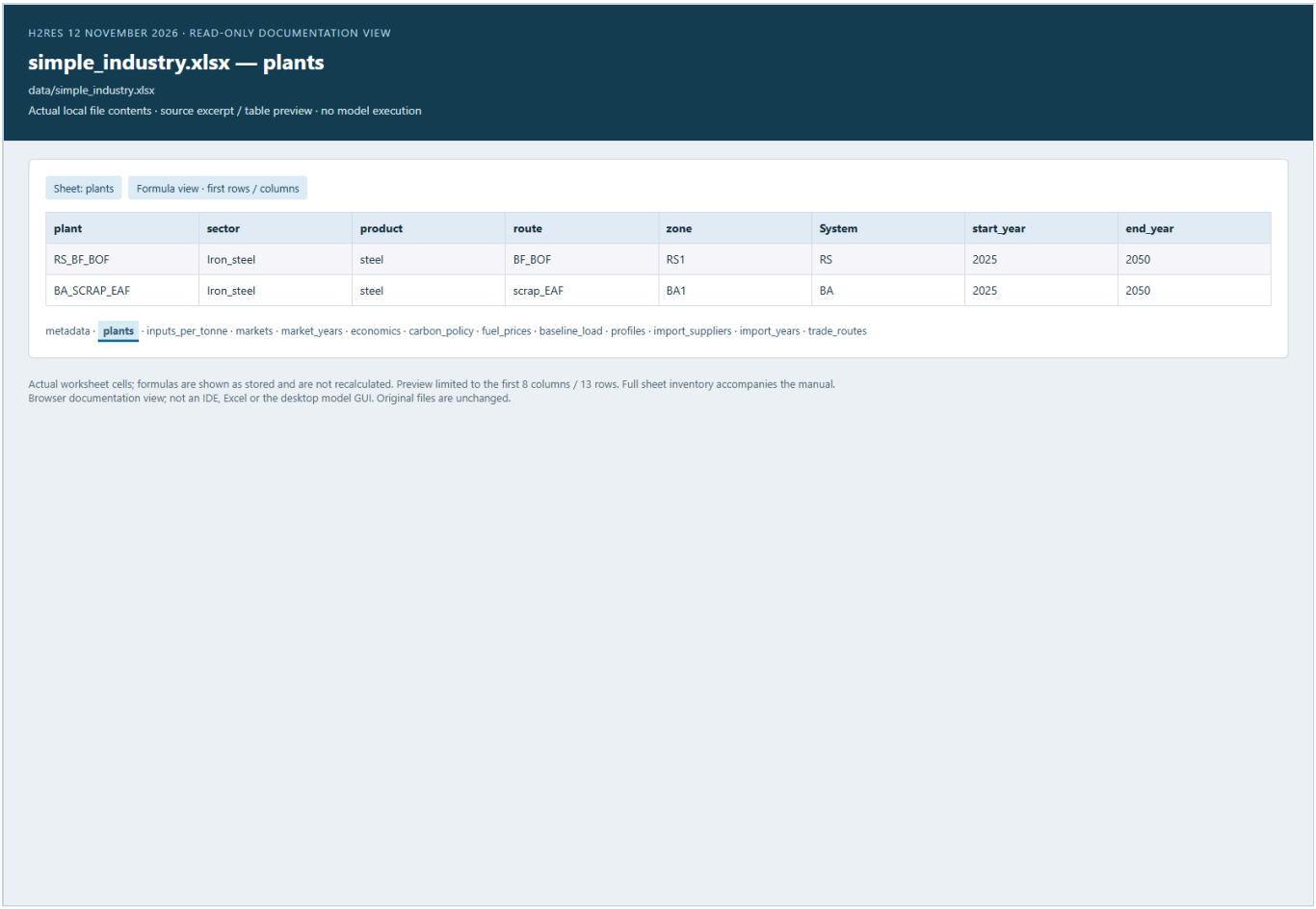
---

11. New product-industry workbook



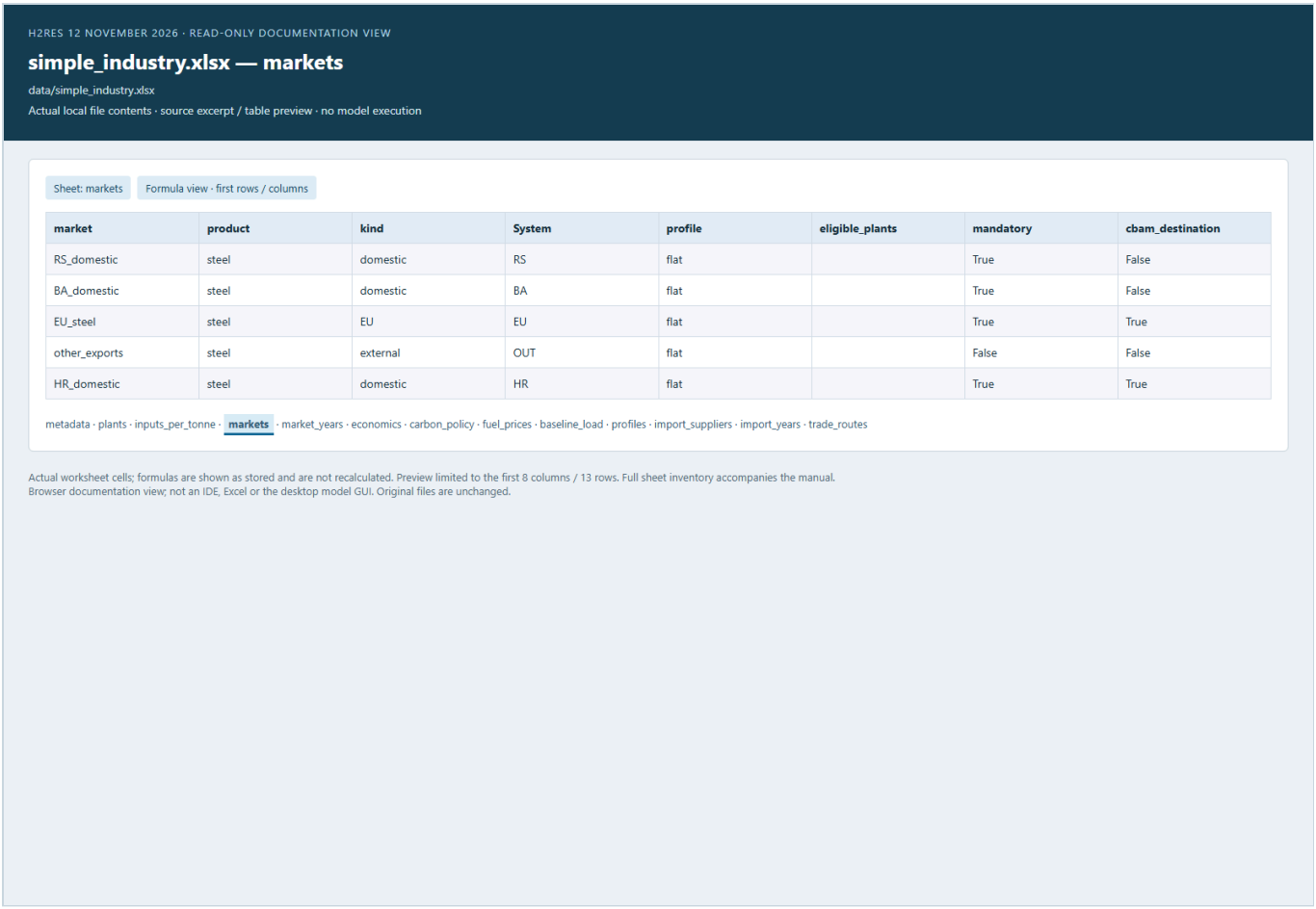
Product-industry reader

`data/simple_industry.xlsx` is a separate product model, selected by `model_simple_industry=True`. It is **not** the `simple_industry` worksheet in flex-tech. Enabling it configures a different run: the legacy industry/transport, e-fuel and overlapping capture/H2 blocks are disabled. Heating keeps its own selection.



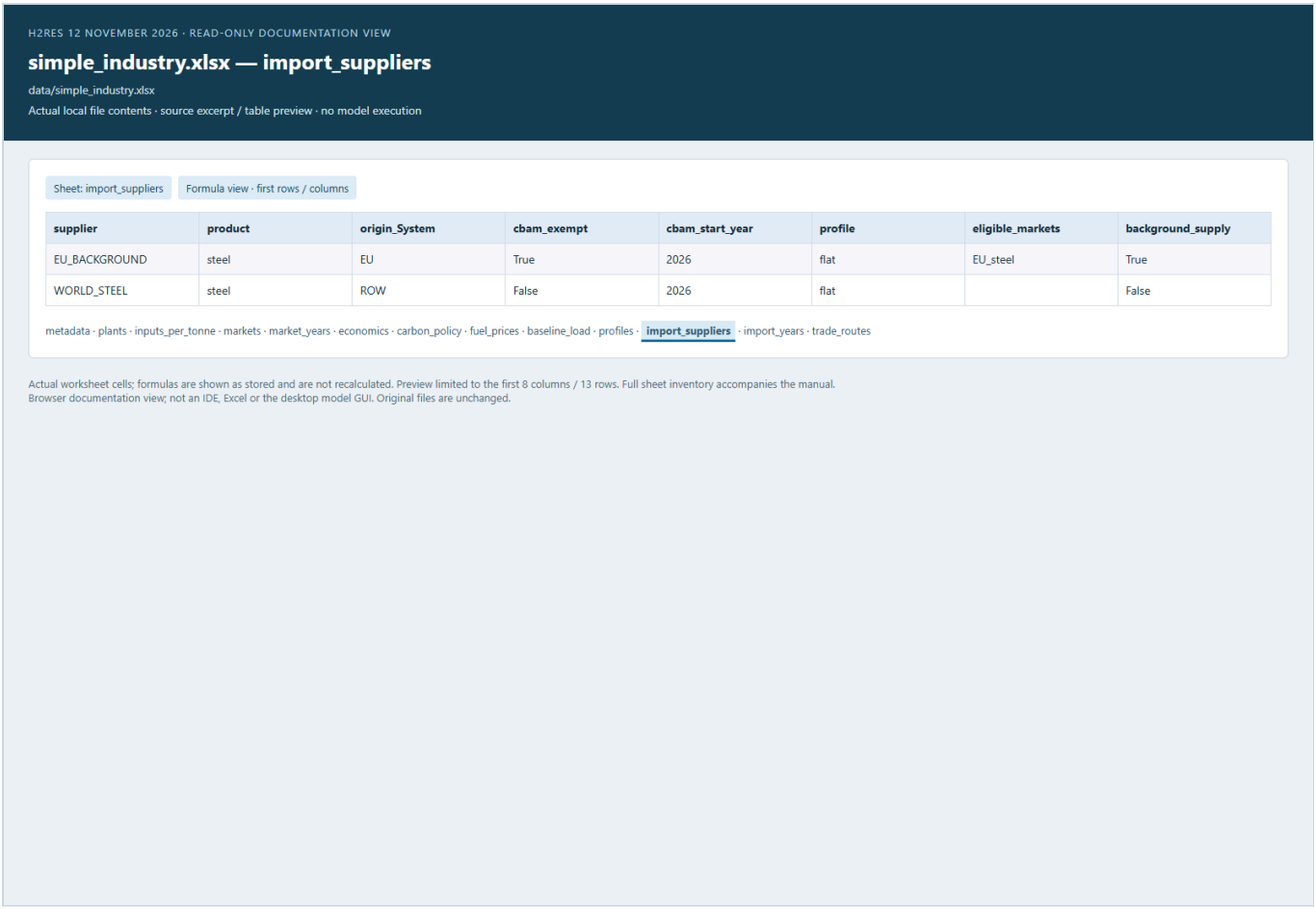
Plants and fixed recipes

Review `plants` for product, route, geography, t/h capacity, net MWh/t electricity, connection power, operating years, flexibility and capital inputs. Review `inputs_per_tonne` for fixed GJ/t recipes and emissions. The example RS BF-BOF and BA EAF cases are illustrative plants, not calibrated Smederevo or Zenica facilities.



Product markets

`markets` defines destinations and eligibility; `market_years` supplies annual tonnes, sale prices and hourly delivery ceilings. Mandatory domestic/Europe markets must be served. Product output is a decision bounded by capacity; a plant can be displaced by other modeled or eligible outside suppliers. Demand is not automatically reduced when supply is inadequate.

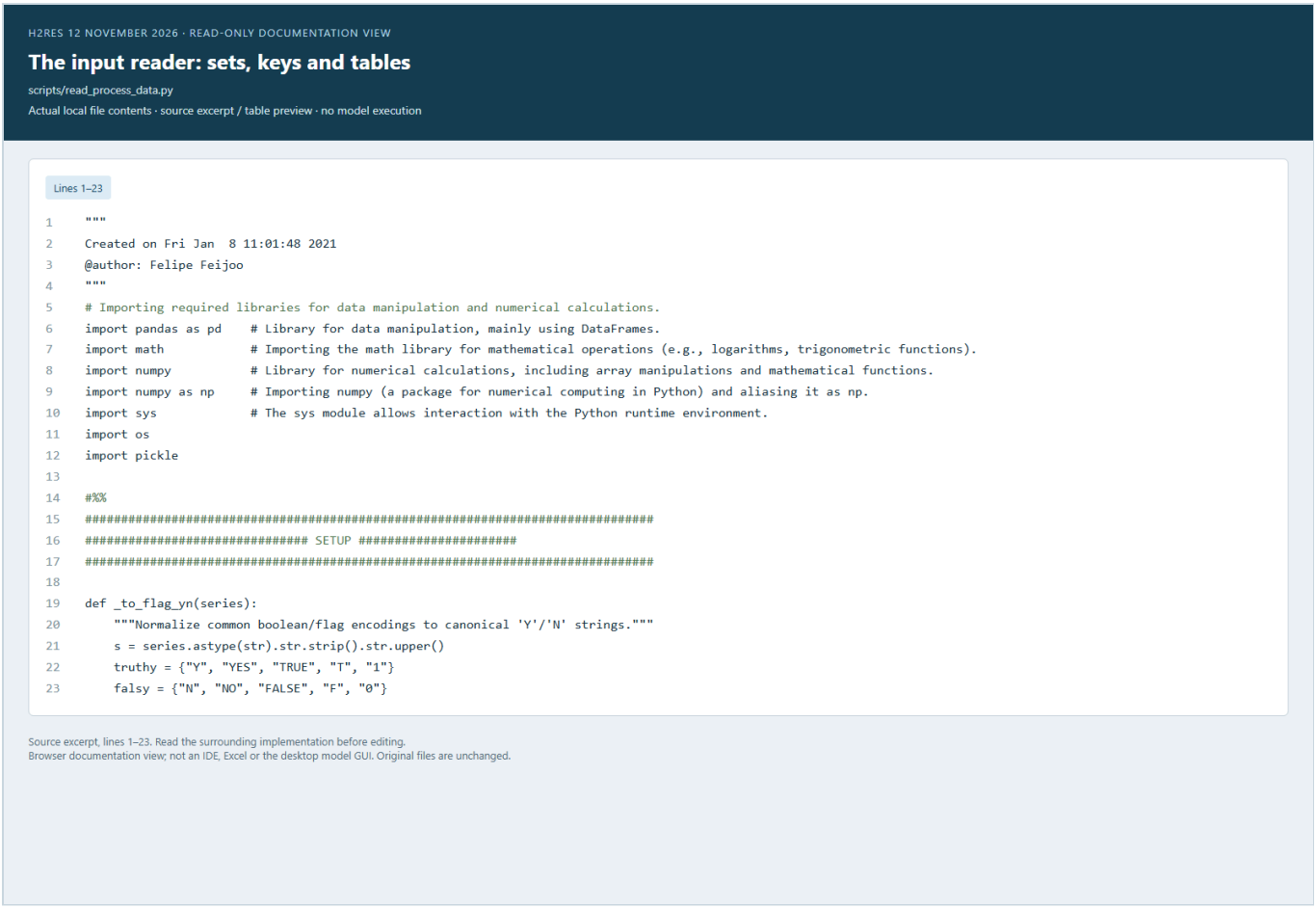


Outside suppliers

`import_suppliers` and `import_years` provide origin, eligibility, bounded availability, cost and embedded emissions. `trade_routes` may further restrict or price supplier/market pairs. An outside supplier has one annual budget shared across its destinations. The EU background offer is restricted to the rest-of-Europe market; it is not unrestricted domestic backup.

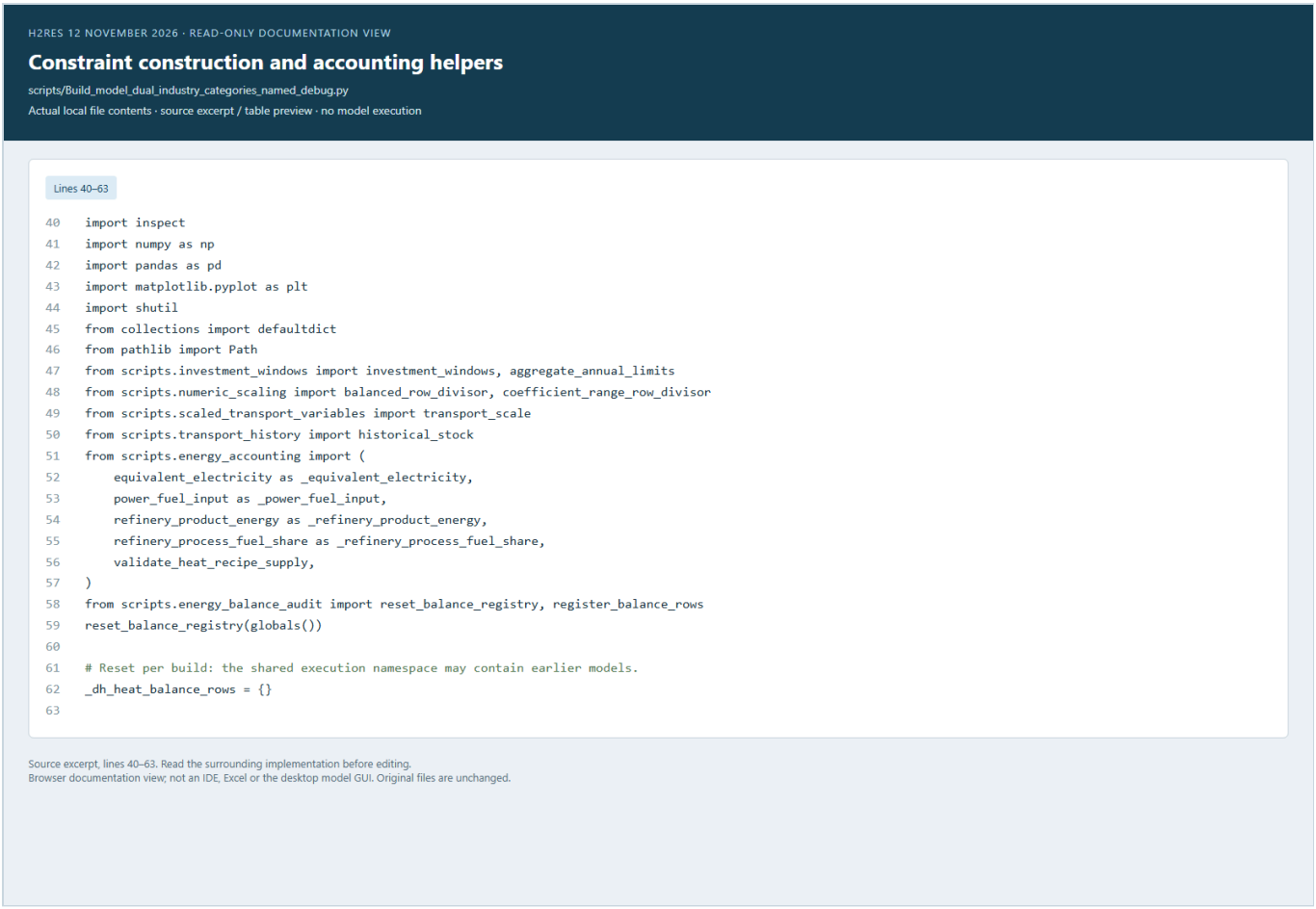
Industrial CBAM, domestic carbon pricing and outside import/export controls are separate `main.py` switches. Electricity CBAM Options 1–4 are separate again and mutually exclusive. Baseline load removal must be calibrated if a plant's electricity is already included in the basic demand CSV. Zero removal in the example means its plant electricity is added. See model manual section 20 and the supplied product-industry notes.

12. Source files used during execution



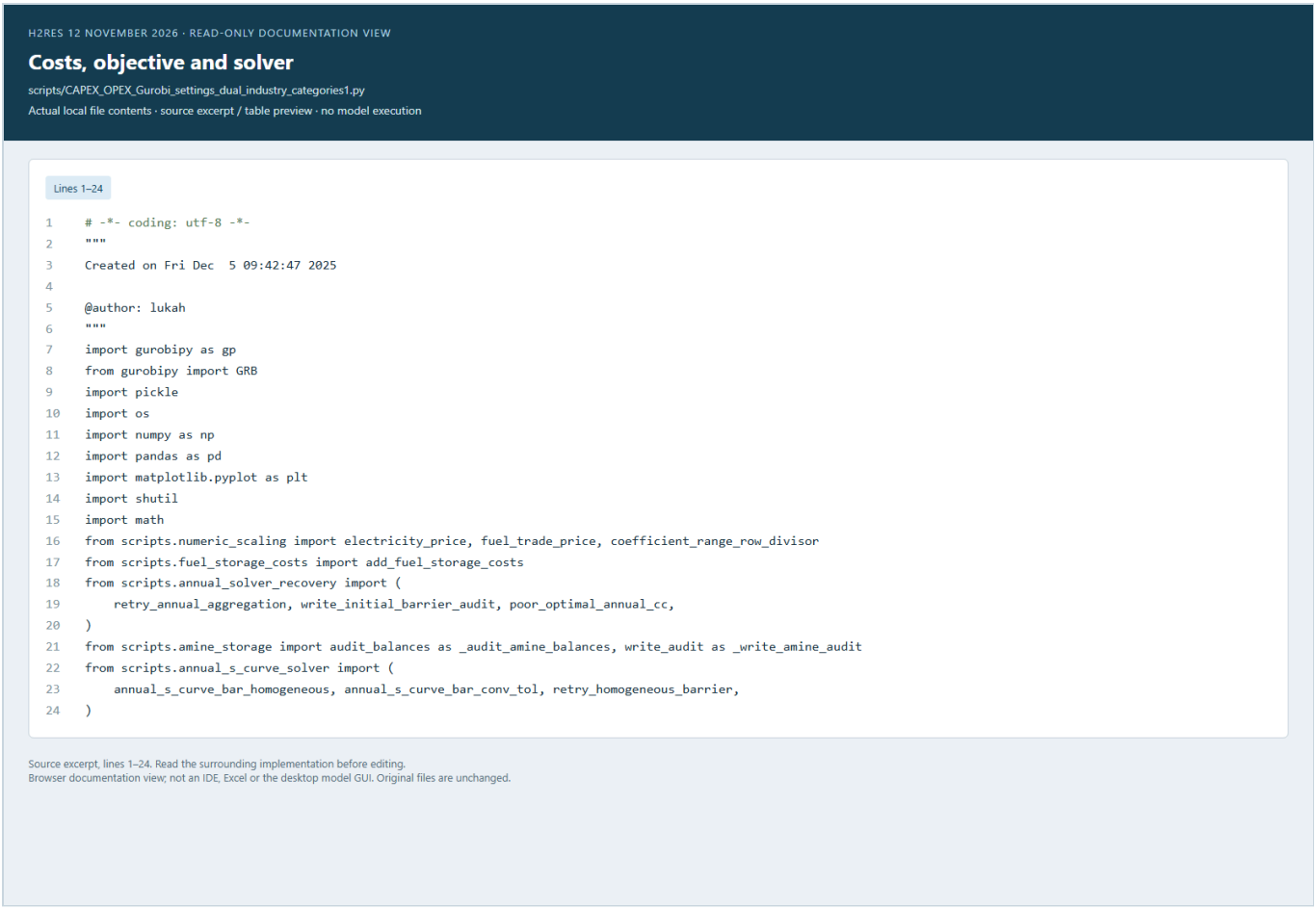
Input reader source

`scripts/read_process_data.py` reads native files, scopes active systems/zones, interprets table layouts and creates sets and numerical dictionaries. Open this file when checking what a workbook column means or which sheet is read under a flag. An error naming a sheet or annual trajectory usually needs inspection of the selected workbook and reader, not an edit to the solver.



Constraint builder source

`scripts/reading_prev_and_vars_def_dual_industry_categories1.py` declares variables and loads historical decisions. `scripts/Build_model_dual_industry_categories_named_debug.py` constructs physical and policy constraints. Helper modules implement shared accounting, industry transition, carbon, DSM and product industry. Read the relevant helper as well as the calling equation.



Cost and solver script

`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py` constructs cost terms, the objective and solver/recovery behavior. Export and plotting scripts run after the applicable solve path. For line-level locations use the code reference; static call-site counts are not the size of a constructed optimization matrix.

---

13. Before and after a run

Before running, verify a fresh/coherent scenario name, the intended interpreter/license, active zones, actual selected workbooks, decision years and complete applicable cost/limit/profile coverage. Close Excel before modifying or replacing its workbooks. Review service remainders, sector electricity accounting and any historical baseline removals. Keep a copy of the selected code, inputs and policies with their hashes.

Run `main.py` with the selected environment. Read the selected input paths and first planning-year messages. Preserve the console log if input validation, model construction or optimization fails; these are different stages. The default resume option can skip compatible completed years, so use a new scenario name after changing geography, accounting, recipes, costs or policies.

After a run, inspect solver status, provisional/recovery notices, physical-balance audits, carbon/capture checks, accepted decision years and exported annual/hourly balances before interpreting plots. The delivered configuration allows provisional solutions and warning-only postoptimal checks. A completed CSV export is therefore not automatically a strict optimality/feasibility certificate.

For DSM inspect the hourly, class, summary and assumption exports. For product industry inspect production, demand satisfaction, imports, sales, inventories, supplier origin, territorial/embedded emissions and fiscal/CBAM accounts. Compare scenarios on the same energy, geographic, economic and time boundaries.

---

14. Common issues and the first files to inspect

Symptom	First inspection
Missing module or Gurobi failure	Selected interpreter, model requirements and Gurobi license
Different workbook used than expected	`main.py` flags, electricity case and printed `tech_dat` /policy path
Missing or ambiguous investment/cost trajectory	Exact generator ID, relevant System/zone/year and all applicable blocks in `cap_inv_lim` /cost sheets
Old or missing industry remainder	Detailed inclusion in `Industry_data`, pooled `simple_industry`, formula cache/recalculation and units
`#REF!`, formula error or blank numeric input	Native workbook and cached value; preserve the original and recalculate a working copy using a spreadsheet application
Wrong country or zero regional demand	Active zones, System/Zone mapping, reader scoping and hourly coverage
New study unexpectedly resumes old years	`scen`, resume controls and existing history compatibility
Infeasible product demand	Mandatory tonnes, eligible plants/suppliers, annual/hourly capacity and import switches
DSM has no effect	Main switches, rollout year, energy budgets, hourly cap and whether the load is basic electricity
Long solve or recovery	Runtime time limits, selected mode, solver log, matrix size and numerical diagnostics
Run finishes with warnings	Provisional acceptance settings, actual solver status and physical-quality reports

Do not bypass a missing trajectory with zero cost or delete an unsupplied demand merely to get past validation. Review identifiers, units and the intended scenario first. The documentation inventory and screenshots are evidence of inspected content, not proof that every retained workbook or run mode is feasible.

---

15. References, provenance and sharing

Use the scientific and data references, BibTeX bibliography, assumption register, exact source/input inventory and workbook schema. The 2004 H2RES island-model lineage and later planning/optimization papers provide background; later local DSM and product-trade extensions are identified separately in this edition.

This documentation was built from static inspection of the specified 12 November source/input snapshot. No optimization was run to prepare it. The separate data extractor still documents its own native/refactored export contracts; this manual does not certify its adapter against every new model extension. Share the PDF for reading, or the documentation ZIP for offline HTML navigation, source notes, screenshots and machine-readable inventories.

Configuring main.py: complete workflows and option interactions

Applies to: the inspected `H2RES_12.11.2026` model. This guide explains the available structures, how to combine their controls and how to interpret the resulting study. The complete option reference explains every recorded assignment in `main.py` and the three loaded policy files. The equation chapter explains the physical, economic and policy mechanisms. Desktop GUI operation is excluded.

The examples are configuration instructions, not reports of new solved cases. Enabling a technology allows its modeled activity; it does not guarantee that the optimizer will build or dispatch it. Results depend jointly on demand, prices, stock, retirement, resources, investment limits, networks and all active policies.

---

1. Five decisions define a study

Choose the following independently, then check their dependencies:

Decision	Principal choices	Where configured
Physical boundary	Electricity, heat, legacy industry/transport, product industry, H2/e-fuels, capture, trade	Main module switches plus eligible data rows
Industry representation	Pooled/mixed detailed services, temperature categories, selected complex processes, separate product model	Main industry flags, Industry_data inclusion and companion workbooks
Planning information/time	Joint foresight, ordinary sequential planning, partial foresight, representative capacity, rolling fixed dispatch	Main time-mode flags, runtime policy and policy setup
Policy experiment	RPS, emissions caps, removals, transition rules, electricity CBAM, product CBAM, domestic carbon prices	Main switches, model policy and policy/workbook trajectories
Numerical acceptance/output	Algorithms, retries, time limits, provisional/strict checks, diagnostics, histories, plots	Main/runtime settings and exporters

A meaningful comparison normally changes one documented group at a time. For example, compare e-fuel and no-e-fuel cases using an explicitly matched demand/service boundary; otherwise different workbooks may change more than the carrier representation.

---

2. How to make a configuration change

1. Preserve the original model/input snapshot. Select the model's Python/Gurobi environment.
2. Set a fresh `scen` in `main.py`, identifying the intended experiment.
3. Edit the original declarations in `main.py`. Do not append settings below the execution section: the model may already have solved by then.
4. Edit numerical and transition parameters in the appropriate policy module. Main loads these before most switches and later derives filenames.

5. Edit the selected workbook's `setup`, active geography, inclusion flags, limits and policy paths. Saving a different workbook does not change the selected path.
6. Check the normalized configuration and selected input paths. A flag may be overridden, ignored outside its parent module, or rejected.
7. Run `main.py`; inspect the first input/read/build messages before interpreting the eventual result.

The following snippets show the settings to change, not complete standalone Python programs. Apply them at their original declarations. A scenario JSON from the data extractor is not automatically loaded by native `main.py`. The source uses a shared namespace, so policy modules and helper normalization matter as much as the visible main switches.

---

3. Dependency and normalization map

Requested combination	Effective behavior / requirement
Categories plus complex industry	Rejected; these are alternative representations
`industry_storage=True`	Requires transport/industry, categories and multi-sector operation
Complex industry	Requires legacy transport/industry and `industry_single_sector=False`; selected sectors come from policy
Either complex alias True	Both aliases are normalized to the same effective complex selection
E-fuels plus standalone H2 pipelines	Main disables standalone `model_H2_transmission`; carrier transmission already handles H2
Explicit Oil/Gas supply without e-fuels	Main disables that supply flag
E-fuels plus no base-year vehicle investment	Main permits base-year additions to meet the e-fuel case's initial fleet floors
Capture off with amine storage on	The variable/setup path turns amine storage off
Capture and e-fuels both off	The applicable setup turns post-generation CO2 storage off
Standalone terminals without e-fuels	Requires legacy transport/industry, the H2 balance and valid H2 terminal/routes/cost data
Terminals with e-fuels	Uses the product-terminal, route and carrier tables instead of assuming the standalone schema
Lightweight representative mode	Enables automatic mode but skips its stage-2 rolling dispatch
Fixed capacity without rolling dispatch	Main normalizes fixed capacity off for a normal capacity solve
Rolling dispatch alone	Does not select the standalone dispatch runner; fixed capacity must also be True
More than one electricity CBAM option	Rejected
Electricity CBAM with `Myopic=False`	Rejected by the inspected builder
Product industry enabled	Configure step disables overlapping legacy industry/transport, e-fuel/H2/capture blocks and selects its active example systems
Product carbon pricing/CBAM with product mode off	Invalid as an independent product-policy configuration
Building heat technology flag with heating off	Does not create an active heating subsystem
Legacy industry with heating off	Not presented as a fully costed preset: current industry CAPEX definitions include heating-nested paths

Requested combination	Effective behavior / requirement
No active/eligible input row or zero build allowance	A main technology flag does not create the missing asset or permit forbidden additions

These include explicit errors, automatic normalization and reviewed support limitations; they are not all the same kind of check. The absence of a blanket `ValueError` does not prove a combination has complete costs, variables and balances. Keep experimental/debug settings separate from production scenarios.

---

4. Time modes: select one execution structure

The calendar-year list in `policy.xlsx` or `policy_efuel.xlsx` is separate from hourly sampling. The inspected e-fuel setup lists 2025, 2030, 2035, 2040, 2045 and 2050, with `chunk_size=1`, `step_size=1`, and hours 1–8760. Thus a name containing `5y` does not itself select five-year chunks.

Workflow	Myopic	Partial	Auto two-stage	Lightweight	Fixed / rolling
Joint full-hour planning	False	False	False	False	False / False
Ordinary sequential planning	True	False	False	False	False / False
Sequential planning with sampled future lookahead	True	True	False	False	False / False
Representative capacity screening	Either	False	True	True	Initially False / False; controller manages stages
Capacity planning followed by chronological fixed dispatch	Either	False	True	False	Initially False / False; controller manages stages
Dispatch from a trusted existing bundle	Match the intended study	False	False	False	True / True

Leave `representative_capacity_mode=False` for the ordinary full-hour and partial-foresight recipes. The automatic controller sets representative/fixed/rolling modes internally. Do not manually turn on every time-related flag.

4.1 Joint full-hour perfect foresight

Use this to optimize the provided decision years jointly with their full configured hourly periods:

```
Myopic = False
partial_perfect_foresight = False
two_stage_auto_mode = False
lightweight_representative_weeks_mode = False
representative_capacity_mode = False
fixed_capacity_mode = False
rolling_dispatch_mode = False
```

The optimizer knows the supplied future costs, demands and policies. This is an information assumption, not a forecast of real decision-maker knowledge. Size can grow rapidly with zones, years, carriers and process detail. Electricity Options 1–4 must remain off in this source. Ordinary sector choices, RPS, carbon caps, DSM and eligible product policies have their own compatibility requirements.

Expected evidence: the joint solve log, accepted capacities across the horizon, full configured operational coverage and physical-quality checks. A five-year decision grid still requires explaining how annual investment and operating costs are represented between decisions.

4.2 Ordinary sequential myopic planning

Start with the preceding settings, change `Myopic=True`, keep partial foresight False and use a new scenario. `chunk_size` and `step_size` count positions in the decision-year list. With one year per chunk and one-position advance, each point is solved sequentially and saved history constrains later decisions. On a 2025/2030/... grid, one position is five calendar years; on an annual grid it is one year.

Increasing chunk size exposes more decision points jointly to each solve. The exact exporter and history behavior matters for overlapping chunks; do not assume every ordinary multi-year chunk commits only its first year. That first-year restriction is explicit in the partial-foresight path. Prefer a clearly documented chunk/step structure and inspect accepted-year exports.

Use this for limited-information or path-dependence experiments, while recognizing that a one-point solve can make choices that later demand/policy makes expensive or infeasible. Myopic results are not necessarily as low-cost as joint foresight under otherwise matched assumptions.

4.3 Partial foresight with a detailed decision year

```
Myopic = True
partial_perfect_foresight = True
two_stage_auto_mode = False
lightweight_representative_weeks_mode = False
representative_capacity_mode = False
fixed_capacity_mode = False
rolling_dispatch_mode = False
```

In `policy/runtime_policy.py`, review:

```
partial_foresight_representative_days = 4
partial_foresight_day_hours = 24
partial_foresight_representative_day_start_hours = (337, 2521, 4705, 6889)
partial_foresight_lookahead_years = 10
```

The current year uses the full configured calendar; future decision years use four sampled days. Ten is a calendar-year reach, not ten decision points. From 2030 on a five-year grid, the allowed points are 2030, 2035 and 2040. `None` exposes the remaining tail. The current decision-year result is exported as committed history; sampled future solutions guide that decision and are revisited later.

This can reduce solve size relative to giving every future point a full year. It does not establish seasonal storage feasibility for the sampled future years. A shorter lookahead can change investments materially. Report the actual sampled starts and weights with the result.

4.4 Representative capacity screening only

```
Myopic = True
partial_perfect_foresight = False
two_stage_auto_mode = True
lightweight_representative_weeks_mode = True
fixed_capacity_mode = False
rolling_dispatch_mode = False
```

Choose `Myopic=False` instead if a joint representative capacity study is intended. The controller performs stage 1 and skips stage 2. Its historical setting names are misleading in this snapshot: `representative_weeks_count=4` and `representative_week_hours=24` mean 96 sampled hours, not 672 hours. Equal sample expansion is 8760/96 = 91.25 where that annualization convention applies.

Use it to screen assumptions and inspect indicative capacity/cost changes. Do not call it a full-year operational validation. The annualized energy totals and sample metadata must accompany plots. If changing to actual week-long blocks, set compatible lengths/starts/counts and reassess all module chronology; changing the label alone has no effect.

4.5 Automatic capacity planning plus rolling dispatch

```
Myopic = True
partial_perfect_foresight = False
two_stage_auto_mode = True
lightweight_representative_weeks_mode = False
representative_capacity_mode = False
fixed_capacity_mode = False
rolling_dispatch_mode = False
```

The controller samples stage-1 capacity planning, exports a dispatch bundle, then freezes capacity for rolling operation. Use `Myopic=False` for the joint stage-1 variant, subject to the selected policies. Keep `two_stage_stage1_export_timing='before_stage2'` so investment/history outputs exist. Turning those exports off can affect later sequential chunks.

This asks whether sampled planning capacities can operate under the configured chronological dispatch assumptions. It is not a single all-hours capacity optimization. Forecast deviations, annual budgets, guidance, stock end conditions and penalized shortages all affect stage 2. Inspect the original window files and stitched annual balances, not only representative stage-1 plots.

4.6 Daily versus weekly intra-annual dispatch

In runtime policy, `rolling_dispatch_granularity='day'` sets 48 optimized hours, 24 committed hours and a 24-hour advance. `'week'` sets 336 optimized hours, 168 committed hours and a 168-hour advance. `'custom'` uses the explicit window, step and realized-horizon settings. With day/week selected, editing only `rolling_window_size` is overwritten by the granularity function.

The second half of a daily window provides forecast/lookahead; it is not committed twice. The next window inherits the last committed physical states and accepted budgets. Storage, maritime transit, annual imports and carbon guidance require correct handoff metadata. Longer windows can reduce short-sighted dispatch while increasing memory and solve time.

The author reports improved intra-annual myopic stability in a more recent no-e-fuel working version. That report should not be generalized to all e-fuel, capture, terminal and process combinations or treated as a new regression result for this documentation snapshot.

4.7 Fixed dispatch from an existing capacity bundle

```
two_stage_auto_mode = False
lightweight_representative_weeks_mode = False
representative_capacity_mode = False
fixed_capacity_mode = True
rolling_dispatch_mode = True
export_stage2_dispatch_bundle = False
dispatch_stage_data_path = './results/REFERENCE_SCENARIO/stage2_dispatch_bundle.pkl'
```

First obtain a trusted bundle through planning with `export_stage2_dispatch_bundle=True` or automatic stage 1. The file contains model state/metadata, not just capacities. Match units, active geography, years, technology/accounting conventions and module settings. This recipe freezes supported assets; it does not mean setting every `cap_inv_lim` cell to zero has the same effect.

A zero build limit allows surviving existing capacity but may not pin every state, retirements or previously planned investment. Use the dedicated fixed-capacity mechanism for a dispatch experiment. Report any unsupported or unpinned asset family explicitly.

---

5. Electricity-only and integrated-energy boundaries

For a deliberately reduced electricity-only structure, disable the legacy coupled modules consistently:

```
model_simple_industry = False
model_heating = False
model_cooling = False
model_transport_and_industry = False
model_industry_categories = False
model_complex_industry = False
model_complex_non_metallic_minerals = False
industry_storage = False
model_efuel = False
model_oil_and_gas_products_supply = False
model_H2_supply_demand_balance = False
model_H2_transmission = False
model_terminals = False
model_CC = False
model_CO2_ammine_storeage = False
model_CO2_post_gen_storeage = False
```

This is a reduced-boundary configuration for validation, not a claim that the existing input demand has been calibrated to an electricity-only study. Inspect the reader's decomposition and demand CSV: demand already assigned to explicit sector blocks must not disappear unintentionally, nor be counted twice. Retain the intended generation, storage, network, RPS and carbon policies separately.

For the delivered integrated case, keep heating and legacy transport/industry on and select the appropriate industry/carrier representation. Disabling heating alone while retaining legacy industry is not a reliable shortcut because current cost paths include heating-nested industry definitions. Model-size reduction should preserve a complete accounting boundary.

---

6. Heating technologies and three different thermal stores

`model_heating=True` activates the heating subsystem. The individual and DH boiler flags select their eligible boiler lists; the individual and DH HP flags select their HP lists. Solar flags select solar thermal, not PV. `model_cooling` is a separate legacy request requiring its cooling data and verification.

Intended change	Controls and input review
Permit individual building HPs	`model_HP_units=True`; individual heat demand, COP/efficiency, capacity, retirement and cost trajectories
Permit DH HPs	`model_HP_units_DH=True`; DH demand/losses and district technology rows
Permit solar heat	Applicable solar flag, availability, technology/cost and storage rows
Individual HP heat shifting	`individual_heat_pump_storage=True` with heating and individual HPs
District heat/cooling storage	`DH_Storage=True` with the relevant thermal subsystem and its storage parameters
Industry category thermal storage	`industry_storage=True` with category industry; independent of both building/DH stores
Fix the pool of available boilers	Review capacity/retirement and build limits; the technology-selection flag alone does not freeze stock

Keep `NoResToHeatInv=False` for normal individual HP electricity/capacity equations. Its inverted legacy gate skips those equations when True; it should not be used as a general prohibition of renewable-to-heat investment. Likewise, `ratio_heat_indv` and `heating_shares_regulate` add specific equation branches and do not simply select a national heating fuel split.

Heating transition policies operate in addition to technology eligibility. Current controls include base-year CHP heat calibration, CHP year-to-year change, individual electric/gas heat change, fossil heat nonincrease, boiler phaseout and biomass heat nonincrease. To study a different phaseout, change both the intended switches and their fuel/year/rate data; disabling only `fossil_boiler_phaseout_active` can leave nonincrease or the carbon limit binding.

---

7. Industry without temperature categories: pooled or mixed detail

```
model_transport_and_industry = True
industry_single_sector = False
model_industry_categories = False
model_complex_industry = False
model_complex_non_metallic_minerals = False
industry_storage = False
```

With `model_efuel=False`, the active technology file is `flex_tech.xlsx`; with e-fuels True it is `flex_tech_efuel.xlsx`. The detailed-industry inclusion field determines which native industry services remain explicit. Setting the applicable detailed services to N must transfer their quantities to the pooled `simple_industry` worksheet. It does not mean industrial energy vanishes.

Mixed configurations are possible: retain chosen branches/services explicitly and pool excluded ones. Recompute and reconcile the remainder whenever inclusion changes. Retained detailed plus pooled demand must equal the original intended industrial demand, with capacities and efficiencies on a consistent basis. Do not confuse a stored formula with a current cached numeric value.

This mechanism is distinct from `industry_single_sector=True`, which selects an older branch. The latter is not the recommended route to obtain a pooled integrated e-fuel or category case.

---

8. Industry with temperature/service categories

```
model_transport_and_industry = True
industry_single_sector = False
model_industry_categories = True
model_complex_industry = False
model_complex_non_metallic_minerals = False
industry_storage = True
```

Choose True or False for the last setting depending on whether industrial thermal storage is part of the experiment. The active technology file is the applicable category variant. Review `Industry_data`, category costs, industry thermal storage, HP applicability and capture data. `industry_category_profiles.csv` supplies service timing.

Use `industry_category_electricity_case='conservative'` for the main e-fuel category workbook; use `'technical_2050'` for its nested alternative. This changes applicability assumptions, not the planning-year list or a mandatory share. A fully detailed category case can legitimately leave no pooled remainder. The category, fuel-compatibility and adoption ceilings still interact.

---

9. Process-based complex industry and tested sector scope

```
model_transport_and_industry = True
industry_single_sector = False
model_industry_categories = False
model_complex_industry = True
model_complex_non_metallic_minerals = False
industry_storage = False
```

The compatibility alias will be normalized to the effective complex selection. In `policy/model_policy.py`, select exact supported branch names through `complex_industry_sectors`; the delivered tuple is `('Non_metallic_minerals',)`. Selected branches replace matching ordinary service rows. The remaining sectors retain the ordinary formulation for that configuration.

Use `industry_complex.xlsx` without e-fuels and `industry_complex_efuel.xlsx` with e-fuels. The complex electricity-case selector can choose each companion's technical alternative. Check `Process_data`, product demand, component units, useful service, feedstocks, compatibility and capture mappings together.

The author reports testing non-metallic minerals, while additional process data are prepared but not comprehensively tested across all sectors. The reported order of 150–200 processes describes the broader development scope, not a verified active process count for every case. Extend sector selection progressively and check demand/energy/process-carbon reconciliation before comparing national results.

---

10. Industrial heat pumps and adoption constraints

Industrial HPs use `industry_heat_pumps_active` in model policy, not the building `model_HP_units` flag. They need eligible useful-service temperatures, source heat, profiles, costs and investment limits. Their cumulative low-temperature potentials overlap: several cumulative thresholds must not be added as separate available heat pools.

`industry_electrification_full_applicability=True` permits the implemented full applicability treatment but retains capacity, transition, resource and other process restrictions. It is not a command to electrify all industry. The conservative/technical workbook case remains a separate choice.

General ramps, S-curve selection and annual combined electric/HP share controls are independent. For the delivered finite transition, review the 2045 completion year, seed 0.02, growth 0.5 and five-year reference interval. The finite progress function is a cubic smoothstep; do not describe it as a calibrated logistic diffusion forecast. A direction override is a policy choice distinct from a technical maximum share.

To compare transition-constrained and unconstrained industry, document which of `industry_fuel_share_ramp_active`, `industry_fuel_share_s_curve_active` and `industry_electric_share_yoy_limit_active` changed. Disabling one does not erase the remaining adoption, applicability, stock or carbon restrictions.

---

11. Integrated e-fuel modeling

For the author's supplied integrated e-fuel configuration, use:

```
model_transport_and_industry = True
industry_single_sector = False
model_efuel = True
model_oil_and_gas_products_supply = True
model_H2_transmission = False
```

Select regular, category or complex industry as described above. Keep heating enabled for the documented legacy integrated cost paths. The technology and policy filenames acquire the e-fuel family, with `policy_fuels.xlsx` for applicable carrier prices. Review conversion recipes, primary inputs, carrier storage, compatible final demand, external supply, terminal/routes, reserve/extraction assumptions and all cost/build trajectories.

E-fuels can require H2, CO2, electricity, heat and other feedstocks. Process carbon, DAC and eligible industrial biogenic CO2 can supply CO2 feedstock under their actual source/capture mappings. When process emissions are present, a tight carbon cap also needs a feasible net-emissions treatment; feedstock demand alone does not guarantee enough permanent storage or offsets.

`model_oil_and_gas_products_supply=True` is part of the supplied integrated configuration, and main disables it when e-fuels are off. The code does not impose a blanket error for every hypothetical e-fuel case with this flag off; such an alternative must establish its own complete supply boundary. Do not silently substitute that alternative for the author's tested setup.

Expected outputs include electricity and carrier balances, electrolysis and conversion capacities, feedstock demand, storage/trade, refinery/fossil supply, capture and emissions. Zero e-fuel output can be the economic result of permitted alternatives, absent final demand, limited CO2/H2, expensive electricity or build restrictions; activation is not a production target.

---

12. No-e-fuel scenarios and matched comparisons

Set `model_efuel=False` and `model_oil_and_gas_products_supply=False`, choose the corresponding non-e-fuel technology/policy inputs and review exogenous CO2 demand if capture remains enabled. Retain the intended heating, transport and industry service boundary. Do not compare the result with an e-fuel run while assuming every price, technology or service row is identical across workbook families.

Without e-fuels, standalone H2 supply/demand, pipelines and aggregate H2 terminals can be used under their own requirements. Capture and geological storage can still be relevant to process emissions. No-e-fuel does not automatically mean no H2, no industrial capture or no product carbon policy: each belongs to a separate valid representation.

---

13. Capture, amine flexibility and the CO2 destination

For eligible industrial capture, enable `model_CC=True` and activate the corresponding capture rows and equipment limits in the selected native technology/process inputs. Confirm capturable streams, capture rates, electricity/heat demand, reboiler capacity and costs. A capture switch does not create an available carbon source or a final destination.

`model_CO2_ammine_storeage=True` adds absorber/stripper timing flexibility through amine inventory. With capture off, the setup turns it off. `model_CO2_post_gen_storeage=True` permits the applicable post-capture stock/capacity representation; geological storage additionally needs eligible active geological units and positive capacity/build allowances. Rich-amine stock, reusable CO2 buffer, geological inventory and injection capacity have different physical meanings and units.

Intended CO2 experiment	**Required interpretation**
Capture off	No eligible industrial capture assets; e-fuel DAC may still exist through its own representation
Capture on, amines off	Capture and stripping follow the corresponding immediate/non-storage relationship
Capture with amines	Add inventory flexibility and enforce mass/energy balances and bounded cyclic/rolling states
Captured carbon for e-fuels	Needs permitted source/feedstock mappings and conversion demand; reuse is not permanent sequestration
Geological storage	Needs main switches plus active native storage rows, stock/injection/build capacities and permitted provenance

Intended CO2 experiment	Required interpretation
Mandatory eligible process capture	Set `complex_industry_process_capture_required=True` in a compatible complex case; uncapturable residuals remain possible

Industrial biomass carbon is counted physically for capture/feedstock where eligible but is not charged as fossil combustion in the supplied aggregate budget convention. Document this distinction; it does not establish biomass sustainability or a complete lifecycle balance.

---

14. Process emissions under a binding carbon limit

If `carbonLimit=True`, residual process emissions can make a stringent or zero-carbon endpoint infeasible even when fossil combustion has been eliminated. There are two distinct configured mechanisms:

1. Capture eligible process carbon and send it to enabled geological storage with sufficient capacity, injection and build allowances.
2. Include a justified LULUCF removal allowance in the carbon budget through `lulucf_removals_active=True` and the supported CSV/fixed input convention.

LULUCF is an accounting offset; process CO2 is not physically sent to a land-use technology. It does not create capturable CO2 or satisfy an e-fuel feedstock requirement. Avoid assigning the same removal allowance twice across fixed and scoped inputs.

For process-only geological eligibility, use:

```
carbonLimit = True
model_CC = True
model_CO2_post_gen_storeage = True
separate_industry_process_carbon_balances = True
```

Also activate the required process capture/storage entries in the selected native sheets. The separated balance prevents other combustion, DAC and untagged imported/buffer carbon from acquiring geological process provenance. Without this separation, the applicable aggregate carbon branch has a different eligible-credit boundary.

Before relaxing the carbon cap, check source emissions, capture eligibility/rates, geological rows, annual build limits, energy for capture, final storage capacity and permitted LULUCF. If uncapturable process emissions exceed the remaining budget, removing a numerical warning cannot make the intended physical scenario feasible.

---

15. Electricity transmission, HVDC and outside systems

`use_HVDC=True` includes eligible HVDC link technology and converter requirements in the electricity transmission representation. Existing/new capacity, transformer/converter costs, routes, distance losses and investment limits still come from the network inputs. False filters that technology; it does not turn off all inter-zonal exchange.

`include_secc_links=True` includes secondary routes through intermediate zones where supplied. Review double counting, route keys and primary/secondary interactions. Transmission has a deterministic duplicate policy (`first` in the delivered settings); reconcile conflicting records rather than assume the first is scientifically correct.

Outside-system trade can be limited by hourly supply shares/curves, annual outside-trade calibration, transfer capacity, prices and the selected carbon representation. To study an isolated system, constrain or deactivate its external routes in the native network tables and inspect the actual balance. Turning off `outside_system_availability` removes an availability restriction and can increase trade; it does not isolate the system.

---

16. H2 pipelines and terminals with or without e-fuels

For standalone H2 pipelines without e-fuels:

```
model_efuel = False
model_transport_and_industry = True
model_H2_supply_demand_balance = True
model_H2_transmission = True
```

Keep a compatible industry/heating representation, H2 production/demand/storage and pipeline/compressor/cost data. Pipeline aliases and compressor electricity inputs matter. With e-fuels True, use the product transmission tables for H2; main normalizes the standalone pipeline flag off.

Terminals are possible in either carrier representation. Set `model_terminals=True`, then select active terminal rows and allowed flow directions in the applicable tables. With e-fuels, review product terminal/transmission, terminal activation, import/export, shipping and storage mappings. Without e-fuels, review `h2_terminals.xlsx` with `terminals` and `shipping_routes`, H2 balance and `H2_terminal` cost coverage.

The no-e-fuel terminal model uses aggregate delivered H2 in MW/MWh and annual throughput capacity in TWh/year, buffered by zonal H2 storage. It is not automatically a full liquid-H2/ammonia vessel, liquefaction and cracking model. Terminal throughput, shore/storage energy, vessel travel time and route capacities must not be conflated.

The author reports repeated successful annual runs for Croatia divided into four zones using HVDC, H2 transmission and terminals. This is author-reported case evidence, not a guarantee for other geography, year grids or all combinations. The equations and exact configured inputs determine transferability.

---

17. Transport: fleet, electrification, alternative fuels and infrastructure

Legacy transport is part of `model_transport_and_industry=True`. Use the correct workbook family, sector demand, activity/plug-in/V2G timing and policy targets. Continuous vehicle-equivalent stock, transport service, final energy and battery stock are distinct model quantities.

Control group	What to change for a study
Electrification pathways	Main electrification switch, target sectors, policy floors and applicable nondecreasing stocks
Final service applicability	`transport_max_share_limits_active` and `transport_all.max_share`; a ceiling is not a required share
New purchases/turnover	Investment-limit switch, unit/sector caps, turnover life/ramp, inherited fleet age and forbidden new units

Control group	What to change for a study
Rail	Direct-electric ceiling, optional rail linkage and battery-rail specifications; do not treat direct EL and battery rail as one unbounded stock
Alternative ICE fuels	Pathway corrections, blend/volume-to-energy rules, paid upgrades, costs and remaining life
Marine methanol/ammonia	New equipment CAPEX, retrofit fractions, pilot-fuel energy and eligible existing stock
Fossil transition	Passenger-car 2050 sensitivity, optional ICE/hybrid fossil decline and annual fuel-composition ramp
Infrastructure	Keep `disable_transport_infrastructure=False` for fleet-linked charging/refueling stocks and costs; supply existing infrastructure data

An ICE vehicle running a compatible non-fossil fuel is different from an electric vehicle or a retired vehicle. A paid conversion consumes eligible incumbent stock and does not permit arbitrary fuel mixing. The test-only unconstrained-upgrade bypass is not a production scenario. E-fuel mode overrides the no-base-year-investment flag because its initial floors can require fleet replacement immediately.

To study infrastructure omission as a sensitivity, label the changed boundary explicitly; a cheaper result without charging/refueling CAPEX is not the same scenario solved more efficiently. Similarly, relaxing turnover or blend limits changes the physical/adoption assumptions, not just runtime.

---

18. Daily and weekly basic-load DSM

For both budgeted classes with rollout:

```
DSM_day = True
DSM_week = True
dsm_energy_budget_active = True
dsm_availability_ramp_active = True
```

Review daily/weekly energy fractions, hourly adjustment fraction, full-availability year and enabling/activation costs in runtime policy. The supplied daily-capable 7% includes the weekly 1.5%; with both active, 5.5% balances daily and 1.5% weekly. The hourly power cap is an independent 10% of basic load.

For daily-only or weekly-only studies, activate only that class and retain coherent budget/rollout controls. Both false preserves fixed basic demand. Legacy hourly-only variants exist with budgets disabled, but they are not the same flexibility assumption as the constrained new DSM. Changing budgets/rollout can invalidate saved histories and bundles.

DSM acts on residual basic electricity, not automatically on every separate heat, transport or process demand. Daily and weekly blocks are anchored to model hour 1, including final incomplete closure; they are not ISO calendar weeks or appliance-level deadlines. Inspect moved MWh, capacity, costs and block-energy conservation rather than assume the allowance must be fully used.

19. Separate product industry and industrial CBAM

```
model_simple_industry = True
industry_cbam_active = True
industry_carbon_pricing_active = False
industry_external_imports_active = True
industry_external_exports_active = True
steel_flexible_operation = True
```

This selects the product model through its configure step, disabling overlapping legacy industry/transport, e-fuel, H2 and industrial capture blocks. It is not an additional industry layer placed on top of all category/complex demand. Heating keeps its independent selection. Review the configured active HR/RS/BA systems or provide a calibrated alternative.

The separate workbook contains fixed plant recipes, electricity/grid requirements, prices, product demand, eligibility, bounded outside suppliers and trade routes. Example plants and offers are illustrative. Calibrate `baseline_load` removal if their electricity is already inside basic demand; zero removal adds their load. Demand is mandatory where specified even if local plants become uneconomic.

Industrial CBAM and domestic carbon pricing are independent. Compare both off, domestic only, border only and both on, while keeping supplier/market/recipe data fixed. The policy fields determine benchmarks, free allocation, credits and exporter/importer allocation. Electricity Options 1–4 are separate again and retain their own Myopic requirement.

Turning outside imports off does not remove modeled cross-country trade or turn off required Europe demand. Restricted European background supply remains only where its eligibility allows it. Turning optional outside exports off does not erase mandatory domestic/Europe markets. Track original origin and final destination when interpreting embedded carbon and fiscal payments.

`steel_flexible_operation=False` applies the appropriate within-day constant-operation treatment to eligible plants; it is not a requirement that plants operate at full load or that each country produces its own required product. Domestic production can be displaced by eligible imports under bounded supply.

---

20. Electricity carbon-pricing Options 1–4

Choose at most one of `Option_1`, `Option_2`, `Option_3`, `Option_4`; leave all False for no selected electricity-CBAM option. The builder requires `Myopic=True` when any is active. Carbon prices and the carbon quantity cap are different instruments: turning one off does not necessarily remove the other.

Option 1 represents border-only electricity CBAM. Option 2 combines border and national tax/ETS treatment. Option 3 uses the applicable national ETS/EU-price relationship. Option 4 is the mixed national/border branch; supplied structural notes flag a national-cost auxiliary without its own nonzero definition, so it is not presented as a fully verified policy recipe. Inspect its exact rows before relying on it.

Supply active membership, border/network mappings, carbon prices and valid source-zone intensities. Base-year signals, realized predecessor signals and future lookahead overrides have different roles. A valid historical zero is retained; absent required history must not be silently converted to zero emissions. These are the model's scenario representations, not a legal determination of current CBAM obligations.

21. Renewable targets, carbon caps, biomass and policy sensitivities

`rps_inv=True` activates native renewable-electricity share constraints from `RPS`. `res_inv=True` selects reporting branches. These similarly named flags do different things. Setting the RPS path to 100% does not certify the renewable origin of all imported fuels, process inputs or final-energy use.

`carbonLimit=True` adds the applicable emissions budget. Prices, capture, geological credit and LULUCF can change how that budget is met. A zero territorial carbon outcome is not necessarily zero embedded imports or a full lifecycle net-zero outcome. Separate process provenance changes which geological credits are permitted.

Resource and transition policies also matter: biomass feedstock limits, biomass-heat nonincrease, generation calibration, hydro caps, outside-trade targets, heating turnover and industrial adoption envelopes can all bind. To remove a historical calibration for a future sensitivity, identify its exact scope and flag rather than delete unrelated resource constraints.

For a whole-energy 100%-renewable study, explicitly define covered carriers, imports, stock origin, biomass/waste eligibility, feedstocks and non-energy/process emissions. The extractor's whole-energy preset adds adapter equations with its own compatibility limits. Native RPS alone is insufficient, and this manual did not validate every 12 November extension through that adapter.

---

22. Changing years, geography and investment opportunities

Change decision years and hours in the selected policy workbook's `setup`, not only the scenario name. Extend every required annual cost, build, demand, product market/supplier, carbon and retirement path. Annual and five-year decision grids must preserve the global investment-window convention. Reference-interval S-curve settings must remain compatible with the decision step.

For another country or subnational/custom zones, establish complete `System`/`zone` mappings, selected active rows, generator and curve IDs, sector demand, outside links, network reachability, stock and cost/limit keys. `active_zone_overrides` can change scoping but cannot manufacture missing inputs. A zone with no existing generator can import through modeled links; it should not receive a fabricated plant merely to fill a table.

To allow more investment, adjust the relevant native annual or total capacity bounds in their correct units and review applicable generator multipliers/adoption policies. MW, MWh, vehicles, tonnes/hour and tonnes of storage are not interchangeable. Increasing an investment limit does not supply more biomass, change a COP or create available CO2 feedstock.

---

23. Solver controls and strict versus exploratory acceptance

The delivered numerical selection uses barrier for planning and dual simplex for rolling operation, with explicit scaling and recovery paths. Time limits of zero are treated as unlimited in the applicable wrappers. A different algorithm can change time or numerical behavior; it does not fix missing physical supply or inconsistent demand.

For a stricter acceptance study, review:

```
solver_allow_provisional_solution = False
solver_postoptimal_checks_warn_only = False
write_solver_diagnostics = True
solver_diagnostics_on_problem_only = True
```

Retain `solver_require_annual_cc_quality=True` where capture is modeled, and review physical energy/amine tolerances. These settings strengthen the applicable acceptance path; they do not remove explicit rolling shortage, carbon or guidance slack variables. If strict policy feasibility is required, also review the relevant slack switches/bounds and actual exported values.

For difficult numerical cases, inspect scaling, coefficients, physical units and history residue before increasing tolerances. The annual aggregation retry and optional simplex fallback are separate from the generic stronger retry. Exact Kappa diagnostics can be expensive. Changing thread count, focus, crossover, aggregation or time limits must be recorded with the solver log.

An infeasible model has no feasible solution under its encoded assumptions. A numerical failure or interrupted solve is a different outcome. A provisional export is not a certified optimum. Report the status and physical residuals independently of whether plot scripts completed.

---

24. Resume, caches, outputs and reproducibility

Ordinary myopic resume uses `myopic_resume_from_results` and the contiguous completed prefix. Automatic two-stage mode separately checks completed dispatch chunks and may reuse a stage-1 bundle only under its settings. Rolling-window resume separately requires saved window files and compatible incoming states. Do not treat these three resume controls as interchangeable.

A new scenario name is appropriate after material changes in geography, identifiers, recipes, accounting, costs, year grid, DSM budgets, product markets or carbon policy. `validate_completed_heating_policy_history=False` preserves old accepted years without proving they satisfy newly changed heating rules. Reusing an old bundle/history can create an internally inconsistent comparison even when it loads successfully.

Local pandas/Excel read caching improves repeated input reads. It is not Open-Meteo download caching, API quota control or proof that online observations were refreshed. Deep-copy settings concern shared-data mutation and memory. Do not use the cache counters to estimate internet requests.

Preserve CSVs needed for investment/stock histories, hourly/annual balances, costs, carbon and module diagnostics. Additional shadow-price exports need their appropriate accepted LP/dual treatment. Archive model and input hashes, policy/configuration files, environment/license configuration without credentials, console logs, representative weights, accepted-year metadata, checkpoint identity and figure-generation scripts.

---

25. Goal-to-configuration checklist

Study question	Required controls / review	Evidence to inspect
Effect of future information	Joint versus ordinary/partial myopic; matched costs, demand and year grid	Investment timing, objective boundary and committed history
Faster indicative planning	Representative capacity-only with declared samples	Sample metadata and annualization; no full-year feasibility claim

Study question	Required controls / review	Evidence to inspect
Chronological operation of planned assets	Automatic two-stage or trusted fixed bundle	Stitched/window balances, stock carryover, shortages and guidance slack
Industrial electrification	Category/regular/complex applicability, HPs, transition and resource limits	Useful-service electric share, final electricity, HP COP and build constraints
Industrial thermal flexibility	Category mode plus industry storage and eligible service data	Charge/discharge, useful heat, energy/power capacity and losses
Alternative industrial process routes	Complex sectors and compatible process/product inputs	Product activity, component energy, legacy replacement and process carbon
E-fuel production	Integrated e-fuel/Oil-Gas recipe plus final demand and feedstocks	H2/CO2/electricity/heat inputs and carrier balances
Fossil-product export sensitivity	`allow_fossil_fuel_exports`, prices, reserves and external routes	Outside fossil exports, refinery activity, revenue and emissions
Process emissions near a zero cap	Capture/geological eligibility or justified LULUCF budget offsets	Gross/net process emissions, injections, inventory and removal accounting
Restrict geological credit to process carbon	Separate process/other carbon balances	Tagged capture/stripping and geological source ledger
Standalone hydrogen network	No e-fuels, H2 balance, pipelines/compressors	H2 production, demand, link flow/losses and auxiliary electricity
Maritime/product imports	Terminals and active allowed routes in the matching schema	Throughput, arrivals, inventory, direction and annual supply bounds
HVDC expansion	Eligible HVDC links, converter/cost/build data	New transfer capacity, delivered flows and losses
Isolated-system study	Deactivate/constrain outside routes, preserve complete domestic balances	Zero outside exchange and resulting shortages/capacity
Accelerated fleet electrification	Floors, turnover, purchases, infrastructure and compatibility	Service shares, physical stocks, energy and charging/refueling
Alternative ICE/marine fuels	Pathways, blends, paid upgrades and pilot fuel	Compatible fuel shares, retrofit stock/cost and emissions
Basic-load flexibility	Daily/weekly DSM plus budgets, rollout and costs	Block conservation, moved MWh, power and costs
Industrial product border pricing	Product model plus industrial CBAM	Origin/destination footprint, liability, credits and fiscal allocation
Industrial domestic pricing	Product module plus domestic-price policy	Direct territorial emissions and domestic payments
Electricity border/national policy	One permitted Option, Myopic and intensity history	Eligible flow, price/intensity signal and separate carbon charges
Renewable electricity target	RPS path and native qualifying generation sets	Native numerator/denominator; distinguish imported/all-energy origin
Tighter solution-quality acceptance	Provisional off, warning-only off, appropriate physical gates	Solver status, physical residuals and every explicit slack

---

26. Evidence and author-supplied implementation notes

The following distinctions are essential when using this guide in a report:

Evidence category	What it supports
Inspected source/input snapshot	Exact defaults, named equations, guards, schema and option coverage recorded here
Supplied module/integration reports	Their stated component and example validation scope, including recorded baseline exceptions
Author's implementation explanation	Intended e-fuel/Oil-Gas pairing, process-carbon handling, pooled/mixed industry, tested non-metallic-mineral scope and network interpretation
Author-reported HR four-zone runs	Repeated annual experience with HVDC, H2 transmission and terminals; no new logs reproduced during documentation
Author-reported newer no-e-fuel intra-annual stability	A report about that working version; not automatic certification of every delivered e-fuel case
This documentation build	Static source linkage, complete option explanations, rendered equations, image/link/package integrity and unchanged original model files

No full national optimization was run to produce the manual. Other complex sectors, incomplete pollutant code, flagged electricity Option 4 behavior, changes in time sampling and combinations outside the reported tested cases require their own validation. This distinction preserves the value of the available evidence without extending it to unsupported results.

Mathematical formulation and equation-to-code guide

Version: the inspected `H2RES_12.11.2026` snapshot. This chapter explains the principal equation families across the available configurations. It complements the configuration guide, complete option reference and line-level constraint inventory. No optimization was executed to prepare this chapter.

Equations marked **reindexed** retain the stated source relationship with shorter symbols. Equations marked **structural** combine or suppress asset-specific branches, bounds or terms to explain the mechanism; they must not be substituted for the executable rows. Derived identities explain accounting or interpretation rather than claim an additional solver constraint. Every displayed equation has a source anchor resolved to actual line positions in equation_sources.json. The inventory includes all matching lines where a symbol occurs more than once.

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1. Sets, symbols and physical units

Symbol	Meaning
`s` , `z` , `l`	System, active zone and directed transfer route
`y` , `a`	Decision year and investment vintage
`t` , `T_y` , `B`	Modeled hour, year-specific modeled-hour set and daily/weekly block
`g` , `k` , `f`	Generator, conversion/storage technology and fuel/carrier
`i` , `b` , `p` , `m`	Industrial service/component, branch, product plant/supplier and product market
`X` , `K`	New and available capacity in the asset's declared native unit
`P` , `Q` , `U` , `F`	Electricity, useful heat/service, carrier input and transfer rates
`S` , `C` , `D`	Stored inventory, charging and withdrawal/discharging
`q` , `v` , `M`	Product tonnes per hourly interval, sales/deliveries and required annual product tonnes
`w` , `eta` , `COP`	Accounting hour weight, efficiency and heat-pump coefficient of performance
`pi` , `EF` , `r`	Price, emission factor and economic interest/discount rate

For one-hour periods MW and MWh per period can be numerically equal; their meanings still differ. Store energy is MWh or tonnes, not MW. Native generator storage fields may be GWh and are converted by the reader. Product fuel recipes are GJ/t while product electricity is MWh/t. The conversion is 1 MWh = 3.6 GJ. CO2 flows and inventories are tonnes/hour and tonnes respectively. Solver row/column scaling is a numerical representation, not a change in these physical quantities.

2. Objective, discounting and capital service

The configured objective contains capital service, fixed operating cost, variable operating cost and applicable recovery/guidance penalties. Product-industry costs and revenues enter through their module hooks. The expression below preserves the main source's explicit `multip` treatment rather than replacing it with an assumed annual sum.

$$\min Z = \sum_{y,z,s} DF_y\left(C^{cap}_{yzs} + mC^{fix}_{yzs} + C^{var}_{yzs}\right) + C^{slack}$$

E01 · scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py:3744 · structural objective, in the source's MEUR basis before numerical scaling.

Source: `scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`; anchor: `total_cost =`; Representation: structural objective, in the source's MEUR basis before numerical scaling.

$$DF_y = (1 + r_d)^{-(y - y_d)}$$

E02 · scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py:160 · reindexed discount factor.

Source: `scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`; anchor: `def _discount_factor`; Representation: reindexed discount factor.

The entered discount rate and capital finance rate are both 0.05 but have different roles. The former compares costs between years; the latter annualizes each asset's upfront capital over its life. A vintage's cost is its build-year price, not the later learning-curve price.

$$A(L, r_f) = \left(\sum_{n=0}^{L-1} (1 + r_f)^{-n}\right)^{-1}, \qquad A(L, 0) = 1/L$$

E03 · scripts/simple_industry_policy.py:50 · reindexed annuity-due identity, also explaining the capital-service convention.

Source: `scripts/simple_industry_policy.py`; anchor: `def capital_service`; Representation: reindexed annuity-due identity, also explaining the capital-service convention.

$$C^{cap}_y = \sum_{a \le y < a + L} c^{cap}_a X_a A(L, r_f)$$

E04 · scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py:147 · structural vintage-payment accounting; paid-back/existing-asset conventions remain asset-specific.

Source: `scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`; anchor: `capital_finance_rate`; Representation: structural vintage-payment accounting; paid-back/existing-asset conventions remain asset-specific.

The reader's sparse-grid multiplier depends on the years in the applicable solve. It is not automatically equal to the distance between every pair of decisions.

$$m = \frac{\max(Y) - \min(Y) + 1}{|Y|}$$

E05 · scripts/read_process_data.py:219 · reindexed solve-grid multiplier.

Source: `scripts/read_process_data.py`; anchor: `multip =`; Representation: reindexed solve-grid multiplier.

Variable costs include fuel, conversion operation, trade, carbon payments and configured shortages/surplus penalties. A representative-hour weight belongs in annual flow accounting exactly once. Restoring cost units requires following each implemented conversion and the overall objective multiplier.

$$C_y^{flow} = 10^{-6} \sum_{t \in T_y} w_{ty}\, \pi_{ty}\, U_{ty}$$

E06 · scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py:58 · structural EUR-to-MEUR hourly-flow cost term, not the complete variable-OPEX equation.

Source: `scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`; anchor: `_period_weight`; Representation: structural EUR-to-MEUR hourly-flow cost term, not the complete variable-OPEX equation.

---

3. Available capacity, retirement and investment windows

Available capacity consists of surviving legacy stock plus surviving new vintages. Each asset family has its own retirement curve, historical handoff and possible remaining-capacity fraction. A nominal lifetime should therefore not be replaced by an assumed abrupt retirement unless that is the actual branch.

$$K_{ky} = K_{ky}^{old} + \sum_{a \le y} \sigma_{kay} X_{ka}$$

E07 · scripts/Build_model_dual_industry_categories_named_debug.py:7000 · structural capacity/vintage accounting; sigma denotes the family's implemented survival rule.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `available_capacity_year`; Representation: structural capacity/vintage accounting; sigma denotes the family's implemented survival rule.

An annual build ceiling and a total installed ceiling are different constraints. For decisions in 2025 and 2030, the standard global investment windows are 2025 and 2026–2030 respectively. Resuming in 2030 must not redefine that global interval.

$$X_{ky} \le \sum_{c \in W(y)} \overline{x}_{kc}, \qquad K_{ky} \le \overline{K}_{ky}$$

E08 · scripts/investment_windows.py:4 · structural annual-limit aggregation and installed-capacity bound.

Source: `scripts/investment_windows.py` ; anchor: `def investment_windows` ; Representation: structural annual-limit aggregation and installed-capacity bound.

This summation is not universal for every legacy limit row: use the relevant family reader. In the supported annual industry-limit convention, a year-specific entry takes precedence over a scalar fallback; missing/blank fallback may mean unrestricted, while zero is binding. Generic and fuel-store readers have their own defaults. Never replace all missing investment limits with zero or a large arbitrary number without checking the selected sheet.

---

4. Electricity production and zonal balance

Dispatchable generation is bounded by available plant capacity. Reserve margins, ramp limits, CHP regions and fuel supply can impose further bounds.

$$0 \le P_{gty} \le K_{gy}$$

E09 · scripts/Build_model_dual_industry_categories_named_debug.py:7167 · reindexed dispatchable output bound.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py` ; anchor: `max_output_dispatch =` ; Representation: reindexed dispatchable output bound.

The inspected non-dispatchable branch uses an equality to available output. It does not implement the same equation as a generic plant-level inequality with a freely chosen curtailment variable. System surplus/CEEP is handled elsewhere.

$$P_{gty} = K_{gy}\, a_{gt(y)}$$

E10 · scripts/Build_model_dual_industry_categories_named_debug.py:7185 · reindexed non-dispatchable generation equality; the year index on a depends on simple_curves.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py` ; anchor: `max_output_nondisp =` ; Representation: reindexed non-dispatchable generation equality; the year index on a depends on simple_curves.

The zonal electricity balance equates supply plus unserved-load relaxation to consumption, exports and surplus. The compact equation groups many separate source terms; only enabled modules contribute. Product-industry baseline removals and DSM adjust the basic demand term rather than add electricity generation.

$$\sum_{g} P_{gzty} + I_{zty} + D^{el}_{zty} + L^{unserved}_{zty} = B^{adj}_{zty} + L^{sector}_{zty} + C^{el}_{zty} + E_{zty} + CEEP_{zty}$$

E11 · scripts/Build_model_dual_industry_categories_named_debug.py:7448 · structural zonal balance with sector, carrier, conversion, storage and transfer terms grouped.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py` ; anchor: `name="el_balance"` ; Representation: structural zonal balance with sector, carrier, conversion, storage and transfer terms grouped.

The source explicitly constrains changes in outside import/export flows. The compact ramp envelope below describes those flows, using the bound assembled from the applicable outside-route capacity and ramp coefficient. A GENCO column called RampUpRate does not by itself prove that every generator has an active hourly ramp

constraint. Representative predecessor maps are not automatically continuous real-calendar chronology.

$$-R^{down}_{ly} \le F^{outside}_{lty} - F^{outside}_{l,t^-,y} \le R^{up}_{ly}$$

E12 · scripts/Build_model_dual_industry_categories_named_debug.py:6150 · structural outside-flow ramp envelope; initial-period and route-specific coefficients remain in the source.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `max_transfer_systimpexp_outside_import_ramp_up`; Representation: structural outside-flow ramp envelope; initial-period and route-specific coefficients remain in the source.

Unserved-load and CEEP penalties have different physical meanings: the former relaxes shortage, the latter penalizes surplus. Turning off `ceep_limit` removes a cap, not necessarily the objective penalty. No equation in this model establishes nonlinear AC voltage stability, frequency response or N-1 security.

---

5. Electricity links, losses and outside trade

An active route is bounded by its available transfer capacity. HVDC is an eligible link technology with applicable converter requirements, not a separate energy balance. Primary, secondary and external routes use their own mappings and indices.

$$0 \le F_{lty} \le K^{trans}_{ly}, \qquad F^{delivered}_{lty} = \frac{F_{lty}}{1 + d_l \lambda_l}$$

E13 · scripts/Build_model_dual_industry_categories_named_debug.py:7448 · structural route-capacity bound with the delivered-flow loss convention appearing in the electricity balance.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `name="el_balance"`; Representation: structural route-capacity bound with the delivered-flow loss convention appearing in the electricity balance.

Outside-system supply composition and normalized curves can add hourly import/export availability. Annual outside-trade calibration is an independent restriction; removing either one does not imply unlimited imports. Link duplications, different distance multipliers and secondary routes require reading the selected input rows rather than assuming one uniform distance/loss coefficient.

---

6. Stationary electricity storage and hydro

The examined stationary store withdraws `stat_discharge` directly from its inventory and applies `stat_grid_eff` to charging in the stock balance. Keep the electrical delivery convention in the separate electricity-balance terms; do not apply a second guessed round-trip efficiency.

$$S_{ty} = S_{t^-,y} - D_{ty} + \eta^{grid} C_{ty}, \qquad 0 \le S_{ty} \le K^{E}_{y}$$

E14 · scripts/Build_model_dual_industry_categories_named_debug.py:5274 · reindexed stationary stock transition with its energy-capacity bound grouped.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `stat_storage_soc =`; Representation: reindexed stationary stock transition with its energy-capacity bound grouped.

The first period uses the configured initial stock rather than a nonexistent predecessor. Charging/discharging power limits and cyclic conditions are separate. One physical-hour stock transition must not be multiplied blindly by a representative annualization weight.

For the simple-curve reservoir branch, inflow is scaled by the source's plant reference capacity and hydro efficiency, while electrical generation withdraws stock divided by efficiency. Spillage is a stock outflow.

$$S^{dam}_{ty} = S^{dam}_{t^-,y} + i_{gt} K^{ref}_{g} \eta_g - P_{gty}/\eta_g - Spill_{gty}$$

E15 · scripts/Build_model_dual_industry_categories_named_debug.py:12122 · reindexed reservoir balance; first-period initialization and year-indexed curves are separate branches.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `name="storage_level_balance"`; Representation: reindexed reservoir balance; first-period initialization and year-indexed curves are separate branches.

Pumped storage also consumes electricity when pumping. Existing inflow, reservoir volume, turbine/pump power and annual calibration/caps constrain operation jointly. Pumped discharge is not a newly created primary renewable resource.

---

7. Native renewable-electricity share

Let `G_N`, `G_B` and `G_H` denote the raw modeled-period generation sums over the source's non-dispatchable, biomass and hydro collections. Let `G_F`, `G_P`, `G_D` and `G_R` denote its fossil, pumped-hydro, dam and run-of-river collections. The native RPS relationship is:

$$G_N + G_B + G_H \geq \rho_{sy}(G_F + G_P + G_D + G_R + G_N)$$

E16 · scripts/Build_model_dual_industry_categories_named_debug.py:12864 · reindexed source-set summation; this row uses modeled-period sums rather than an assumed all-energy denominator.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `rpscondition`; Representation: reindexed source-set summation; this row uses modeled-period sums rather than an assumed all-energy denominator.

These collection names describe actual source sets, not an assertion that every set is disjoint or that imports have certified renewable origin. The displayed source row does not explicitly apply representative-hour weights. Review the chosen temporal mode and set definitions before interpreting it as a weighted annual statistic. Setting rho to one does not impose 100% renewable energy in heat, transport, industrial materials or imported fuels.

8. Individual heat, district heat, CHP and thermal storage

Boilers and heat pumps convert their input to useful heat, but their coefficients have different physical interpretations. A COP above one represents ambient/source heat transferred into useful heat.

$$Q^{boiler}_{kty} = \eta_k U_{kty}, \qquad Q^{HP}_{kty} = COP_{kt} P^{HP}_{kty}$$

E17 · scripts/Build_model_dual_industry_categories_named_debug.py:1601 · structural boiler/heat-pump conversions, with fuel and COP mappings suppressed.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `heat_pump_ind`; Representation: structural boiler/heat-pump conversions, with fuel and COP mappings suppressed.

Individual and district balances combine enabled boilers, HPs, CHP, solar and thermal storage. District losses and the source's delivered-heat boundary must be retained.

$$\sum_k Q_{kzty} + Q^{CHP}_{zty} + D^{heat}_{zty} = H_{zty} + C^{heat}_{zty} + Loss^{heat}_{zty}$$

E18 · scripts/Build_model_dual_industry_categories_named_debug.py:62 · structural heat-balance family; separate individual/DH and rolling-shortage branches are grouped.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `heat_balance`; Representation: structural heat-balance family; separate individual/DH and rolling-shortage branches are grouped.

The shared CHP helper attributes condensing-equivalent output to both fuel cost and combustion emissions:

$$P^{eq}_{gty} = P_{gty} + \beta_g Q_{gty}, \qquad U^{CHP}_{gty} = P^{eq}_{gty} / \eta^{el}_g$$

E19 · scripts/energy_accounting.py:28 · reindexed shared CHP fuel accounting.

Source: `scripts/energy_accounting.py`; anchor: `def power_fuel_input`; Representation: reindexed shared CHP fuel accounting.

CHP operating regions additionally couple electrical and thermal output. Supplying heat reduces the admissible electricity output in the extraction branch, while a separate power-to-heat relationship restricts feasible operation. The compact envelope below states that mechanism; exact coefficients and extraction/backpressure branches remain input-specific.

$$P_{gty} + \beta_g Q_{gty} \leq K_{gy}$$

E20 · scripts/Build_model_dual_industry_categories_named_debug.py:8531 · structural extraction-CHP upper envelope, not its complete feasible region.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `gen_heat_extraction_max`; Representation: structural extraction-CHP upper envelope, not its complete feasible region.

Thermal stores shift heat between periods using charging/discharging efficiencies, losses, energy/power capacity and initialization/closure rules. Building HP storage, district storage and category industry storage are distinct asset families; enabling one does not activate the others.

---

9. Heating transition policies

Annual useful heat is obtained by aggregating the appropriate hourly service. A relative nonincrease or change limit applies to this scoped annual quantity, not necessarily to each hour.

$$H_{ky}^{annual} = \sum_{t \in T_y} w_{ty} Q_{kty}, \qquad H_y^{fossil} \le H_y^{fossil,-} + \varepsilon$$

E21 · scripts/fossil_heating_nonincrease.py:168 · structural annual heating/history accounting and nonincrease rule.

Source: `scripts/fossil_heating_nonincrease.py`; anchor: `def add_fossil_heating_nonincrease`; Representation: structural annual heating/history accounting and nonincrease rule.

Fuel-specific boiler phaseout makes applicable boiler operation unavailable by its deadline. The source excludes CHP from this boiler-specific rule; other carbon and CHP heat restrictions may still bind. Electric/gas annual changes, CHP calibration and biomass nonincrease have their own scopes and tolerances. Turning one off does not turn the others off.

---

10. Legacy industry, categories and the pooled remainder

Useful industrial demand is allocated across eligible equipment, while final carrier input follows efficiency/COP. The category version adds the service/temperature category index to the same physical idea.

$$\sum_k Q_{kity} = H_{ity}, \qquad U_{kity} = Q_{kity}/\eta_{ki}$$

E22 · scripts/Build_model_dual_industry_categories_named_debug.py:2374 · structural industrial service/efficiency family; share variables and unit-specific carrier recipes are suppressed.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `industries_share_fuel`; Representation: structural industrial service/efficiency family; share variables and unit-specific carrier recipes are suppressed.

Detailed rows excluded from explicit modeling must be accounted for in the legacy pooled `simple_industry` worksheet. Excluding a row changes representation; it must not delete its demand. A valid mixed case can model selected services explicitly and pool the rest.

$$H^{pool}_{zty} = \sum_{i \in I^{excluded}} H_{izty}, \qquad H^{total} = H^{detailed} + H^{pool}$$

E23 · scripts/read_process_data.py:2440 · structural accounting invariant for detailed-versus-pooled services; not a claim that every workbook formula is automatically recalculated.

Source: `scripts/read_process_data.py`; anchor: `simple_industry`; Representation: structural accounting invariant for detailed-versus-pooled services; not a claim that every workbook formula is automatically recalculated.

An all-excluded case pools the applicable legacy industry. A fully detailed case can have an empty remainder. This pooled worksheet is unrelated to the separate product workbook `data/simple_industry.xlsx`.

Applicability ceilings cap a technology or fuel family's useful-service share. Shared substitution budgets prevent several alternatives from independently consuming the same replaceable service pool.

$$Q_{kity} \le \bar{s}_{ki} H_{ity}, \qquad \sum_{k \in A_i} Q_{kity} \le \bar{H}^{replaceable}_{ity}$$

E24 · scripts/Build_model_dual_industry_categories_named_debug.py:2474 · structural individual/shared service substitution limits.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py`; anchor: `industry_low_carbon_fuel_substitution_cap`; Representation: structural individual/shared service substitution limits.

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11. Process-based complex industry

Only selected complex branches replace their overlapping legacy demand. Product activity and component intensities determine useful-service requirements and independent process emissions.

$$H_{ity} = q_{b(i),ty} \alpha_i e_i, \qquad E^{process}_{bty} = \sum_{i \in b} q_{bty} \alpha_i \gamma_i$$

E25 · scripts/complex_industry_support.py:144 · structural product/component service and process-carbon relationships; native activity may be tonnes or a declared index.

Source: `scripts/complex_industry_support.py`; anchor: `activity`; Representation: structural product/component service and process-carbon relationships; native activity may be tonnes or a declared index.

Non-energy feedstocks, combustion and process emissions must not be counted twice. Product demand, component coverage, compatible carriers and process capture are separate input dimensions. The author reports testing the complex formulation for non-metallic minerals; availability of additional process rows does not establish a validated transition in every sector.

12. Annual industry transition, HP applicability and heat storage

The annual direct-electric plus HP share is weighted by delivered industrial service. It is not the electrical input share because HP COPs differ from one and from direct-electric equipment.

$$s^{el}_{iy} = \frac{\sum_{t \in T_y} w_{ty} \sum_{k \in EL \cup HP} Q_{kity}}{\sum_{t \in T_y} w_{ty} H_{ity}}$$

E26 · scripts/industry_annual_s_curve.py:92 · structural demand-weighted annual electric-service share.

Source: `scripts/industry_annual_s_curve.py`; anchor: `def add_annual_industry_electricity_limits`; Representation: structural demand-weighted annual electric-service share.

The finite completion progress is a cubic smoothstep between the original base year and configured deadline. This is distinct from a generic logistic curve. Additional seed, growth, direction and reference-interval rules constrain the endogenous adoption envelope; the smoothstep below is not the complete S-curve implementation.

$$u_y = \min\left(1, \max\left(0, \frac{y - y_0}{y_c - y_0}\right)\right), \qquad p_y = u_y^2(3 - 2u_y)$$

E27 · scripts/industry_annual_s_curve.py:6 · reindexed finite completion-progress function.

Source: `scripts/industry_annual_s_curve.py`; anchor: `def completion_progress`; Representation: reindexed finite completion-progress function.

A relative annual electric-share decrease allowance acts on the previous share. For a one-year step with a 10% reduction allowance, 40% becomes a minimum of 36%, not 30%. Multi-year history/reference-interval formulas must be taken from the helper.

$$s^{el}_{i,y+1} \geq (1 - d) s^{el}_{iy} - \varepsilon$$

E28 · scripts/industry_annual_s_curve.py:107 · structural one-year interpretation of the combined electric-service decrease bound.

Source: `scripts/industry_annual_s_curve.py`; anchor: `industry_electric_share_max_annual_decrease`; Representation: structural one-year interpretation of the combined electric-service decrease bound.

Industrial HPs also share cumulative temperature and source-heat budgets. A 60°C potential and an 80°C cumulative potential are not independent quantities to add. Category thermal stores have separate energy inventory, charging/discharging and investment bounds and require category mode.

13. Transport service, stock and fuel compatibility

Fleet evolution combines surviving existing vehicles, surviving new purchases, retirement and eligible paid conversions. It is continuous vehicle-equivalent stock in this LP, not an integer fleet purchase schedule.

$$N_{vy} = N_{vy}^{old,\,surv} + \sum_{a} \sigma_{vay} X_{va} + R_{vy}^{in} - R_{vy}^{out}$$

E29 · scripts/Build_model_dual_industry_categories_named_debug.py:9132 · structural fleet/history/conversion accounting.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py` ; anchor: `name="veh_stock_balance"` ; Representation: structural fleet/history/conversion accounting.

Service demand and final energy are separate. Stock, driving/activity profiles and primary/secondary fuel efficiency define the admissible hourly service and energy requirements. The notation below groups those technology-specific details in the coefficient `kappa` .

$$\sum_{v} S_{vity} = D_{ity}^{transport}, \qquad U_{fvity} = \kappa_{fvi} S_{vity}$$

E30 · scripts/Build_model_dual_industry_categories_named_debug.py:9448 · structural transport service/energy coupling; coefficients and native passenger-km/tonne-km conversions remain explicit in the reader.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py` ; anchor: `name="transport_energy_profile"` ; Representation: structural transport service/energy coupling; coefficients and native passenger-km/tonne-km conversions remain explicit in the reader.

Blend compatibility can be volume-based while the energy balance is energy-based. The model's conversion and paid-upgrade rules must be retained; a 10% ethanol volume limit must not simply be inserted as a 10% MWh share. Marine pilot fuel, upgrade costs and remaining life create additional requirements.

Annual service ceilings, stock electrification floors, turnover, investment limits and ICE fuel-composition ramps are distinct policies. V2G can discharge only within fleet battery/plug-in/driving constraints; the battery cannot both satisfy arbitrary mobility and arbitrary grid deliveries without its stock balance.

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14. Hydrogen and e-fuel conversion

Hydrogen output is limited by installed conversion capacity. The appropriate input efficiency links electrolysis to electricity consumption and reforming to fuel/carbon inputs.

$$0 \le H_{kty}^{prod} \le K_{ky}^{H2}, \qquad P_{kty}^{electrolysis} = H_{kty}^{prod} / \eta_{k}^{electrolysis}$$

E31 · scripts/Build_model_dual_industry_categories_named_debug.py:1839 · structural electrolyser capacity/input relationship.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py` ; anchor: `name="h2_electrolysis"` ; Representation: structural electrolyser capacity/input relationship.

E-fuel equipment has an output capacity bound. For each enabled conversion recipe, feedstock slots are proportional to output; coefficients have the slot's declared units, which can be MWh carrier, tCO2 or another physical feedstock unit per output unit.

$$0 \le V_{kzty} \le K^{fuel}_{kzy}, \qquad U^{(j)}_{kf,zty} = a^{(j)}_{kf} V_{kzty}$$

E32 · scripts/Build_model_dual_industry_categories_named_debug.py:10470 · reindexed feedstock-slot proportionality with the separately implemented output-capacity bound grouped.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py` ; anchor: `efuel_inputs1 =` ; Representation: reindexed feedstock-slot proportionality with the separately implemented output-capacity bound grouped.

Each carrier must balance production, imports, storage and downstream uses. Hydrogen used to make another fuel is not also available to meet final H2 demand. Heat/electric auxiliaries and refinery process energy are additional coupled loads.

$$V^{prod}_{fzty} + I_{fzty} + D^{f}_{zty} = U^{final}_{fzty} + U^{feedstock}_{fzty} + C^{f}_{zty} + E_{fzty}$$

E33 · scripts/Build_model_dual_industry_categories_named_debug.py:10470 · structural carrier-node balance; transport, industry, refinery, terminal and transfer terms depend on the active carrier.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py` ; anchor: `efuelinput1` ; Representation: structural carrier-node balance; transport, industry, refinery, terminal and transfer terms depend on the active carrier.

Carrier stores have asset-specific losses, efficiency, charging/discharging and annual boundaries. The generic expression indicates those roles rather than assert a universal efficiency placement:

$$S^{f}_{t} = (1 - \ell_{f}) S^{f}_{t-} + \eta^{in}_{f} C^{f}_{t} - D^{f}_{t} / \eta^{out}_{f}$$

E34 · scripts/Build_model_dual_industry_categories_named_debug.py:965 · structural carrier-stock notation; inspect actual family coefficients, initial stock and chronological carryover rules.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py` ; anchor: `fuel_storage_capacity` ; Representation: structural carrier-stock notation; inspect actual family coefficients, initial stock and chronological carryover rules.

The author's supplied integrated e-fuel setup enables both e-fuel modeling and explicit Oil/Gas product supply. Alternative supply exclusions need complete compatible data. Turning e-fuels on makes conversion available; demand, prices, resource limits and carbon constraints determine whether it is built.

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15. Pipeline and terminal capacities

Without e-fuels, the standalone H2 pipeline and aggregate H2 terminal representations can be used where their dependency checks pass. With e-fuels, product transmission already includes the H2 pathway and the standalone H2-transmission flag is normalized off. Terminals can be used with or without e-fuels, but their input schemas and physical detail differ.

The standalone terminal throughput capacity is in TWh/year. Its total allowed loading/unloading throughput must be converted to MWh, while zonal H2 stores provide buffering. Shipping routes impose their own allowed directions, fleet, travel/delay and delivery assumptions.

$$\sum_{t \in T_y} w_{ty} \left(F_{ty}^{load} + F_{ty}^{unload} \right) \leq 10^6 K_y^{terminal}$$

E35 · scripts/h2_terminals.py:3 · structural aggregate shared-throughput limit; include the source's precise flow collection and scope.

Source: `scripts/h2_terminals.py`; anchor: `Terminal capacity is annual TWh`; Representation: structural aggregate shared-throughput limit; include the source's precise flow collection and scope.

Terminal activation and allowed directions are table-level choices in addition to `model_terminals=True`. An enabled terminal with no allowed/eligible route is not an unlimited import source. Aggregate delivered H2 terminals do not represent a detailed vessel/liquefaction/cracking chain unless that chain is supplied by the applicable product formulation.

---

16. CO2 feedstock, capture, amines and geological provenance

Industrial combustion factors used in the emissions budget and physically recoverable carbon used for capture/feedstock can differ for biogenic sources. For a combustion input, the compact relationship is:

$$E_{kty}^{comb} = EF_k^{budget} U_{kty}, \qquad CO2_{kty}^{available} = EF_k^{physical} U_{kty} + E_{kty}^{process}$$

E36 · scripts/complex_industry_accounting.py:73 · structural distinction between budget and physical capture/feedstock carbon.

Source: `scripts/complex_industry_accounting.py`; anchor: `def combustion_factor`; Representation: structural distinction between budget and physical capture/feedstock carbon.

Under the supplied convention, qualifying industrial biomass CO2 can supply feedstock while its biogenic combustion factor does not add fossil emissions to the aggregate budget. This is a bookkeeping convention, not proof of sustainable biomass or full lifecycle neutrality. Process emissions, captured process carbon, DAC and eligible biomass carbon can provide e-fuel CO2 under their actual mappings and capacities.

$$CO2^{process,\,use} + CO2^{DAC} + CO2^{bio,\,use} + CO2^{other,\,eligible} = CO2^{fuel,\,demand} + CO2^{buffer,\,net} + CO2^{eligible,\,geo}$$

E37 · scripts/industrial_carbon_balances.py:52 · structural grouped CO2-bus accounting; source restrictions determine which terms may serve each destination.

Source: `scripts/industrial_carbon_balances.py`; anchor: `def add_industrial_carbon_balances`; Representation: structural grouped CO2-bus accounting; source restrictions determine which terms may serve each destination.

Capture is bounded by both equipment and available flue gas. The separated process component is bounded by process emissions and the total capture; several capture technologies cannot each claim another copy of one branch's process stream.

$$0 \leq C_{kty}^{process} \leq r_k E_{kty}^{process}, \qquad C_{kty}^{process} \leq C_{kty}^{total}$$

E38 · scripts/industrial_carbon_balances.py:106 · reindexed process-capture upper bounds; shared-source bounds are additional.

Source: `scripts/industrial_carbon_balances.py` ; anchor: `captured[key] <= rate * process` ; Representation: reindexed process-capture upper bounds; shared-source bounds are additional.

An amine inventory separates absorber capture timing from stripper timing. The process-tagged inventory must remain a subset of the total physical amine inventory.

$$S_t^{amine,process} = S_{t^-}^{amine,process} + C_t^{process} - R_t^{strip,process}, \qquad S_t^{amine,process} \leq S_t^{amine,total}$$

E39 · scripts/industrial_carbon_balances.py:118 · reindexed process-tagged amine stock relation.

Source: `scripts/industrial_carbon_balances.py` ; anchor: `inventory[key] == previous` ; Representation: reindexed process-tagged amine stock relation.

With separated carbon balances, only eligible locally stripped process CO2 can supply geological injection. Untagged imports, reusable buffer withdrawals, DAC or combustion carbon do not gain process provenance by sharing a bus.

$$G_{zty}^{geo} \leq R_{zty}^{strip,process}$$

E40 · scripts/industrial_carbon_balances.py:57 · structural provenance upper bound; the exact scoped process balance and competing uses remain in the source.

Source: `scripts/industrial_carbon_balances.py` ; anchor: `Only locally` ; Representation: structural provenance upper bound; the exact scoped process balance and competing uses remain in the source.

Geological inventory counts stored tonnes, while injection capacity counts tonnes/hour. Initial stock, total storage capacity and annual build limits are additional. Reusing CO2 in a fuel is not permanent geological sequestration.

---

17. Carbon limit, LULUCF and infeasibility

For the separated-carbon planning branch, non-process emissions plus net industrial process emissions must respect the policy allowance plus permitted removals. The source scales this row by 1000 numerically; the physical relation remains tonnes.

$$E_{sy}^{nonprocess} + E_{sy}^{process,net} \leq B_{sy}^{CO2} + R_{sy}^{LULUCF}$$

E41 · scripts/Build_model_dual_industry_categories_named_debug.py:12915 · reindexed separated planning carbon cap before numerical row scaling.

Source: `scripts/Build_model_dual_industry_categories_named_debug.py` ; anchor: `CO2_total_non_process[year, System] + CO2_net_industry_process` ; Representation: reindexed separated planning carbon cap before numerical row scaling.

Process emissions can make a stringent carbon cap infeasible even after fossil combustion is removed. Capture plus active geological stock/injection capacity can reduce eligible net process emissions, while a defensible LULUCF allowance can offset the budget. These are different mechanisms. LULUCF does not physically receive process CO2 and does not supply e-fuel feedstock.

Both the main storage/capture switches and matching active technology/capture/storage rows must permit geological handling. A True flag cannot overcome zero capacity/build limits, no capturable stream, missing reboiler energy or a forbidden source. The required-capture switch only concerns eligible capturable process carbon; uncapturable residuals can still violate a zero budget.

In rolling dispatch, a cumulative/weighted carbon allocation and optional penalized slack replace the ordinary planning cap treatment. A reported rolling solution with positive carbon slack has relaxed the strict allocation and must be disclosed as such.

---

18. Electricity demand-side management

Let `b` be basic electricity after its applicable decomposition and baseline removals. Signed DSM adjustments change timing but do not create generation. Daily and weekly classes share hourly power, including absolute activity, so opposite actions cannot hide capacity use.

$$b^{adj}_{zty} = b_{zty} + \sum_{c} x_{czty}, \qquad a_{czty} \geq x_{czty}, \quad a_{czty} \geq -x_{czty}$$

E42 · scripts/electricity_dsm.py:295 · structural aggregation and absolute-adjustment linearization.

Source: `scripts/electricity_dsm.py`; anchor: `dsm_total_absolute`; Representation: structural aggregation and absolute-adjustment linearization.

$$\sum_{c} a_{czty} \leq K^{DSM}_{zy}, \qquad \sum_{c} a_{czty} \leq \phi_{zy} b_{zty}$$

E43 · scripts/electricity_dsm.py:298 · structural shared available-power and hourly baseline-fraction limits.

Source: `scripts/electricity_dsm.py`; anchor: `dsm_shared_power_capacity`; Representation: structural shared available-power and hourly baseline-fraction limits.

Each class balances energy within its block. The half-absolute sum counts moved MWh once rather than charge the upward and downward leg as two independent shifted quantities.

$$\sum_{t \in B} x_{ct} = 0, \qquad \frac{1}{2} \sum_{t \in B} a_{ct} \leq f_{cy} \sum_{t \in B} b_t$$

E44 · scripts/electricity_dsm.py:17 · structural daily/weekly block closure and moved-energy allowance.

Source: `scripts/electricity_dsm.py`; anchor: `budget`; Representation: structural daily/weekly block closure and moved-energy allowance.

With both classes and energy budgets enabled, the weekly pool is a subset of daily-capable flexibility. At full rollout the supplied values give 5.5% daily-balanced plus 1.5% weekly-balanced, totalling 7%.

$$f_y^{daily} = r_y(f_D - f_W), \qquad f_y^{weekly} = r_y f_W$$

E45 · scripts/electricity_dsm.py:21 · reindexed nested class-potential interpretation when both budgeted classes are active.

Source: `scripts/electricity_dsm.py`; anchor: `dsm_weekly_energy_fraction`; Representation: reindexed nested class-potential interpretation when both budgeted classes are active.

$$r_y = \min\left(1, \max\left(0, \frac{y - y_0}{y_{full} - y_0}\right)\right)$$

E46 · scripts/electricity_dsm.py:40 · structural linear rollout for a full-availability year after the global base year; boundary validation remains in the helper.

Source: `scripts/electricity_dsm.py`; anchor: `dsm_availability_full_year`; Representation: structural linear rollout for a full-availability year after the global base year; boundary validation remains in the helper.

$$C_y^{DSM} = c_K K_y^{DSM} + c_E \frac{1}{2} \sum_{t,\,c} w_{ty} a_{cty}$$

E47 · scripts/electricity_dsm.py:313 · structural DSM enabling/activation resource cost in EUR before the model's cost conversions.

Source: `scripts/electricity_dsm.py`; anchor: `def operating_cost`; Representation: structural DSM enabling/activation resource cost in EUR before the model's cost conversions.

The hourly 10% symmetric cap can limit moved energy to 5% of a block's load even if the energy allowance is 7%. An unexhausted energy allowance is therefore not by itself evidence that the modeled flexibility is uneconomic. Block balancing is not an appliance-specific delay/deadline model.

---

19. Product industry: recipes, stocks and mandatory demand

The separate product workbook supplies fixed plants/recipes and markets. Physical capacity is a production ceiling, not a requirement to operate at nameplate output. Electricity follows tonnes produced and the plant-specific MWh/t coefficient; non-electric inputs follow GJ/t recipes.

$$0 \le q_{pty} \le \overline{q}_{py} a_{pt}, \qquad P_{pty}^{product} = e_p q_{pty}$$

E48 · scripts/simple_industry_model.py:114 · structural production-capacity/availability bound and reindexed electricity conversion.

Source: `scripts/simple_industry_model.py`; anchor: `simple_industry_electricity_balance`; Representation: structural production-capacity/availability bound and reindexed electricity conversion.

$$U^{GJ}_{fpty} = u_{fp} q_{pty}, \qquad E^{direct}_{pty} = q_{pty}\left(\gamma^{process}_{p} + \sum_{f} u_{fp} EF_{f}\right)$$

E49 · scripts/simple_industry_inputs.py:35 · reindexed fixed-recipe direct-carbon factor and structural fuel use.

Source: `scripts/simple_industry_inputs.py`; anchor: `def direct_factor`; Representation: reindexed fixed-recipe direct-carbon factor and structural fuel use.

Inventory preserves origin. A plant's output can be stored and then delivered to compatible destinations; delivery uses up inventory, and bounded initial/final stock rules prevent free product.

$$S_{pty} = S_{p,t^-,y} + q_{pty} - \sum_{m} v_{pmty}$$

E50 · scripts/simple_industry_model.py:173 · reindexed plant material balance.

Source: `scripts/simple_industry_model.py`; anchor: `stock[key] == previous + output[key]`; Representation: reindexed plant material balance.

Mandatory markets must receive their prescribed annual tonnes under the applicable sample normalization and hourly demand/delivery limits. Optional markets impose sales ceilings rather than a compulsory sale.

$$\sum_{p,t} w_{ty} v_{pmty} = M_{my} \quad (m \in Mandatory), \qquad \sum_{p,t} w_{ty} v_{pmty} \le M_{my} \quad (m \in Optional)$$

E51 · scripts/simple_industry_model.py:124 · structural annual mandatory/optional market interpretation; hourly profile constraints remain additional.

Source: `scripts/simple_industry_model.py`; anchor: `mandatory`; Representation: structural annual mandatory/optional market interpretation; hourly profile constraints remain additional.

Outside suppliers share one annual budget across all destinations and have hourly caps. Optional route rows can restrict eligibility and route throughput/cost. There is no unlimited emergency product supply.

$$\sum_{m,t} w_{ty} v_{pmty} \le \overline{M}^{supplier}_{py}, \qquad \sum_{m} v_{pmty} \le \overline{q}^{supplier}_{py} a_{pt}$$

E52 · scripts/simple_industry_model.py:121 · structural outside-supplier annual/hourly shared limits.

Source: `scripts/simple_industry_model.py`; anchor: `import_suppliers`; Representation: structural outside-supplier annual/hourly shared limits.

Calibrated baseline electricity removal avoids counting a real plant twice. Removals are year-specific MWh distributed by their profile; the example uses zero removals, so product plants add load.

$$b_{zty}^{residual} = b_{zty}^{original} - b_{zty}^{removed}, \qquad L_{zty}^{total} = b_{zty}^{residual} + \sum_{p \in z} e_p q_{pty}$$

E53 · scripts/simple_industry_model.py:29 · structural baseline-removal and added product-electricity accounting; excessive removals are rejected by the input logic.

Source: `scripts/simple_industry_model.py`; anchor: `def baseline_removed`; Representation: structural baseline-removal and added product-electricity accounting; excessive removals are rejected by the input logic.

Flexible eligible plants can vary hourly. With flexibility disabled, the corresponding operation is constant within each calendar day after availability adjustment; it can change between days. It is not equivalent to imposing a full-year flat production level or requiring local plants to satisfy all demand.

---

20. Industrial CBAM and domestic carbon pricing

For a modeled plant, direct emissions and entered precursor carbon determine the embedded footprint. Imported offers supply their own embedded factor. The domestic-price payment concerns direct emissions after its configured allowance; it is independent of the product border charge.

$$e_p^{embedded} = e_p^{direct} + e_p^{precursor}, \qquad c_{py}^{domestic} = \pi_{py}^{domestic} \max(0, e_p^{direct} - a_{py}^{domestic})$$

E54 · scripts/simple_industry_policy.py:5 · reindexed modeled-plant embedded factor and domestic per-tonne payment; imported suppliers use their entered footprint/cost treatment.

Source: `scripts/simple_industry_policy.py`; anchor: `def coefficients`; Representation: reindexed modeled-plant embedded factor and domestic per-tonne payment; imported suppliers use their entered footprint/cost treatment.

CBAM liability uses the supplied footprint, benchmark/free-allocation factor, EU price and eligible foreign credit. These max operations are performed on exogenous input coefficients before building the LP; they do not create a nonlinear variable-times-variable carbon-price calculation.

$$e_{py}^{liable} = \max(0, e_p^{embedded} - f_{py}^{free} b_{py}^{benchmark})$$

E55 · scripts/simple_industry_policy.py:16 · reindexed exogenous liable-footprint coefficient.

Source: `scripts/simple_industry_policy.py`; anchor: `liable = max`; Representation: reindexed exogenous liable-footprint coefficient.

$$c_{py}^{CBAM} = \max(0, \pi_{py}^{EU} e_{py}^{liable} - credit_{py})$$

E56 · scripts/simple_industry_policy.py:20 · reindexed exogenous CBAM coefficient conditional on origin/start-year/exemption and destination eligibility.

Source: `scripts/simple_industry_policy.py`; anchor: `cbam = max`; Representation: reindexed exogenous CBAM coefficient conditional on origin/start-year/exemption and destination eligibility.

The charge applies once to eligible delivery to an EU destination using original supply provenance. Fiscal/exporter/importer allocation can differ from territorial emissions. A cleaner modeled territory can simply be importing more embodied carbon; report both accounts. These equations describe the implemented scenario representation, not a legal compliance calculator or endogenous global steel-price equilibrium.

---

21. Electricity CBAM intensity and historical signals

Electricity Options 1–4 use their own membership, border and national-payment rows. At most one is permitted, and the inspected builder requires `Myopic=True`. They are independent of product CBAM and a quantity carbon cap.

An import-charge term is structurally transferred electricity times an exogenous source-zone intensity and applicable carbon price/credit. The precise border/national split depends on the selected option.

$$C_y^{border} = \sum_{l,t} w_{ty} F_{lty}^{eligible} e_{origin(l),y}^{signal} \pi_{ly}^{applicable}$$

E57 · scripts/electricity_cbam.py:42 · structural cost-coefficient interpretation; option-specific price and membership rows remain in the builder/cost script.

Source: `scripts/electricity_cbam.py`; anchor: `def intensity`; Representation: structural cost-coefficient interpretation; option-specific price and membership rows remain in the builder/cost script.

The intensity helper uses historical/base-year signals for the base year, realized predecessor signals for subsequent decision years, and permitted exogenous overrides/held signals in lookahead years. A valid zero intensity is retained. Missing or duplicate required realized history is an error, not a reason to assume zero-carbon imports.

---

22. Planning windows, representative accounting and rolling chronology

Ordinary myopic chunks are slices of the decision-year list. `chunk_size` and `step_size` count entries, whereas the partial-foresight lookahead is measured in calendar years.

$$Y_j = Y[j : j + c], \qquad j = 0, h, 2h, \ldots$$

E58 · main.py:663 · reindexed ordinary chunk construction with c=chunk_size and h=step_size.

Source: `main.py`; anchor: `def _build_myopic_chunks`; Representation: reindexed ordinary chunk construction with c=chunk_size and h=step_size.

Ordinary partial foresight instead selects future decision points up to the calendar lookahead. The decision year retains full hours; later years use their sampled blocks.

$$Y_j^{partial} = \{y \in Y : y_j \le y \le y_j + L\}$$

E59 · main.py:670 · reindexed finite-lookahead year selection; None uses the remaining tail.

Source: `main.py`; anchor: `partial_foresight_lookahead_years`; Representation: reindexed finite-lookahead year selection; None uses the remaining tail.

The delivered representative capacity settings are four blocks of 24 hours, even though their identifiers contain the word week. A uniform 8760/96 factor is an accounting illustration for the supplied equal-weight sample; actual `_period_weight` and helper annualization must be followed in every family.

$$\widehat{A}_y = \frac{8760}{96} \sum_{t \in T_y^{sample}} A_{ty} = 91.25 \sum_{t \in T_y^{sample}} A_{ty}$$

E60 · policy/runtime_policy.py:25 · derived equal-weight annualization example for four 24-hour blocks; not a universal storage transition multiplier.

Source: `policy/runtime_policy.py`; anchor: `representative_week_hours`; Representation: derived equal-weight annualization example for four 24-hour blocks; not a universal storage transition multiplier.

In daily rolling mode, optimize 48 hours, commit 24, then advance by 24. In weekly rolling mode the corresponding values are 336, 168 and 168. The next solve receives the last committed state rather than the uncommitted end of the forecast window.

$$S_{j+1}^{initial} = S_j^{committed,\,end}, \qquad K_y^{dispatch} = K_y^{accepted,\,stage1}$$

E61 · scripts/run_stage2_rolling_dispatch.py:22 · structural rolling carryover and fixed-capacity principle; stock, route-delay, carbon and checkpoint metadata remain family-specific.

Source: `scripts/run_stage2_rolling_dispatch.py`; anchor: `rolling_realized_horizon`; Representation: structural rolling carryover and fixed-capacity principle; stock, route-delay, carbon and checkpoint metadata remain family-specific.

Guidance, terminal targets, reserve/shortage relaxations and forecast perturbations affect the rolling result. Full-year chronology with fixed capacity is different from reoptimizing capacity using full-year foresight. The author's report of improved no-e-fuel intra-annual stability is not evidence that every e-fuel/terminal/process combination has passed the same test.

---

23. Numerical scale, residuals and interpretation

Explicit row and column scales aim to improve conditioning without changing physical units. An equation divided by a large row scale can have a small solver residual while its restored physical residual is material.

$$\widetilde{x} = x/s_x, \qquad \widetilde{r} = r/s_r, \qquad r = s_r\widetilde{r}$$

E62 · scripts/numeric_scaling.py:12 · structural scaling identity; actual divisors vary by family.

Source: `scripts/numeric_scaling.py`; anchor: `def balanced_row_divisor`; Representation: structural scaling identity; actual divisors vary by family.

Output completion is not a strict solution certificate when provisional acceptance, warning-only postoptimal checks or explicit slack are allowed. Report solver status, objective/gap where meaningful, physical MW/MWh/tonne residuals, nonzero shortages/carbon slack and which years were committed. The mathematical formulation, inherited assumptions and local solver outcome are distinct pieces of evidence.

---

24. How to use these equations in a study

Choose the physical representation first, then the time/information assumption, then the policy experiment. Record every departure from the delivered configuration. Associate each reported result with the relevant equation family, active input rows and physical unit. For example, new e-methanol output needs conversion capacity, H2, CO2, electricity/heat auxiliaries and compatible demand; a renewable-electricity share alone does not guarantee it appears.

Use R02 for the published H2RES optimization background and the reference register for data, technology and DSM sources. Cite the local version/hash and these later extensions separately. The source map is a reproducible documentation aid, not proof that all configurations or a complete national transition are feasible.

H2RES model manual

Documented model: `D:\POSAO\H2RES_12.11.2026`. **Documentation edition:** 12 November 2026. The source/version inventory identifies the exact files reviewed. The model is separate from the data-preparation interface.

The expanded mathematical formulation supplies 62 numbered equations with source anchors and physical interpretation. Use the main configuration guide for detailed workflow/combination instructions and the complete option reference for every recorded main/policy setting. The equations below retain the original narrative overview and complement the more detailed formulation chapter.

---

1. Model purpose, lineage and scientific use

H2RES is used to study integrated energy systems and transition pathways across electricity, heat, transport, hydrogen and other energy carriers. Its historical island-energy modelling lineage and the later optimization formulation are distinct developments. The 2004 island study and the 2022 planning formulation are appropriate background references; the newer local code contains additional processes and workflows that those articles do not document R01, R02.

The formulation examined here is a linear investment-and-operation model implemented in Python with Gurobi. The static inventory of active runtime source files found continuous variable declarations, including the `_add_vars` wrapper. Vehicle numbers may therefore be fractional optimization quantities; they are not automatically integer purchase decisions. A fuel called a binary choice in ordinary language does not imply a binary solver variable. Confirm the constructed model if additional user modules are added.

The model minimizes a configured economic objective subject to physical, stock, investment, service, network, resource and policy constraints. It can help compare pathways under alternative assumptions. It does not discover future policy targets, determine a unique forecast, prove causation in a real market, or by itself establish full lifecycle sustainability. Published H2RES applications are examples of model use, not a validation certificate for this source snapshot R03, R04.

---

2. Architecture and execution flow

2.1 Main runtime files

File	Responsibility
`main.py`	Working directory, scenario name, feature flags, file selection, caching, planning/dispatch execution and exports
`policy/model_policy.py`	Entered economic, transition, resource and policy assumptions
`policy/runtime_policy.py`	Numerical tolerances, sampling, solver and recovery settings
`policy/transport_pathways.py`	Transport conversion/compatibility assumptions and default migration values
`scripts/read_process_data.py`	Reads/scopes input tables, constructs sets/dictionaries, annual limits and processed demand

File	Responsibility
`scripts/reading_prev_and_vars_def_dual_industry_categories1.py`	Loads historical decisions/stocks and declares optimization variables
`scripts/Build_model_dual_industry_categories_named_debug.py`	Core physical/policy constraint construction
`scripts/CAPEX_OPEX_Gurobi_settings_dual_industry_categories1.py`	Economic expressions, objective, solver configuration, optimization and recovery
`scripts/complex_industry_support.py`	Validates/adapts process activity and replaces selected ordinary industry services
`scripts/industry_annual_s_curve.py`	Demand-weighted adoption envelopes and shared electric-service transition policies
`scripts/transport_pathways.py`	Fuel compatibility, paid conversion pathways and stock/energy links
`scripts/energy_accounting.py`	Shared CHP fuel-equivalent and refinery physical accounting
`scripts/physical_solution_audit.py`	Physical-unit checks for solutions and history
`scripts/run_stage2_rolling_dispatch.py`	Rolling operational checks with fixed stage-1 capacities
`scripts/export_plot*.py` and `combine_plot.py`	Investment/history exports, balances, summaries and visualizations

Scripts are executed in a shared namespace through `exec_script`. A variable can therefore be created by a reader/helper and consumed later without a conventional function argument. Configuration provenance must include the imported policy modules as well as `main.py`. Static code review must account for this execution pattern.

2.2 Runtime sequence

```
main + three policy modules
    -> select technology/policy/profile files
    -> read and normalize inputs
    -> choose planning/dispatch mode and time grid
    -> load previous accepted investment/stock history
    -> declare variables
    -> build balances and constraints
    -> define costs/objective and solve
    -> check solution/status/physical quality
    -> export accepted results and historical handoffs
```

A separate data-preparation run stops before the model sequence above. Its optional bundle launcher configures the separate model in memory, redirects data paths and inserts the whole-energy renewable constraints before solving. The original model's `main.py` does not automatically read the preparation GUI settings.

---

3. Exact configuration and precedence

`main.py` changes its working directory to its own model directory, then executes the three policy modules in its namespace. It subsequently sets feature flags and derives file names. Reader/workbook settings supply additional numerical inputs. A preparation-bundle adapter can override selected flags and data paths at its supported execution boundary. The resulting precedence is contextual rather than a single universal JSON configuration layer.

The configuration reference lists source assignments and line positions. Values described below are the examined local snapshot, not universal H2RES defaults. Historical README files include other scenario names and mode selections.

3.1 Observed local selection

The reviewed `main.py` uses scenario name `industry_CC_5y`, `Myopic=True`, `partial_perfect_foresight=False`, heating and transport/industry enabled, e-fuels enabled, category industry enabled and complex industry disabled. Category electricity is `conservative`, industrial category storage is enabled, and industry fuel-share/S-curve controls are enabled. Solar individual/DH heating and cooling switches are disabled in this snapshot; model availability and a currently enabled option are different matters.

The snapshot also has transport pathway corrections/electrification/share ceilings enabled, standalone H2 transmission and standalone terminals disabled, and `disable_transport_infrastructure=False`. Transport infrastructure is therefore not suppressed by that debug switch. Product based simple industry and both daily/weekly electricity DSM are disabled in the delivered selection; they are documented in sections 19 and 20.

3.2 Mutually exclusive or dependent modes

Category and complex industry are alternatives. Complex mode requires transport/industry and cannot use the legacy single-sector industry option. Category industrial storage requires category mode. Standalone hydrogen pipelines are disabled when e-fuels handle carrier transmission. Standalone H2 terminals use a separate non-e-fuel implementation. Oil/gas product supply depends on e-fuels. These guards are executed, not merely suggestions.

The model policy defaults complex sector selection to `Non_metallic_minerals`; preparation may generate all eleven sectors. An exported selection must be applied explicitly in the model configuration or supported adapter. Otherwise a workbook containing all sectors can still run only the selected subset.

3.3 Input workbook selection

Technology file selection follows regular/category/complex and e-fuel flags. Category e-fuel conservative mode now selects `data/flex_tech_efuel_categories.xlsx`; its technical alternative selects `data/variants/technical_2050/flex_tech_efuel_categories.xlsx`. Complex electricity variants select the corresponding `industry_complex*_electricity_technical_2050.xlsx` process workbook while retaining their normal flex-tech filename. Categories and complex industry are alternatives. See the visual user guide and data reference for the exact mapping.

---

4. Spatial and temporal representation

Let s denote a model System, z a zone belonging to it, y a modeled decision year, t an operational period, g a generation asset, k another technology, f a carrier, i an industrial service/process and a an investment vintage. Assets and flows are created for active mappings; country names elsewhere in a workbook do not imply active modeled systems.

The full input calendar is normally 8,760 hourly positions. Operational subsets and representative weights depend on mode. A constraint using `_periods_for_year(y)` need not cover the same periods in every optimization year. Temporal metadata must be exported with results.

4.1 Perfect foresight

With `Myopic=False` and the applicable mode switches disabled, the model jointly chooses investments and operations across the modeled horizon. It knows the deterministic future input trajectories provided to it. Perfect foresight is an information assumption, not a claim that actual planners possess those forecasts.

4.2 Myopic planning

With `Myopic=True`, overlapping windows are built from the planning grid and setup controls. The model solves a window, accepts/exports its decision-year results, and carries historical decisions into later windows. Already accepted years are not reoptimized as freely as future years. New scenario names and coherent histories are necessary after material changes to costs, units or policies.

4.3 Partial perfect foresight

This mode combines detailed current-year operations with sampled future-year operations in each myopic solve. Future samples guide current investment, but they are not an independently validated full-year dispatch for those future years. The current source uses explicit per-year period sets and sparse variable creation. Read the configuration snapshot for representative-day settings; do not assume all years in a chunk contain 8,760 variables per asset.

4.4 Representative capacity planning and two-stage operation

Representative-capacity mode can sample configured weeks, with weights representing the full year. Lightweight mode exports only the capacity stage. Two-stage mode can subsequently export a fixed-capacity dispatch bundle and perform rolling operation. These approximations have different abilities to represent storage chronology, seasonal scarcity and ship arrivals. A representative-week optimum does not prove that a full chronological year is feasible.

4.5 Rolling dispatch

Rolling mode advances through operational windows with capacity fixed from a stage-1 bundle. It carries stocks, relevant fuel/terminal states and guidance into subsequent windows. Resume and stitching options can reuse accepted window files. Review terminal conditions, guidance penalties and automatic slack settings; they can relax a pure hard-feasibility interpretation.

---

5. Mathematical notation and decision variables

The following equations summarize implemented families. Asset-specific dimensions, history maps and flag branches are suppressed. The code reference and machine-readable call inventories give executable locations and conditional context.

Symbol	Meaning	Typical physical basis
X(k,z,s,y)	New capacity decision	MW, MWh, vehicles, tCO2/h or another explicitly declared asset unit
K(k,z,s,y)	Available installed capacity	Same unit as the asset's capacity
P(g,z,s,t,y)	Electricity generation	MW for one-hour periods

Symbol	Meaning	Typical physical basis
Q(k,z,s,t,y)	Delivered heating/industrial service	MW useful heat/service
U(f,k,z,s,t,y)	Carrier/fuel input	MWh of carrier per hour
E(k,z,s,t,y)	Storage stock	MWh, tCO2 or other carrier stock unit
C,D	Storage charging/discharging flows	MW or corresponding hourly carrier flow
N(v,z,s,y)	Transport stock	Continuous vehicle-equivalent count in the LP
F(f,z,z',t,y)	Carrier transfer	MW-equivalent flow
H(i,z,s,t,y)	Industrial delivered-service demand	Useful MWh per hour
A(b,z,s,t,y)	Product activity	Tonnes or index-activity units per hour
w(t,y)	Representative period weight	Represented hours

Variable bounds and scales differ by family. Scaled transport variables have physical-value wrappers; numerical solver column values must be converted back before comparing with raw vehicles, stock or energy data. State and investment symbols in this manual denote physical quantities.

---

6. Economic objective and investment accounting

6.1 Objective implemented in this source

The production objective has the following structure, in million currency units before an explicit solver scaling factor:

```
min  sum_(y,z,s) DF(y) × [CAPEX_year(y,z,s)
                        + multip × fixed_OPEX_year(y,z,s)
                        + variable_OPEX_MEUR(y,z,s)]
     + rolling_carbon_slack_penalty
     + rolling_terminal_fuel_SOC_slack_penalty
     + stage2_quantity_guidance_slack_penalty
```

`DF(y)=(1+discount_rate)^(-(y-discount_base_year))`. The reader defines `multip=(max(years)-min(years)+1)/len(years)` for its solve grid. Variable-OPEX expressions also apply the configured annualization/multiplier and EUR-to-MEUR conversion. Those exact conventions matter on sparse decision-year grids. They must not be replaced in a report by an unexplained sum over every calendar year or by a different textbook objective.

The examined policy has discount rate 0.05, discount base year 2025, capital finance rate 0.05 and numerical objective scale 1000. Numerical scaling is intended to change conditioning, not physical economic units. Read exported objective/cost variables consistently with that scale.

6.2 Capital service and vintages

The cost script treats capital service as an annuity-due payment for each investment vintage. The first payment is in the build year, and payment continues while that vintage is within technical life. Its unit capital cost is the build-year cost, not each later year's learning-curve price. Payments after the modeled horizon are omitted under this accounting convention.

```
annuity_due_factor(L,r) = 1 / sum_(n=0 to L-1) (1+r)^(-n)
annuity_due_factor(L,0) = 1/L
payment(vintage a, year y) is due when a <= y < a + lifetime
```

Costs may be MEUR per MW, per MWh of stock, per vehicle, per tCO2/hour or per terminal TWh/year. Applying one unit cost to another basis can dominate the result while leaving the LP numerically feasible. The live source includes literature-linked annual fixed-OPEX data and generic/fallback technology mappings. Matched rows and fallback paths should be audited separately.

6.3 Operating costs, trade and penalties

Variable OPEX includes the appropriate fuel/commodity purchases, non-fuel operation, ramping, external trade and carbon charges, with special accounting for e-fuel recipes. Do not add a fuel bill a second time merely because its energy passes through several conversion assets. Revenue signs depend on import/export terms and market configuration. The code contains separate national/external-carbon options and fuel-specific prices; study documentation must identify the active option.

Unserved electricity, rolling heat/service shortages and selected guidance or carbon shortfalls can be penalized. These are scarcity/slack representations, not real zero-cost supplies. The configured electricity unserved-load cost is a scenario penalty, not a nationally estimated VOLL. A low objective obtained with nonzero slack must be reported as such.

---

7. Capacity, lifetime, decommissioning and build constraints

For an asset class, the general available-capacity relation is:

```
K(y) = surviving_existing_capacity(y)
       + sum_a survival(a,y) × X(a)
```

Survival/retirement is asset-specific. Existing capacity uses remaining-life/decommissioning inputs; new vintages use technical life and the implemented retirement shape. Some families retain a final-life fraction. The exact helpers and history conventions should be used when interpreting the asset's investment CSVs.

Capacity may have a total ceiling and an annual build limit. They are different restrictions:

```
X(y) <= sum_(c in investment_window(y)) annual_build_limit(c)
K(y) <= total_installed_ceiling(y)
```

Investment windows use the global decision-year grid. For endpoints 2025 and 2030, the first endpoint represents 2025 only and the next represents 2026–2030, including on resume. Blank annual limits use the legacy scalar if present; absent/blank scalar means unrestricted. Zero disallows additions but does not delete existing capacity. A period with an unrestricted year remains unrestricted under the relevant industry-limit reader. Negative/nonfinite supplied limits are rejected.

Generation-specific runtime build multipliers can further scale workbook limits. In the reviewed policy four Croatian wind/PV names have multiplier 2; that is a scenario configuration, not a universal country potential or model constant. Transport, terminal, capture and storage additions use their native physical units and may have separate

active flags.

---

8. Electricity generation, networks and storage

8.1 Dispatch and renewable availability

Dispatchable output is bounded by available capacity and applicable technical constraints. Renewable output uses matching availability factors. Hydro run-of-river has flag-dependent equality/inequality behavior in the source. Curtailment or excess electricity is represented where enabled; do not assume every renewable MWh available must be delivered.

The principal electricity balance includes generators, fuel cells, relevant storage discharge, vehicle-grid services, net imports and an explicit unserved-load variable. Demand includes baseline electricity plus endogenous heating, industry, transport, electrolysis, synthesis/CC and storage charging, adjusted under the code's baseline conventions.

```
generation + fuel_cells + discharge + delivered_imports + unserved_load
 = baseline_load + endogenous_sector_electricity + charging + exports + excess
```

The exact terms and signs are in `el_balance`. This is a zonal energy balance, not a full nonlinear AC network. Do not infer voltage, frequency, transient stability or N-1 security from hourly energy feasibility alone.

8.2 Network transfers

Primary/secondary links, outside links and carrier networks use input mappings, transfer capacities, losses and costs. One delivered-flow conversion appearing in the electricity balance uses `1/(1+distance_multiplier*loss)`; substituting a different loss formula in a methodology description would misstate this implementation. Outside-system availability can restrict imports/exports using hourly supply-share/curve tables.

Transmission duplicates can be resolved by the configured `first` policy. The log reports removed rows and conflicts. A deterministic rule does not prove that conflicting network data are correct. New country/system aggregation requires deliberate network and outside-trade calibration.

8.3 Electricity and generic storage

An illustrative physical stock relation is:

```
E(t) = retained_stock(t-1) + eta_charge × C(t) × dt
                           - withdrawal_for_discharge(t) × dt + natural_inflow(t)
0 <= E(t) <= storage_capacity
0 <= C(t) <= charge_power_limit
0 <= D(t) <= discharge_power_limit
```

Efficiency may be applied to the stock transition or to the delivered balance depending on family; the source is the authority. Representative weights used for annual accounting are not automatically multi-hour storage transition durations. Initial/final/cyclic conditions can materially change apparent flexibility. The reviewed policy has `storage_year_continuity_mode="cyclic"`, with additional rolling/fuel boundary controls.

8.4 Hydro reservoirs and pumped storage

Hydro includes plant-specific turbine capacity, storage, inflow and pumping where applicable. Water-energy stock balances and available generation are constrained separately. Pumped storage consumes electricity when charging; it is not a new primary renewable source merely because its fuel label is Hydro. The source/profile year and reservoir assumptions need to accompany any annual generation interpretation.

---

9. Buildings, district heating and CHP

The heating block meets individual and district-heating service demand using enabled fuel boilers, electric heaters/heat pumps, solar heat, CHP and storage. Each class has capacity, efficiency/COP and investment/retirement constraints. Cooling is a separate optional representation and is disabled in the examined selection.

For a boiler or heat pump, the conceptual conversion is `useful_heat = efficiency_or_COP × input_energy`. COP above one represents heat transfer from an external source. It is not electricity generation. Heat-source constraints, DH integration and storage options must be accounted for when comparing technology outputs.

CHP is coupled production. The shared physical helper uses condensing-equivalent electricity `P + beta*Q`; its fuel input divides that quantity by electrical efficiency. That expression is used consistently for the relevant costs and emissions. CHP feasible operating regions additionally use input ratios/parameters. Treating CHP heat as cost/emission-free while retaining the full electric output would double-allocate the fuel benefit R30.

Current heating policies include base-year plant heat calibration, annual limits on CHP heat change, electric/gas heating transitions, fossil-heat non-increase and an explicit boiler phaseout. The reviewed entered phaseout years are coal 2030 and gas/oil 2035; the source notes that CHP is excluded from that boiler rule. These are local scenario settings, not automatic national legislation. Solar and storage options remain subject to their switches.

---

10. Industry

10.1 Ordinary and category formulations

Industrial equipment supplies useful service while carrier demand depends on efficiency. Regular mode uses sector/service keys; category mode adds temperature/service categories. `NON_HEAT` is retained where applicable. Excluded detailed services can be represented in a pooled remainder with its own input row. The remainder must be consistent with selected detailed demand to avoid omission or duplication.

Initial useful-service shares, maximum shares, capacity and annual build limits interact. A maximum share is applicability, not a mandatory target. An investment ceiling does not change a fuel's compatibility with the process. An asset can be technically eligible but unable to expand quickly because turnover or adoption constraints bind.

10.2 Process/activity formulation

Complex industry replaces ordinary demand/technology rows only for its selected services. It uses branch product activity and component intensities:

```
useful_service(i,t,y) = product_activity(branch(i),t,y)
                        × activity_factor(i) × energy_intensity(i)
process_CO2(branch,t,y) = sum_i product_activity(branch,t,y)
                                × activity_factor(i) × process_intensity(i)
```

Output can be tonnes or an index. The generic per-activity fields are authoritative. Process emissions are not added again as combustion of a non-energy feedstock. The adapter validates selected technology/process/product coverage, activity units, annual product demand, positive-service shares and heat-pump eligibility fields.

The supplied material convention treats non-energy gas/oil labels as H2-fed demand. That is a model convention rather than a universal chemical process representation. The Croatian ammonia exception and non-HR unresolved reclassification are documented in the preparation manual and assumptions register.

10.3 Fuel substitution budgets

The method distinguishes common low-enthalpy/steam/drying heat, process-specific heat, electrical/mechanical/cooling services, material feedstocks and reducing chemistry. Compatible substitute families include H2 and enabled synthetic fuels, with local/shared limits so the same replaceable pool is not repeatedly allocated without restriction. Biomass has a baseline-share ceiling, and lignin expansion is constrained to at most two additional percentage points of eligible local service in the supplied screening method.

These are replacement-equipment assumptions. They are not certification of blending in an unchanged incumbent burner. The authoritative latest fuel-compatibility handoff supersedes older zero-ammonia/narrow-hydrogen/unlimited-lignin comments R20, R21, R22.

10.4 Adoption trajectories and S-curves

Demand-weighted annual fuel/electric-service adoption constraints are separate from hourly dispatch. Technology switching can be optimized hour by hour within the accepted annual transition and applicability envelope. The electric service share combines direct electric equipment and industrial HP service; it is not simply their electrical input fraction.

The reviewed S-curve policy uses an endogenous envelope with initial fraction 0.02, growth parameter 0.50 and reference milestones every five years. Shorter model steps interpolate toward those milestones so seeds/tolerances are not restarted every year. Directions can be explicitly increasing/decreasing by fuel, unit or asset tuple; gas/coal/oil/biomass directions are entered policies. These parameters are scenario choices, not nationally measured adoption rates.

10.5 Industrial heat pumps and storage

Industrial HP units share cumulative temperature and source-heat budgets. If eligible shares below 60, 80, 125 and 150 °C are cumulative, their sum is not independent addressable demand. The builder also respects source and service restrictions. Category industrial heat storage is a separate optional asset family; enabling it outside category mode is rejected.

11. Transport and fleet transition

Annual transport service is met by stocks of ICE, hybrid, electric and fuel-cell equipment in road, rail, shipping and aviation classes represented in the selected workbook. Service/activity profiles, fuel input, stock, battery availability and infrastructure constraints have different units and purposes.

The stock identity is conceptually:

```
stock(y) = surviving_legacy_stock(y)
           + surviving_new_stock(y) + accepted_conversions_in(y)
           - retired_or_converted_out_stock(y)
```

Calendar aging and historical stock are explicit. The reviewed transport settings include turnover periods/ramp policy and maximum technology service shares. Some investment/rail-link/infrastructure switches are disabled in the local snapshot. Document those switches before interpreting an optimized transport pathway as an infrastructure plan.

ICE/hybrid primary-fuel recipes and paid upgrades use `compat_*` metadata. Drop-in hydrocarbons, FAME, alcohol blends, marine methanol/ammonia and pilot fuel have distinct configured paths. The source's default volume-share ceilings and retrofit premiums are screening assumptions; this documentation makes no claim that they constitute current legal approvals. Marine conversion parameters drawn from container-vessel literature are transferred to model proxy classes and require case-specific calibration R25.

EV plug-in availability constrains charging/V2G opportunities. Minimum battery state and activity-derived travel energy restrict storage use. The model can couple EV batteries to the power system, but a country-level normalized profile is not a location-specific charging-network observation.

Transport solver scaling is handled by physical wrappers. Very small fractional legacy stocks and saved residuals can cause degeneracy or infeasible histories. The code cleans tiny numerical residue using configured tolerances and checks physical history; changing tolerances can affect acceptance and should be reported.

12. Hydrogen, electrofuels, feedstocks and terminals

12.1 Hydrogen

Electrolyzers consume electricity according to conversion efficiencies and available capacity. Steam reforming can use gas with associated costs/emissions. Fuel cells convert hydrogen back to electricity. Zonal H2 service includes applicable transport/industry demand, conversion/feedstocks, storage and network exchanges. The active balance and supply switches must be enabled together for a meaningful carrier accounting.

12.2 E-fuel conversion recipes

The e-fuel representation includes technology rows and recipes for enabled carriers such as H2, ammonia, methanol, synthetic methane, diesel, kerosene and other workbook fuels. Electricity, heat, CO2, nitrogen and upstream carrier inputs are coupled through the recipe fields. Generic form:

```
input_j(k,t,y) = recipe_coefficient_j(k) × production(k,t,y)
production(k,t,y) <= available_conversion_capacity(k,y)
```

Energy-carrier inputs and material inputs have different units. Nitrogen and CO2 must not be counted as energy simply because they appear in a feedstock table. Recipe coefficients and output energy bases require explicit checking; do not infer a balanced reaction solely from labels.

Fuel inventory states, conversion capacities, commodity purchases and trade can interact. The current code has separate initial stock and boundary/continuity options. An initial storage fraction is an assumption about available inventory, not a free replenishable primary-energy resource.

12.3 Oil and gas product supply

An optional e-fuel-associated block represents extraction/imports, crude refining and product supply. The shared helper accounts for product energy and a configured refinery process-fuel share. The reviewed share is 0.08. Whether fossil products may be exported is separately configurable and can materially change economic incentives. Assess exports and process fuel in both energy and carbon balances.

12.4 Standalone H2 terminals

The separate terminal module is enabled in non-e-fuel mode by its terminal/H2-balance switches. Inputs come from `h2_terminals.xlsx`, with asset and shipping-route tables. Capacity is annual TWh/year, converted to an hourly limit using 8,760 hours. Loading, unloading, imports and exports share capacity; losses and handling electricity are charged on the appropriate flows.

This is aggregate delivered hydrogen. It does not separately model every port tank, vessel, liquefaction or ammonia-cracking process. Local H2 storage supplies buffering. Carrier-specific project calibration therefore belongs in the aggregate assumptions or an explicitly extended model.

Detailed internal shipping has departure schedules, delay, cargo size and fleet occupancy. Representative planning can use an instantaneous/average-capacity approximation. The source prevents departures that would arrive beyond the annual boundary; such cargo does not automatically cross calendar years. The terminal's standalone representation and e-fuel terminal representation should not be conflated.

---

13. Carbon accounting, capture and renewable policies

13.1 Emissions scope

The model separately accounts for power/CHP, heating, industry, transport, reforming and refining emissions where active. Combustion coefficients, process CO2 and capture/storage flows are not interchangeable. The physical electricity/CHP helpers and stream-specific industrial emission factors avoid particular double-counting errors, but a complete study still needs a stated system boundary.

The ordinary budget has the conceptual form:

```
sum(enabled sector emissions) - credited geological_storage
 <= policy_CO2_limit + permitted_LULUCF_removal
```

Exact coefficients/scaling and alternate branches are in the carbon-limit rows. LULUCF can be disabled or parameterized. The code's CO2 energy-system accounting is not a comprehensive product lifecycle, upstream methane, land-use, N2O, ammonia-slip or biodiversity assessment. `non_CO2_polutants` is explicitly marked incomplete.

13.2 Capture and geological storage

Capture capacity, rate, heat/electricity demand, reboiler capacity and optional intermediate storage impose technical and economic constraints. `model_CC`, post-generation CO2 storage and e-fuel CO2 feedstock are different switches. One can retain parts of the e-fuel CO2 handling even when a broader CC setting is disabled.

The separated industry-process/other-carbon representation restricts which captured streams can enter geological storage. It is enabled in this snapshot through `separate_industry_process_carbon_balances=True`; `model_CC`, amine storage and post-generation CO2 storage are also enabled.
`complex_industry_process_capture_required=False` leaves process-capture deployment to the applicable carbon and technical constraints rather than universally forcing it. Untagged imported CO2 does not automatically become a certified process stream. Reuse of CO2 in e-fuels should not be described as permanent sequestration.

13.3 Renewable electricity share

The native RPS policy constrains qualifying electricity generation/share according to the source's numerator/denominator definitions. It is not a universal all-energy renewable fraction. Reservoir/pumped-storage treatment, imports, curtailment and generation calibration must be interpreted consistently with the actual equations.

13.4 Whole-energy renewable preset from preparation

The preparation adapter adds stricter endpoint restrictions for the selected `res100` / `all-energy` scenario. At and after the target, it blocks fossil/nuclear energy use in covered generation, heating, industry and transport, relevant reforming/refining flows, unverified outside energy imports and imports through routes whose endpoints are outside the active model geography. It also prevents unserved electricity from serving as a zero-origin substitute.

Because generic stores lack a renewable-origin tracer, affected initial stocks are cleared at the target boundary; subsequent internally renewable stocks may carry forward. Material CO2/nitrogen remain separate. Mixed fossil-waste industrial streams are blocked under the conservative origin treatment. Earlier non-electric transition is chosen subject to the other policies rather than a claimed linear whole-energy renewable trajectory.

This policy is an energy-origin definition, not net-zero CO2. Biogenic sustainability, upstream lifecycle burdens and independent process emissions require separate decisions. Its current launcher supports the standard myopic/perfect-foresight workflows and rejects the unsupported whole-energy two-stage combination. Feasibility may be prevented by fixed processes, feedstock conventions, build rates or resource limits.

---

14. Policy calibration and scenario design

Scoped policy workbooks can contain generation targets, hydro caps, transport targets, biomass caps and outside trade calibration. Rows are selected by active System/zone and modeled years. Some calibration applies only to the global base year; resource caps can apply in every year. Calibration targets are not all equivalent to future investment opportunities.

Before creating a new national study, review demand forecasts, technology prices, currency/base year, lifetimes, reserves, resource potential, fuel/commodity availability, network/outside trade, heat-share policies, transport turnover, industrial compatibility and process routes. Automatically sourced historical data do not determine those future scenarios.

Use one coherent baseline and document each changed assumption. A suggested sensitivity matrix varies demand growth, resource potentials, grid expansion, storage chronology, external trade, capital/operating costs, industrial transition rates, biomass eligibility and renewable-origin definitions. Optimized technology shares should always be interpreted conditional on this matrix.

---

15. Preparing and running a study

15.1 Environment and input validation

The separate model needs a suitable Python/Gurobi environment. Its supplied local reference lists Python-tested numerical packages and `gurobipy==12.0.3`; these are recorded local versions rather than a requirement that another compatible release is scientifically identical. Large national models need appropriate solver licensing and sufficient memory. No license credentials belong in published inputs.

For a native model run, work in the model's own folder, set a fresh scenario name and review the active flags, policies and file names. `python main.py` executes that model, including optimization; it is not a data-preparation command. Run an input-reader-only check before a costly solve, using a supported validation harness.

For a preparation-generated bundle, read `bundle_manifest.json`, `validation.json`, `scenario.json` and process-generation metadata. The generated integration launcher points at an existing model installation. Its default is input-reader-only; `--solve` explicitly requests construction/optimization. The preparation GUI does not invoke it.

15.2 Separate preparation integration

This validation command belongs to the model integration workflow, after preparation:

```
python outputs_all_HR/data/run_model_scenario.py --model-root "D:\POSAO\H2RES_12.11.2026"
```

This command reads inputs without requesting optimization. Do not substitute a stale bundle merely because filenames exist. Adapter compatibility is version-specific; inspect model/source hashes when moving to another release.

The preparation contract was developed against the 2 November native layout. A current legacy bundle is not evidence that the new product-industry workbook, DSM assumptions or every later model extension has been generated or validated. Use the adapter validation output before considering a solve; this documentation did not validate that adapter against a full 12 November solve. Run the native model directly for the script-based workflow described in the visual guide.

15.3 Resume and historical decisions

Set resume options only for coherent, accepted results. Investment handoffs, fuel/vehicle/heat histories and terminal/stock files must belong to the same input convention and scenario. Changing costs, unit names, decision grid, physical scaling or technology representation while reusing histories can produce mathematically valid but

inconsistent comparisons or an infeasible model.

A partial foresight run exports the accepted decision year, not every sampled future-year solution as a committed history. Failed/provisional outcomes need their quality gates before they can be used as later stock constraints. Preserve the exact logs and history files when investigating a transition failure.

---

16. Solver behavior and numerical quality

The cost script selects the primary solver method and applies numerical scaling/tolerances from runtime policy. The examined settings favor barrier for planning, with specialized annual-S-curve behavior and recovery options. A configured time limit of zero can mean unlimited in the implemented wrappers. Recovery calls may differ from the primary solve and can run for a long time.

Solver status and engineering quality must be reported separately. `OPTIMAL` means the solver's optimality criteria were met for the constructed problem and tolerances; it does not validate input realism. `TIME_LIMIT` or `SUBOPTIMAL` may have a usable incumbent but are not equivalent to a certified optimum. `INFEASIBLE`, `INF_OR_UNBD` and numerical failures require different diagnoses R28, R29.

Inspect primal constraint/bound violations, dual/residual information where meaningful, objective/bound gap and the model's physical balance audit. Numerical row scaling can make a small solver residual correspond to a different MW/MWh/tCO2 violation. Read the physical-unit gates before accepting exports. Do not use a higher tolerance merely to remove an inconvenient policy or stock inconsistency.

The delivered `main.py` has `solver_allow_provisional_solution=True` and `solver_postoptimal_checks_warn_only=True`. Consequently, a finished export can contain warnings or a provisional solution: completion alone is not a strict quality certificate. Runtime policy also requires annual capture-quality checks and sets a 0.1 physical-energy tolerance and a 0.001-tonne amine-mass tolerance in their applicable checks. Read the exported status and physical audits together with the actual branch that produced them; a warning is not proof that a gate stopped the run.

A barrier solution without crossover can lack a conventional simplex basis; shadow prices or recovery behavior need careful interpretation. Memory exhaustion can occur during building, presolve, factorization or export, and should not be labelled proof of physical infeasibility. Gurobi's numerical guidelines explain the general distinctions; the local configured thresholds are in the configuration inventory.

---

17. Results and interpretation

Result family	Interpretation and checks
Investment/capacity CSVs	New versus surviving capacity, native units, vintage/retirement and decision windows
Hourly/yearly electricity balances	Generation, demand components, charging/discharge, losses, imports/exports and unmet load
Heat and industry outputs	Useful service versus final fuel input; direct-electric/HP shares; process replacement accounting
Transport stocks/flows	Vehicle-equivalent stock, demand service, primary/secondary fuel recipes and conversions
Hydrogen/e-fuel outputs	Production, upstream inputs, storage, trade and terminal throughput

Result family	Interpretation and checks
Carbon outputs	Sector emissions, independent process emissions, capture/reuse/storage and budget/slack
Cost outputs	Unit bases, annualization, discounting, source price year and solver scale
Solver/quality logs	Status, method, runtime, recovery, objective/bound and physical residuals
Combined dashboards/plots	Derived presentation of accepted exports; not a substitute for balance tables

Myopic results are commonly organized under `results/<scenario>/<decision-year>` with annual investment/history files; detailed paths depend on exporter and mode. Stage-2 bundles and rolling window/stitching outputs have their own metadata. Terminal outputs can include `h2_terminal_flows.csv`, annual capacity/summary tables and `transfer_terminal_new_YEAR.csv`.

Check that representative-period additive quantities are annualized exactly once. Installed capacities and shares are not summed like hourly energy. Compare runs on matching economic/energy boundaries and time grids. A shadow price is conditional on the active constraints and an accepted economic solution, not a forecast of a regulated tariff.

---

18. Validation evidence, inherited warnings and known limits

This documentation was produced by static source/input review and reference checking. It did not execute a fresh national solve. Local preparation reports establish bounded reader/schema/source-reconciliation results; they do not independently establish all model modes or years.

The supplied folder contains the unified-model implementation notes, component regression tools, and a historical critical-correction/rebuild harness. The current `docs/UNIFIED_MODEL.md` describes the integration of simple industry, electricity CBAM history handling and DSM. These are distinct from older verification files describing unpromoted candidates. In particular its `verification.json` distinguishes component tests, synthetic workflow fixtures and user-reported remote transition evidence, with no real local checkpoint promoted in that report. Historical run instructions and comments must not be cited as proof that those candidate fixes are active or that a full transition passed locally.

Validate the exact production objective and physical constraints in the documented source, then compare any proposed correction with its checkpoint/diff and acceptance certificate. A source hash, accepted transition log and scenario/input identity are stronger evidence than a filename such as `corrected` or a comment claiming a previous optimal case.

Principal study limitations are deterministic future inputs, spatial aggregation/proxies, sampled chronology in applicable modes, continuous transport stock, inherited technology/fuel compatibility assumptions, incomplete lifecycle/non-CO2 accounting, terminal aggregation, possible data-duplicate reconciliation and country-specific calibration gaps. The assumptions register makes these limitations explicit.

For a publishable run, archive the exact model/inputs, environment, flags/policies, source references/hashes, solve logs, quality certificates, decision/history files, representative weights and scripts producing reported tables/figures. Cite the H2RES scientific formulation and the actual data/technology references, while stating that this local version includes later extensions.

19. Basic electricity demand-side management in this version

`scripts/electricity_dsm.py` adds lossless load shifting to basic electricity demand. It does not automatically shift all heating, transport or industrial service demands. `DSM_day`, `DSM_week`, `dsm_energy_budget_active` and `dsm_availability_ramp_active` are all false in the delivered `main.py`. See the visual guide for activation and the supplied detailed DSM implementation notes for time-mode behavior and regression scope.

Runtime policy defines a 7% daily-capable energy pool that includes a 1.5% weekly-capable subset. With both classes enabled, 5.5% balances daily and 1.5% balances weekly. These are scenario upper bounds on moved energy, not measured Croatian loads or compulsory dispatch. The default rollout reaches its target in 2030 from the global base year. The separate shared hourly adjustment cap is 10% of baseline load; available-power cost is EUR 10,000/MW/year and activation cost EUR 5 per MWh moved.

For original hourly load `b[h]`, signed class adjustments `x[c,h]`, absolute adjustments `a[c,h]` and available power `K`, the main restrictions are:

```
new_basic_load[h] = b[h] + sum_c x[c,h]
a[c,h] >= x[c,h]; a[c,h] >= -x[c,h]
sum_c a[c,h] <= K
sum_c a[c,h] <= hourly_shift_fraction * b[h]
sum_(h in block) x[c,h] = 0
0.5 * sum_(h in block) a[c,h] <= class_energy_fraction * sum_(h in block) b[h]
```

The half-absolute sum counts each moved MWh once. Daily blocks have 24 hours and weekly blocks 168, anchored at model hour 1; the last incomplete block also closes. Opposite class shifts share power and cannot cancel their capacity use. A symmetric 10% hourly cap can limit moved energy to 5% of block load even when the energy allowance is 7%. These are different constraints. This model does not represent individual appliance deadlines, load shedding or cross-zone demand relocation.

Review `dsm_hourly_YEAR.csv`, `dsm_classes_YEAR.csv`, `dsm_summary_YEAR.csv` and `dsm_assumptions.json`. Compare original and adjusted demand, net block energy, moved MWh, available power and enabling/activation costs. Changing budgets or rollout can invalidate existing histories and capacity bundles.

Demand-response background is provided by R35, R36 and R37; these sources do not validate the local energy-pool percentages or costs.

20. Product based simple industry and regional trade

This is a separate optional representation, selected by `model_simple_industry=True` and `simple_industry_dat='./data/simple_industry.xlsx'`. It is different from the worksheet named `simple_industry` inside a flex-tech workbook: that worksheet is the legacy pooled-service remainder. The new external workbook contains fixed industrial product recipes, physical plants and product markets. The delivered model leaves this mode off.

Enabling product industry configures the electricity-and-product-industry run and disables overlapping legacy transport/industry, category/complex, e-fuel, hydrogen and industrial capture/storage blocks. Heating retains its switch. Active example systems are HR, RS and BA through `simple_industry_active_zones` in model policy.

Independent switches control industrial CBAM, domestic carbon pricing, outside imports, optional outside exports and flexible steel operation. Only one electricity CBAM option may be selected; electricity CBAM and product CBAM are separate accounts.

The workbook uses `plants`, `inputs_per_tonne`, `economics`, `fuel_prices`, `markets`, `market_years`, `carbon_policy`, `import_suppliers`, `import_years`, `trade_routes`, `profiles`, `baseline_load` and `metadata`. Production is tonnes per one-hour interval, electricity is MWh/t, non-electric recipes GJ/t and carbon factors tCO2/GJ. Supply ceilings, grid limits and inventories limit dispatch; nameplate capacity does not require production. Fixed-operation plants remain constant within each calendar day after availability adjustment; flexible plants may vary hourly under their limits.

Mandatory product markets must be supplied. Regional plants can trade; bounded outside offers are available only under their applicable switches and eligibility. One supplier's annual budget is shared across destinations. No unlimited emergency source or automatic demand reduction is introduced. Origin tracking preserves supplier footprints for final-destination carbon charging and separates territorial direct emissions from embedded trade emissions. Prices are prescribed; there is no endogenous steel-price equilibrium.

The supplied RS BF-BOF and BA scrap-EAF plants, external supplier prices/footprints and rest-of-Europe supply are illustrative scenario inputs, not calibrated named installations. The example requires RS 800,000, BA 300,000 and HR 1,000,000 tonnes/year, plus 134,000,000 tonnes/year for Europe outside HR. The European background offer serves that Europe market, not a bypass for domestic import restrictions. Check every price, quantity and route before drawing policy conclusions.

`baseline_load` specifies already-existing plant electricity to remove from the basic profile. Example removals are zero, so example plant loads are additions. Calibrating a real plant requires a defensible removal to avoid counting the same demand twice. Likewise, do not enter coke carbon in both the fuel recipe and process-carbon field.

Review `simple_industry_hourly.csv`, `simple_industry_fuels.csv`, `simple_industry_sales.csv`, `simple_industry_markets.csv`, `simple_industry_summary.csv`, `simple_industry_imports.csv`, `simple_industry_demand.csv`, `simple_industry_cbam_borders.csv`, `simple_industry_fiscal.csv` and the applicable levelized-cost summary. For field-level operation, trade budgets and capital/CBAM allocation, use the supplied simple-industry implementation notes. The unified-model notes report bounded component/8760-hour example evidence and known baseline regression exceptions; they do not certify a full national transition solve.

Complete configuration option reference

Scope: every assignment in the inspected top-level `main.py` and its three policy modules. This reference covers all 381 recorded assignments, including repeated/derived declarations and internal cache objects; the coverage report is generated from the actual configuration inventory. Function-local solver implementation variables are not additional user switches.

Each row gives the entered expression, its source line and its operational meaning. The entered default is not always the effective value: normalization, parent modules, workbook eligibility and execution-stage overrides still apply. Consult the configuration guide for dependencies and complete workflows, and the equation chapter for physical/economic interpretation.

Boolean controls accept Python `True` or `False`. Dictionary keys, exact unit labels, native units and supported string values matter. An empty mapping supplies no extra overrides for that mapping; it does not necessarily disable other model constraints. A setting marked diagnostic, internal, incomplete or test-only is not a supported physical scenario selector.

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main.py

Setting and source	Entered value / expression	Meaning and effect
`scen` (`main.py:20`)	`'industry_CC_5y'`	Scenario and results/history identifier. Change it for a new experiment; the text does not select years, sectors or policies.
`model_policy_dat` (`main.py:25`)	`'./policy/model_policy.py'`	Physical/economic policy module executed before most main switches. Its values are visible in the shared model namespace.
`runtime_policy_dat` (`main.py:26`)	`'./policy/runtime_policy.py'`	Temporal and numerical policy module. Change sampling, rolling windows and solver parameters here, not in an unrelated workbook.
`rps_inv` (`main.py:37`)	`True`	Add the native renewable-electricity generation-share constraint using the selected policy workbook. False removes this constraint; it does not remove renewable technologies or carbon prices.
`carbonLimit` (`main.py:38`)	`True`	Apply the planning emissions cap or applicable rolling budget. False removes this budget, while ETS/CBAM costs and technical capture constraints may remain.
`lulucf_removals_active` (`main.py:39`)	`False`	Include the configured land-use removals allowance in carbon accounting. It offsets the permitted net budget and is not a CO2 capture, transport or storage technology.
`lulucf_removals_dat` (`main.py:40`)	`'./data/lulucf_removals.csv'`	Optional System/year removal trajectory. Check tCO2-equivalent/year and its interaction with the fixed removal setting.

Setting and source	Entered value / expression	Meaning and effect
`res_inv` (`main.py:41`)	`True`	Select renewable-investment reporting/plot branches. This is not the RPS switch and does not independently permit or forbid renewable investment.
`hydro_storage` (`main.py:42`)	`True`	Construct reservoir/pumped-storage operating equations for eligible hydro units. Turning it off is a different hydro formulation, not proof that hydro can meet the same chronology.
`save_csv` (`main.py:43`)	`True`	Request applicable CSV result exports. Preserve these for history handoffs and audits; plots alone do not replace the required histories.
`ceep_limit` (`main.py:44`)	`False`	Add the surplus-electricity cap relative to the modeled demand boundary. The surplus penalty can remain in the objective when this cap is off.
`Myopic` (`main.py:45`)	`True`	True selects sequential planning with accepted historical decisions. False selects joint foresight unless a higher-priority automatic or fixed-dispatch workflow applies. Electricity CBAM Options 1–4 require True.
`partial_perfect_foresight` (`main.py:46`)	`False`	In ordinary myopic planning, retain full-hour current-year operation and sampled future years within the configured lookahead. It does not mean all those future years have been validated hourly.
`simple_curves` (`main.py:47`)	`True`	True reads reusable period-based inflow/availability curves. False expects the supported year/period schema. It is an input-layout choice, not a representative-time or spatial-resolution selector.
`outside_system_availability` (`main.py:48`)	`True`	Apply additional hourly outside-grid import/export availability from the supplied shares and normalized curves. Turning it off does not remove every transfer-capacity or annual-trade limit.
`outside_system_supply_shares_dat` (`main.py:49`)	`'./data/outside_system_supply_shares.csv'`	Outside-system supply composition used to construct available trade. Keep System and technology mappings consistent.
`outside_system_normalized_curves_dat` (`main.py:50`)	`'./data/outside_system_normalized_curves.csv'`	Hourly normalized availability by outside supply type. Values constrain trade rather than create domestic generator stock.
`generation_source_calibration_active` (`main.py:54`)	`True`	Apply supplied source/fuel generation calibration targets in their scoped years. Removing calibration frees that restriction but does not alter the generator inventory.
`hydro_generation_cap_active` (`main.py:55`)	`True`	Apply the extra hydro generation cap from policy. Reservoir energy and turbine limits still have their own equations.
`outside_trade_calibration_active` (`main.py:56`)	`True`	Apply configured annual outside-trade calibration targets. Distinguish these targets from hourly availability and network capacity.

Setting and source	Entered value / expression	Meaning and effect
`dh_chp_base_year_heat_calibration_active` (`main.py:57`)	`True`	Match scoped district-CHP base-year heat to entered plant targets. Requires matching generator IDs and active heat demand.
`dh_chp_heat_yoy_limit_active` (`main.py:58`)	`True`	Constrain annual changes in CHP heat according to the model-policy rate, historical tolerance and reduction-limit horizon.
`model_heating` (`main.py:61`)	`True`	Enable heating demand, technologies and balances. Several legacy industry cost definitions are nested within heating in this source, so an industry run with heating disabled is not a documented fully costed preset.
`ratio_heat_indv` (`main.py:62`)	`True`	Enable the legacy individual-boiler production/capacity constraint branch. Do not interpret this as a simple population-allocation or fixed fuel-share option.
`heating_shares_regulate` (`main.py:63`)	`False`	Add links between heating technology shares, installed capacity and scaled peak demand. This is additional to annual energy balances.
`individual_heat_pump_storage` (`main.py:64`)	`False`	Enable local heat storage in the individual-HP branch. Requires heating and eligible individual HPs; it is separate from district or industrial thermal storage.
`DH_Storage` (`main.py:65`)	`False`	Enable the applicable district heat/cooling storage branch. Storage requires its own energy/power parameters and boundary conditions; it is not the industrial category store.
`NoResToHeatInv` (`main.py:66`)	`False`	Legacy inverted gate: False permits the normal individual-HP electricity/capacity equations; True skips those gated equations. Do not use True as a general ban on renewable electricity feeding heat or as a safe freeze-investment option.
`model_Thermal_units` (`main.py:67`)	`True`	Include eligible individual fuel-boiler technologies when heating is enabled. Existing/retirement and investment tables still determine available capacity.
`model_Thermal_units_DH` (`main.py:68`)	`True`	Include eligible district fuel-boiler technologies when heating is enabled. CHP and district heat pumps remain separately controlled.
`model_HP_units` (`main.py:69`)	`True`	Include individual heating HP technologies and their COP/electricity links. Requires coherent thermal demand, costs and COP data.
`model_HP_units_DH` (`main.py:70`)	`True`	Include district-heating HP technologies. Industrial HPs have an independent policy switch.
`model_Solar_indv_heating` (`main.py:71`)	`False`	Include individual solar-thermal supply when heating is enabled. This is solar heat, not PV generation.

Setting and source	Entered value / expression	Meaning and effect
`model_Solar_DH` (`main.py:72`)	`False`	Include district solar-thermal supply when heating is enabled. Availability, conversion and storage inputs must exist.
`model_cooling` (`main.py:73`)	`False`	Request the legacy cooling block and cooling inputs. It is off in the delivered case and was not validated by this documentation exercise.
`electric_heating_yoy_limit_active` (`main.py:74`)	`True`	Apply annual increase/decrease limits to the configured individual electric-heating technologies. These restrict transition rates, not each hour's dispatch directly.
`individual_gas_heat_yoy_limit_active` (`main.py:75`)	`True`	Apply the entered annual gas-heating transition limits and reduction horizon to individual boilers.
`fossil_heating_nonincrease_active` (`main.py:76`)	`True`	Prevent annual fossil heat from increasing in the applicable individual/DH scopes. Disabling it does not automatically cancel the boiler phaseout.
`fossil_boiler_phaseout_active` (`main.py:77`)	`True`	Apply fuel-specific boiler retirement/zero-output deadlines. CHP is excluded from this boiler-specific rule; carbon policy remains separate.
`Option_1` (`main.py:80`)	`False`	Electricity CBAM at the border only. Select at most one electricity option and use Myopic=True; supply membership, prices, transfers and valid carbon-intensity history.
`Option_2` (`main.py:81`)	`False`	Electricity CBAM with a national carbon tax/ETS representation. Review domestic-versus-border accounting, membership and price trajectories; do not enable alongside another electricity option.
`Option_3` (`main.py:82`)	`False`	Electricity national ETS representation with the applicable EU-price/border relationship. It is a distinct scenario branch, not an automatic calibration to current law.
`Option_4` (`main.py:83`)	`False`	Mixed electricity national/border carbon-tax branch. Inspect the national-cost auxiliary and equations before use: the supplied structural notes flag an auxiliary without its own nonzero defining equation. Do not present this as an independently validated policy preset.
`include_secc_links` (`main.py:84`)	`False`	Include eligible secondary electricity transfer routes through intermediate zones. Requires complete secondary route mappings and compatible capacities/losses.
`model_simple_industry` (`main.py:90`)	`False`	Enable the separate fixed-recipe product-industry module. Its configure step disables overlapping legacy industry/transport, e-fuels, H2 and capture blocks and changes active example systems. It is not the pooled flex-tech worksheet.
`industry_cbam_active` (`main.py:91`)	`False`	Apply industrial product CBAM in simple-industry mode, using supplier origin, final destination and embedded footprint. It is separate from electricity Options 1–4.

Setting and source	Entered value / expression	Meaning and effect
`industry_carbon_pricing_active` (`main.py:92`)	`False`	Apply the simple-industry domestic carbon-price policy. Independent of industrial CBAM; requires the product module and complete plant/year policy rows.
`industry_external_imports_active` (`main.py:93`)	`False`	Allow eligible additional outside product suppliers in simple-industry mode. Their annual/hourly budgets remain binding. Europe background supply retains its restricted Europe-market treatment.
`industry_external_exports_active` (`main.py:94`)	`False`	Allow optional non-EU product sales in simple-industry mode. Mandatory domestic/Europe demand is not disabled by turning this off.
`steel_flexible_operation` (`main.py:95`)	`True`	Permit hourly operation of product plants marked flexible. False uses the applicable within-day constant-operation formulation, not mandatory full-load production.
`simple_industry_dat` (`main.py:96`)	`'./data/simple_industry.xlsx'`	Separate product-industry workbook with plants, recipes, markets, suppliers, prices, carbon policy and baseline load removals.
`model_transport_and_industry` (`main.py:99`)	`True`	Enable legacy transport/industry data, equipment and associated carrier blocks. E-fuel recipes and industry CAPEX have additional dependencies; turning this off is not a supported way to retain the same integrated e-fuel case.
`industry_single_sector` (`main.py:100`)	`False`	Legacy single-sector branch. Keep False for the documented e-fuel, category and complex-industry cases. Pooled Industry_data rows are a separate mechanism.
`model_industry_categories` (`main.py:101`)	`True`	Use temperature/service categories for detailed legacy industry and select the category technology/profile layout. Mutually exclusive with complex industry.
`industry_category_electricity_case` (`main.py:102`)	`'conservative'`	Choose conservative or technical_2050. In the e-fuel category case, technical_2050 selects the nested technology workbook; it is an applicability sensitivity, not a target year selector.
`industry_complex_electricity_case` (`main.py:103`)	`'conservative'`	Choose conservative or technical_2050 for the complex companion process workbook. Selects the corresponding process electricity-assumption file.
`industry_storage` (`main.py:104`)	`True`	Enable category-specific industrial thermal storage. Requires transport/industry, categories and multi-sector operation; turn it off for regular or complex recipes.
`industry_fuel_share_ramp_active` (`main.py:106`)	`True`	Enable general industrial transition envelopes in planning. They constrain annual adoption; hourly dispatch can still respond to prices and availability.

Setting and source	Entered value / expression	Meaning and effect
`industry_fuel_share_s_curve_active` (`main.py:107`)	`True`	Permit the configured endogenous S-curve mode when general ramps are active. False forces the applicable flat-ramp interpretation; the independent annual electric-share control can remain active.
`industry_electric_share_yoy_limit_active` (`main.py:108`)	`True`	Apply the combined direct-electric plus industrial-HP delivered-service share policy. This is a useful-service share, not an electricity-input fraction.
`model_complex_industry` (`main.py:116`)	`False`	Select process-based replacement for configured industrial sectors. Categories must be False and legacy multi-sector industry must remain enabled. Non-metallic minerals is the author's tested scope; other sectors are not equally evidenced.
`model_complex_non_metallic_minerals` (`main.py:118`)	`False`	Backward-compatible alias normalized together with model_complex_industry. Set both consistently when switching recipes so an old True alias cannot reactivate complex mode.
`model_complex_industry` (`main.py:119`)	`bool(model_complex_industry or model_complex_non_metallic_minerals)`	Select process-based replacement for configured industrial sectors. Categories must be False and legacy multi-sector industry must remain enabled. Non-metallic minerals is the author's tested scope; other sectors are not equally evidenced.
`model_complex_non_metallic_minerals` (`main.py:120`)	`model_complex_industry`	Backward-compatible alias normalized together with model_complex_industry. Set both consistently when switching recipes so an old True alias cannot reactivate complex mode.
`complex_industry_dat_efuel` (`main.py:127`)	`'./data/industry_complex_efuel.xlsx'`	Base companion process workbook for complex industry with e-fuels. The electricity-case selector can choose its technical_2050 variant.
`complex_industry_dat_no_efuel` (`main.py:128`)	`'./data/industry_complex.xlsx'`	Base companion process workbook for complex industry without e-fuels. It must match the no-e-fuel technology and carrier representation.
`two_stage_auto_mode` (`main.py:132`)	`False`	Run representative capacity planning followed by fixed-capacity rolling dispatch. The controller sets internal stage flags; supports sequential or joint capacity planning through Myopic.
`lightweight_representative_weeks_mode` (`main.py:136`)	`False`	Run the automatic controller's representative capacity stage only and skip rolling dispatch. It enables automatic mode and is a screening workflow, not an hourly feasibility certificate.
`two_stage_skip_stage1_shadow_exports` (`main.py:137`)	`True`	Skip the expensive stage-1 shadow-price export path in automatic mode. Ordinary investment/history outputs still need to be retained.

Setting and source	Entered value / expression	Meaning and effect
`two_stage_resume_skip_completed_chunks` (`main.py:138`)	`True`	Skip chunks whose required stage-2 completion outputs are found. This is a resume convenience, not proof that changed inputs are compatible with old results.
`two_stage_resume_reuse_stage1_bundle` (`main.py:139`)	`False`	Allow reuse of an existing stage-1 capacity bundle where its checks permit. Keep False after structural changes unless bundle identity and chronology are established.
`myopic_resume_from_results` (`main.py:142`)	`True`	Resume ordinary myopic planning after the contiguous accepted completed prefix. False starts at the base year; a new scenario is safer than overwriting a prior experiment.
`validate_completed_heating_policy_history` (`main.py:143`)	`False`	Request validation of completed heating histories against current rules. False does not certify old policies; it lets existing accepted years remain while unsolved years use current rules.
`two_stage_timing_logs` (`main.py:144`)	`True`	Emit timing information for the automatic stage sequence. It changes diagnostics rather than the physical feasible set.
`representative_capacity_mode` (`main.py:145`)	`False`	Internal/manual capacity-planning time-sampling selection. Automatic mode controls it itself. The legacy week-named parameters currently describe four 24-hour blocks.
`fixed_capacity_mode` (`main.py:146`)	`False`	Freeze capacity from the dispatch bundle in the dedicated rolling path. Without rolling_dispatch_mode, main normalizes it back to False.
`dispatch_stage_data_path` (`main.py:147`)	`'./results/' + scen + '/stage2_dispatch_bundle.pkl'`	Path to the stage-2 pickle capacity bundle. Use a trusted bundle from the same model/data conventions; it is not an Excel input file.
`export_stage2_dispatch_bundle` (`main.py:148`)	`False`	Export a capacity/state/metadata bundle after planning for later fixed dispatch. Automatic mode sets this during stage 1.
`rolling_dispatch_mode` (`main.py:149`)	`False`	Select rolling-operation behavior. The standalone runner requires fixed_capacity_mode=True; True alone does not select that execution branch.
`rolling_write_solution_quality` (`main.py:150`)	`False`	Write additional rolling solution-quality artifacts. Their presence is diagnostic; inspect whether the solve was provisional or warnings were allowed.
`rolling_write_window_files` (`main.py:151`)	`True`	Save individual rolling-window outputs/checkpoints. Needed for an auditable interruption/resume workflow.
`rolling_keep_results_in_memory` (`main.py:152`)	`False`	Retain rolling results in memory instead of relying exclusively on streamed/window files. Consider RAM use for many zones, carriers and hours.
`data_read_cache_enabled` (`main.py:153`)	`True`	Cache repeated CSV/Excel reads inside a run using path, modification time, size and reader arguments. This is separate from internet/weather download caching.

Setting and source	Entered value / expression	Meaning and effect
`data_read_cache_copy_deep` (`main.py:154`)	`True`	Return deep copies of cached frames to prevent later mutations from contaminating reused inputs. False changes memory/copy behavior and requires care with shared dataframes.
`transport_investment_limits_active` (`main.py:155`)	`False`	Activate the applicable extra transport investment-limit policy. Vehicle service shares, stock turnover and electrification targets are independent restrictions.
`transport_pathway_corrections_active` (`main.py:156`)	`True`	Apply fuel compatibility, paid physical conversions and supported drop-in fuel rules. Turning this off is not a calibrated unrestricted-conversion scenario.
`transport_passenger_car_zero_fossil_2050_sensitivity` (`main.py:157`)	`True`	Apply the named zero-fossil passenger-car sensitivity. This is not automatically zero fossil use in every transport sector.
`transport_allow_unconstrained_paid_upgrades_for_test` (`main.py:160`)	`False`	Test-only bypass for persisted paid-upgrade rows when pathway constraints are disabled. Do not use it for a reported policy case without establishing the missing stock/fuel constraints.
`transport_no_base_year_vehicle_investment` (`main.py:161`)	`False`	Prevent base-year fleet additions in the applicable ordinary case. E-fuel mode overrides this to False to allow its nonzero initial electrification floors.
`transport_max_share_limits_active` (`main.py:164`)	`True`	Enforce annual service ceilings from transport_all.max_share. A service-share ceiling is different from a vehicle-stock share.
`transport_ice_hybrid_stock_share_non_increasing_active` (`main.py:169`)	`False`	Add per-family ICE/hybrid stock-share nonincrease. The delivered case leaves this off because aggregate electrification/history constraints already restrict transition and tiny redundant anchors can harm conditioning.
`transport_electrification_limits_active` (`main.py:170`)	`True`	Apply selected electrification target paths. Requires the applicable sector/technology mapping and workbook targets.
`transport_rail_electrification_link_active` (`main.py:171`)	`False`	Apply the optional linkage between rail electrification representations. Independent of the direct-electric share ceiling and battery-rail specifications.
`transport_rail_direct_electric_share_limit_active` (`main.py:172`)	`True`	Cap direct-electric service for passenger and freight rail under the configured maximum share. It does not impose that share as a minimum.
`rolling_auto_carbon_budget_slack` (`main.py:173`)	`True`	Allow the configured penalized automatic rolling carbon-budget recovery. Nonzero slack means the strict carbon allocation was relaxed and must be reported.
`rolling_terminal_fuel_soc` (`main.py:174`)	`True`	Apply terminal/end-of-horizon fuel-inventory guidance or targets. This refers to end conditions, not only physical maritime terminals.

Setting and source	Entered value / expression	Meaning and effect
`rolling_terminal_fuel_soc_hard_balance` (`main.py:175`)	`False`	Use hard end-inventory constraints in the applicable rolling branch. False permits the configured soft treatment; check actual slack use.
`rolling_terminal_fuel_soc_link_sequential_years` (`main.py:176`)	`True`	Carry/coordinate terminal fuel-inventory targets across sequential years in the rolling workflow. Distinguish this from independent cyclic annual storage.
`rolling_terminal_fuel_soc_ramp_targets` (`main.py:177`)	`True`	Ramp toward terminal inventory targets over the configured lead interval. Controls guidance timing rather than physical storage capacity.
`rolling_resume_from_window_files` (`main.py:178`)	`False`	Reuse compatible solved rolling windows after interruption. Requires saved files, consistent metadata and coherent incoming stocks/budgets.
`rolling_zero_stage1_fuel_soc0` (`main.py:179`)	`True`	Prevent representative planning storage capacity from receiving free initial fuel inventory. This is an energy-accounting boundary condition.
`rolling_fuel_soc_stage1_floor` (`main.py:180`)	`False`	Permit the applicable stage-1 SOC floor guidance in rolling operation. Leaving it off avoids lifting chronological carryover to an unearned representative-week stock.
`stage2_quantity_guidance_active` (`main.py:181`)	`True`	Guide stage-2 annual/rolling quantities toward stage-1 results with configured tolerance and penalties. Guidance is not identical to an unconstrained chronological reoptimization.
`oil_refinery_product_generation_decline_active` (`main.py:182`)	`False`	Apply the extra refinery-product output-decline policy. Separate from product reserves, carbon limits and outside export permission.
`transport_ice_hybrid_fossil_share_decline_active` (`main.py:183`)	`False`	Apply the extra ICE/hybrid fossil-fuel share-decline policy in its configured scope. Compatible non-fossil fuel operation can differ from retiring the entire vehicle.
`transport_ice_fuel_share_ramp_active` (`main.py:184`)	`True`	Limit changes in ICE fuel composition using the entered change limit. This acts alongside conversion/blend compatibility.
`biomass_heating_nonincrease_active` (`main.py:185`)	`True`	Prevent applicable annual biomass-heating expansion. It is a scenario restriction, not a statement that biomass cannot serve existing demand.
`biomass_feedstock_limit_active` (`main.py:186`)	`True`	Apply the scoped biomass feedstock/resource limits. Do not equate carbon-neutral bookkeeping with unlimited sustainable supply.
`model_efuel` (`main.py:189`)	`True`	Enable the e-fuel carrier, conversion/feedstock, storage and trade representation and select e-fuel input families. Use the author's integrated recipe with explicit Oil/Gas supply and matching CO2 handling; production remains an economic decision.

Setting and source	Entered value / expression	Meaning and effect
`complex_industry_dat` (`main.py:196`)	`complex_industry_dat_efuel if model_efuel else complex_industry_dat_no_efuel`	Derived active companion process path after e-fuel and electricity-case selection. Do not edit this while leaving conflicting selectors unchanged.
`use_HVDC` (`main.py:206`)	`False`	Include eligible HVDC electricity links and converter requirements. HVDC is a network technology under transmission, not a separate energy sector or a switch for all cross-border trade.
`model_terminals` (`main.py:207`)	`False`	Enable terminals. With e-fuels, use the product-terminal/route tables; without e-fuels, use the standalone H2 terminal workbook and H2 balance. Activate individual assets and permitted flow directions in the applicable tables.
`h2_terminal_dat` (`main.py:208`)	`'./data/h2_terminals.xlsx'`	Standalone no-e-fuel H2 terminal workbook. Requires its terminals/routes schema, eligible active geography and H2_terminal cost years.
`model_H2_transmission` (`main.py:209`)	`False`	Enable standalone H2 pipelines/compressors when e-fuels are off. Main disables this flag with e-fuels because carrier transmission already handles H2 there.
`model_oil_and_gas_products_supply` (`main.py:210`)	`True`	Enable explicit Oil/Gas/refinery-product supply in the e-fuel representation. Main disables it without e-fuels. Enable it for the author's supplied integrated e-fuel recipe; review reserves, imports, storage and processing inputs.
`allow_fossil_fuel_exports` (`main.py:214`)	`True`	Allow eligible fossil refinery products to leave the modeled systems. False prevents that outside revenue channel while retaining applicable internal deliveries.
`exogeneous_NH3_demand` (`main.py:215`)	`False`	Add prescribed ammonia demand in e-fuel mode. Requires matching nh3_demand.csv and year/geography policy; endogenous ammonia feedstock demand is distinct.
`non_CO2_polutants` (`main.py:217`)	`False`	Request an explicitly incomplete legacy pollutant branch. It is not a complete non-CO2 or lifecycle environmental assessment.
`model_H2_supply_demand_balance` (`main.py:218`)	`True`	Request the standalone H2 supply-demand balance used by eligible non-e-fuel configurations. Other integrated flags can also build H2 balances; False does not necessarily erase every H2 term.
`model_CC` (`main.py:245`)	`True`	Enable eligible industrial capture/stripper/reboiler equipment and constraints. Capture capacity, active rows, energy inputs and a CO2 destination are still required. DAC in e-fuel mode is a distinct source.
`model_CO2_ammine_storeage` (`main.py:246`)	`True`	Enable capture-linked rich/lean-amine inventory flexibility. The historical misspelling is the actual configuration name. Capture off normalizes this option off.

Setting and source	Entered value / expression	Meaning and effect
`model_CO2_post_gen_storeage` (`main.py:247`)	`True`	Enable post-capture CO2 capacity/inventory handling where capture/e-fuels permit it. Geological storage additionally requires eligible active storage rows, capacity/build limits and allowed provenance.
`complex_industry_process_capture_required` (`main.py:248`)	`False`	Require the capturable share from eligible complex process sources to be captured. False lets budgets/economics decide; True does not remove uncapturable residual emissions.
`separate_industry_process_carbon_balances` (`main.py:249`)	`True`	Track process and other carbon separately and restrict geological injections to eligible captured process carbon. DAC/combustion/imported CO2 cannot acquire process provenance merely by entering a common buffer.
`DSM_day` (`main.py:259`)	`False`	Enable the daily basic-load shifting class. Combine with budget/rollout controls for the new constrained DSM formulation; does not automatically shift all sector-specific loads.
`DSM_week` (`main.py:260`)	`False`	Enable weekly basic-load shifting. With both classes and budgets enabled, the weekly pool is a subset of the daily-capable potential, not an additional unrestricted allowance.
`dsm_energy_budget_active` (`main.py:261`)	`False`	Apply moved-MWh budgets per daily/weekly block, counted once by half the absolute adjustment. These are separate from hourly power limits.
`dsm_availability_ramp_active` (`main.py:262`)	`False`	Scale DSM energy availability from the global base year toward its configured full-availability year. Turning it off removes that rollout restriction, not every DSM bound.
`shadow_prices` (`main.py:265`)	`False`	Request the applicable marginal/shadow-price export workflow. An LP solution without an accepted basis/dual treatment is not automatically an interpretable market-price forecast.
`solver_stage1_auto_retry_suboptimal` (`main.py:275`)	`False`	Allow the optional stronger generic retry for problematic stage-1 outcomes. Off in the delivered case; the separately enabled annual aggregation retry can still operate.
`solver_stage1_annual_pf_aggregation_retry` (`main.py:276`)	`True`	Retry eligible annual-myopic problems using the configured alternate aggregation setting. Despite its name it applies with or without partial foresight in this snapshot.
`solver_allow_provisional_solution` (`main.py:277`)	`True`	Permit acceptance of an eligible non-certified/provisional point within configured checks. True does not convert SUBOPTIMAL or TIME_LIMIT into OPTIMAL.
`solver_postoptimal_checks_warn_only` (`main.py:278`)	`True`	Let applicable postoptimal balance/bound/diagnostic checks warn instead of abort. For a stricter study set False and inspect the independent annual-CC gate and all remaining slack mechanisms.

Setting and source	Entered value / expression	Meaning and effect
`solver_enable_dual_simplex_fallback` (`main.py:279`)	`True`	Permit Method=1 recovery after the applicable failed/poor barrier outcome. Recovery can have its own focus, threads and unlimited time limit.
`solver_finalize_stalled_barrier_with_crossover` (`main.py:280`)	`False`	Attempt crossover finalization after an eligible barrier stall. It changes recovery behavior, not the physical model.
`solver_skip_dual_simplex_after_acceptable_stall` (`main.py:281`)	`False`	Skip the simplex recovery after an accepted stall in the applicable path. Leaving False preserves the fallback attempt.
`write_solver_diagnostics` (`main.py:282`)	`True`	Write available scale/quality diagnostic artifacts. This does not itself enforce strict acceptance.
`solver_diagnostics_on_problem_only` (`main.py:283`)	`True`	Restrict expensive diagnostics to problematic cases. False can substantially increase time and files on otherwise clean runs.
`solver_diagnostics_compute_kappa_exact` (`main.py:284`)	`False`	Opt into exact conditioning calculations. These can be very expensive and are not a prerequisite for an ordinary run.
`disable_transport_infrastructure` (`main.py:285`)	`False`	Inverted debug control: False retains fleet-linked EV charging/H2 refueling stock and cost constraints; True suppresses them and changes the study boundary.
`transport_infrastructure_dat` (`main.py:286`)	`'./data/transport_infrastructure.csv'`	Technology/cost/connection requirements for fleet-linked transport infrastructure.
`transport_infrastructure_existing_dat` (`main.py:287`)	`'./data/transport_infrastructure_existing.csv'`	Existing infrastructure stock by active geography. Keep it consistent with the fleet and technology input conventions.
`fuel_price_dat` (`main.py:291`)	`'./data/fuel_prices.csv'`	Primary fuel-price input; distinguish it from e-fuel policy carrier prices and electricity trade prices.
`inflows_dat` (`main.py:292`)	`'./data/hydro_inflows.csv'`	Hydro inflow profiles using exact reservoir/pumped-hydro IDs and the selected simple_curves schema.
`heat_demand_dat_indv` (`main.py:293`)	`'./data/heating_individual.csv'`	Individual heating service profiles. Annual growth is applied through the applicable policy paths.
`cool_demand_dat_indv` (`main.py:294`)	`'./data/cooling_individual.csv'`	Individual cooling profiles for the optional cooling representation.
`heat_demand_dat_DH` (`main.py:295`)	`'./data/heating_district.csv'`	District-heating service profiles; use the expected delivered/network-loss boundary.
`cool_demand_dat_DH` (`main.py:296`)	`'./data/cooling_district.csv'`	District cooling service profiles for the optional cooling block.
`ev_transpload_dat` (`main.py:297`)	`'./data/transport_profiles.csv'`	Transport activity, plug-in and V2G timing input. These are not interchangeable with annual service quantities.
`transploaddemand_dat` (`main.py:298`)	`'./data/transport_demand.csv'`	Annual transport-service demand by the native sector/geography keys.
`importcost_dat` (`main.py:299`)	`'./data/import_prices.csv'`	Electricity import-price input for the selected trade formulation. Fuel purchases have separate prices.

Setting and source	Entered value / expression	Meaning and effect
`heat_dem_industry_dat` (`main.py:300`)	`'./data/industry_heat.csv'`	Industry useful-service timing path. Later assignment selects category profiles when categories are enabled, otherwise industry_heat.csv.
`genco_dat` (`main.py:301`)	`'./data/generators.csv'`	Generator inventory with exact asset names, stock, location, efficiency, retirement, CHP and storage parameters.
`policy_dat` (`main.py:302`)	`'./data/policy_efuel.xlsx' if model_efuel else './data/policy.xlsx'`	Derived choice of policy_efuel.xlsx or policy.xlsx. The workbook horizon and trajectories must match the selected technology representation.
`demand_dat` (`main.py:307`)	`'./data/electricity_demand.csv'`	Basic electricity demand by period/System/Zone. Sector-accounting adjustments must avoid double counting.
`demand_industry_el` (`main.py:308`)	`'./data/industry_electricity.csv'`	Industry electricity baseline/accounting input used by the reader's sector decomposition.
`demand_heating_el` (`main.py:309`)	`'./data/heating_electricity.csv'`	Heating electricity baseline/accounting input used by the reader's sector decomposition.
`solar_thermal_aval` (`main.py:310`)	`'./data/solar_thermal_availability.csv'`	Solar-thermal availability input. It is separate from renewable generator/PV availability.
`flex_tech_name` (`main.py:315`)	`'flex_tech'`	Initial basename used to construct the technology workbook. Flags append e-fuel/category/complex suffixes before reading.
`flex_tech_directory` (`main.py:316`)	`'./data'`	Technology directory; the e-fuel category technical case changes it to the nested variants/technical_2050 folder.
`flex_tech_dat` (`main.py:329`)	`flex_tech_directory + '/' + flex_tech_name + '.xlsx'`	Derived final technology-workbook path. Check the reader's selected path rather than assuming the open Excel tab is active.
`heat_dem_industry_dat` (`main.py:331`)	`'./data/industry_category_profiles.csv' if model_industry_categories else './data/industry_heat.csv'`	Industry useful-service timing path. Later assignment selects category profiles when categories are enabled, otherwise industry_heat.csv.
`el_hourly_cost_dat` (`main.py:343`)	`'./data/electricity_prices.csv'`	Hourly electricity price input for applicable cost/trade operations; keep its period and geography schema consistent.
`_data_read_cache` (`main.py:370`)	`globals().setdefault('_data_read_cache', OrderedDict())`	Internal ordered cache created at runtime. Do not use it as a user-facing scenario parameter.
`_excel_file_cache` (`main.py:374`)	`globals().setdefault('_excel_file_cache', OrderedDict())`	Internal cached Excel reader handles. Do not change it to select another workbook.
`_data_read_cache_stats` (`main.py:378`)	`globals().setdefault('_data_read_cache_stats', {'hits': 0, 'misses': 0, 'read_time': 0.0})`	Internal hit/miss/read-time counters. They describe local file-reading performance, not API requests or model feasibility.

policy/model_policy.py

Setting and source	Entered value / expression	Meaning and effect
`generation_annual_build_limit_multipliers` (policy/model_policy.py:11)	`{'HR_Wind': 2.0, 'HR_Wind_1': 2.0, 'HR_PV': 2.0, 'HR_PV_1': 2.0}`	Multiply scoped generator annual build allowances by exact unit ID. Values above one permit faster additions, not higher weather availability or free capacity.
`dh_chp_heat_max_annual_change` (policy/model_policy.py:19)	`0.25`	Relative annual CHP heat-change allowance; 0.25 denotes 25% under the implemented historical/interval rule.
`dh_chp_heat_reduction_limit_years` (policy/model_policy.py:20)	`10`	Number of years over which the extra CHP heat-reduction restriction is applied; separate from plant technical lifetime.
`dh_chp_heat_history_tolerance_mwh` (policy/model_policy.py:21)	`1.0`	MWh numerical allowance when comparing saved/current annual CHP heat. Do not use it as an intended heat-supply slack.
`individual_gas_heat_max_annual_change` (policy/model_policy.py:22)	`0.25`	Relative annual individual gas-boiler heat-change allowance.
`individual_gas_heat_reduction_limit_years` (policy/model_policy.py:23)	`5`	Duration of the extra individual-gas heat reduction restriction.
`fossil_heating_history_tolerance_mwh` (policy/model_policy.py:24)	`1.0`	MWh tolerance on historical fossil-heating comparisons, not a policy exemption.
`fossil_boiler_phaseout_years` (policy/model_policy.py:25)	`{'coal': 2030, 'gas': 2035, 'oil': 2035}`	Fuel-to-calendar-year boiler phaseout mapping. The delivered coal/gas/oil years are scenario choices; CHP is handled separately.
`dh_chp_base_year_heat_targets_twh` (policy/model_policy.py:26)	`{('TE-TO_Zagreb', 'HR1', 'HR'): 0.9, ('EL-TO_Zagreb', 'HR1', 'HR'): 0.55, ('TE_-TO_Osijek', 'HR1', 'HR'): 0.232}`	Exact CHP asset/zone/System base-year heat targets in TWh. Requires matching identifiers and calibration enabled.
`lulucf_removals_fixed_tco2e_per_year` (policy/model_policy.py:32)	`0.0`	Fixed annual removal allowance in tCO2-equivalent for the supported accounting path. Default zero; inspect its relationship to CSV/scoped removals before changing it.
`electric_heating_max_annual_decrease` (policy/model_policy.py:33)	`0.1`	Relative annual reduction allowance on the configured individual electric-heating service scope.
`electric_heating_max_annual_increase` (policy/model_policy.py:34)	`0.1`	Relative annual increase allowance on the configured individual electric-heating scope.
`electric_heating_week_hours` (policy/model_policy.py:35)	`24 * 7`	Block length for applicable electric-heating history comparisons; 168 hours is a week, not a representative sample count.
`electric_heating_individual_units` (policy/model_policy.py:36)	`('Air_to_water_HP', 'Ground_to_water_HP', 'Electric_heater', 'Water_to_water_HP')`	Exact technologies included in the individual electric-heating transition policy.
`industry_fuel_share_max_change_per_year` (policy/model_policy.py:44)	`0.1`	General fuel/service share-change allowance in the selected ramp formulation. Its interpretation differs from an equipment MW build limit.
`industry_fuel_share_ramp_mode` (policy/model_policy.py:45)	`'endogenous_s_curve'`	Select the supported flat/endogenous_s_curve transition interpretation, subject to main switches. This is a scenario adoption rule.

Setting and source	Entered value / expression	Meaning and effect
`industry_fuel_share_s_curve_initial_fraction` (`policy/model_policy.py:46`)	`0.02`	Seed fraction for adoption envelopes; not an additional physical service demand or measured adoption rate.
`industry_fuel_share_s_curve_growth_rate` (`policy/model_policy.py:47`)	`0.5`	Growth parameter in the adoption envelope. Increasing it relaxes the permitted transition speed under that formula.
`industry_fuel_share_s_curve_reference_years` (`policy/model_policy.py:51`)	`5`	Reference interval used so annual versus sparse decisions do not repeatedly restart seeds/tolerances. Must meet the helper's integer/grid compatibility checks.
`industry_fuel_share_s_curve_completion_year` (`policy/model_policy.py:55`)	`2045`	Finite transition deadline. Default 2045 completes the applicable fossil-combustion phaseout envelope; None retains the historical open-ended treatment.
`industry_fuel_share_s_curve_directions` (`policy/model_policy.py:59`)	`{'Gas': 'decrease', 'Coal': 'decrease', 'Lignite': 'decrease', 'Oil': 'decrease', 'fuel_oil': 'decrease', 'diesel': 'decrease', 'petrol': 'decrease', 'kerosene': 'decrease', 'LPG': 'decrease', 'Biomass': 'decrease', 'lignin': 'increase'}`	Explicit increase/decrease/auto directions by asset tuple, unit or fuel. Applicability ceilings remain separate; asset/unit overrides take precedence.
`industry_electric_share_max_annual_decrease` (`policy/model_policy.py:66`)	`0.1`	Relative maximum decrease of combined direct-electric/HP useful-service share; 10% of a 40% share gives 36%, not 30%.
`industry_electric_share_history_tolerance` (`policy/model_policy.py:67`)	`1e-06`	Small share-unit comparison allowance. It is not a percentage-point adoption budget.
`transport_decom_max_share` (`policy/model_policy.py:70`)	`None`	Optional additional transport decommissioning-share parameter. None leaves the implemented default treatment; do not infer unlimited vehicle replacement without the other turnover constraints.
`transport_turnover_years` (`policy/model_policy.py:71`)	`25`	Reference fleet turnover life used for the selected vehicle groups.
`transport_turnover_units` (`policy/model_policy.py:72`)	`('EV_car', 'FCEV_car', 'HYBRID_car', 'ICE_car', 'EV_truck', 'FCEV_truck', 'HYBRID_truck', 'ICE_truck', 'EV_MARINE_SHORT', 'FCEV_MARINE_SHORT', 'ICE_MARINE_SHORT')`	Exact vehicle/transport units included in the extra turnover policy.
`transport_turnover_ramp_start_year` (`policy/model_policy.py:85`)	`2025`	Start calendar year for the turnover-policy ramp.
`transport_turnover_ramp_full_year` (`policy/model_policy.py:86`)	`2030`	Calendar year when the turnover ramp reaches its full setting.
`transport_turnover_ramp_initial_share` (`policy/model_policy.py:87`)	`1.5`	Initial multiplier/share used by the turnover ramp; inspect its formula before interpreting a value above one.
`TRANSPORT_TURNOVER_MAX_SHARE` (`policy/model_policy.py:88`)	`0.3`	Upper turnover-policy share in the applicable implementation; separate from a vehicle's transport_all service ceiling.
`transport_existing_fleet_age_spread_years` (`policy/model_policy.py:89`)	`10`	Age-spread assumption for distributing retirement of inherited fleet stock.

Setting and source	Entered value / expression	Meaning and effect
`transport_forbidden_investment_units` (policy/model_policy.py:90)	`()`	Exact units prohibited from receiving new investment by the applicable transport policy. An empty tuple imposes no exclusions here.
`transport_inv_max_share_by_unit` (policy/model_policy.py:91)	`{'all': 0.2, 'EV_car': 0.42, 'FCEV_car': 0.25, 'ICE_AVIO_SHORT': 0.4, 'ICE_AVIO_LONG': 0.4, 'EL_RAIL': 0.4, 'EL_RAIL_CARGO': 0.4, 'ICE_MARINE_LONG': 0.4, 'ICE_MARINE_SHORT': 0.2}`	Unit-specific new-investment share ceilings, with the supported all fallback. Does not directly set final service shares.
`transport_inv_total_max_by_sector` (policy/model_policy.py:102)	`{}`	Absolute new-fleet investment caps by sector. An empty mapping supplies no extra sector-specific caps here.
`transport_inv_max_ramp_start_year` (policy/model_policy.py:103)	`2025`	Start year for investment-share ceiling rollout.
`transport_inv_max_ramp_full_year` (policy/model_policy.py:104)	`2028`	Year when investment-share ceilings reach their full configured values.
`transport_inv_max_ramp_initial_share` (policy/model_policy.py:105)	`0.5`	Initial rollout fraction for the applicable investment ceilings.
`transport_inv_declining_max_share_by_unit` (policy/model_policy.py:106)	`{}`	Optional units whose investment ceilings decline along the configured path.
`transport_inv_decline_start_year` (policy/model_policy.py:107)	`2025`	Start year of optional declining transport investment ceilings.
`transport_inv_decline_end_year` (policy/model_policy.py:108)	`2050`	Final year of optional declining investment ceilings.
`transport_inv_decline_final_share` (policy/model_policy.py:109)	`0.1`	Endpoint fraction for the configured declining investment ceiling, not an automatic zero-fossil rule.
`transport_stock_max_share_year` (policy/model_policy.py:110)	`2050`	Reference year for optional stock-share ceilings.
`transport_stock_max_share_by_unit` (policy/model_policy.py:116)	`{}`	Optional final/target stock-share caps by unit. Empty supplies no additional caps in this mapping.
`transport_electrification_target_sectors` (policy/model_policy.py:117)	`('Passenger_cars', 'commercial_vehicles', 'Motor_coaches_buses_and_trolley_buses', 'Rail_transport', 'Rail_metro_and_tram', 'Domestic_shipping', 'Domestic_aviation', 'International_aviation')`	Sectors to which the selected electrification target policy applies. International shipping is not automatically included because aviation is listed.
`transport_non_decreasing_stock_units` (policy/model_policy.py:127)	`('EL_RAIL', 'EL_RAIL_CARGO')`	Exact transport units with the extra nondecreasing-stock requirement.
`transport_rail_direct_electric_max_share` (policy/model_policy.py:133)	`0.9`	Maximum direct-electric rail service share in the separate passenger/freight cap; default 0.9 means a ceiling of 90%.
`transport_rail_battery_specs` (policy/model_policy.py:136)	`{'BAT_RAIL': {'average_stor': 700, 'average_ch_rate': 1000}, 'BAT_RAIL_CARGO': {'average_stor': 1400, 'average_ch_rate': 2000}}`	Battery-rail storage/charging specifications by unit, using native equipment units. Distinct from overhead/direct electric rail.

Setting and source	Entered value / expression	Meaning and effect
`stage2_carbon_budget_mode` (`policy/model_policy.py:142`)	`'cumulative_weighted'`	Rule for allocating carbon across rolling operation. The supplied cumulative_weighted mode uses cumulative budgets with configured weights, not independent unconstrained hourly caps.
`stage2_carbon_hour_weight` (`policy/model_policy.py:143`)	`0.25`	Contribution of uniform time to the applicable rolling carbon-allocation weights.
`stage2_carbon_heat_weight` (`policy/model_policy.py:144`)	`0.75`	Contribution of heat activity to rolling carbon-allocation weights.
`stage2_carbon_el_weight` (`policy/model_policy.py:145`)	`0.0`	Contribution of electricity activity to rolling carbon-allocation weights. Zero removes this weighting component, not electricity emissions.
`stage2_carbon_auto_slack_cost` (`policy/model_policy.py:146`)	`1000000.0`	Penalty for automatically introduced rolling carbon-budget slack. The native cost scale must be preserved when reporting it.
`stage2_carbon_slack_cost` (`policy/model_policy.py:147`)	`0.0`	Additional configured rolling carbon-slack cost. Zero here does not imply the auto-slack mechanism is cost-free.
`rolling_refinery_hourly_cap_multiplier` (`policy/model_policy.py:148`)	`1.0`	Scale the applicable stage-2 refinery hourly capacity allowance. It is a dispatch restriction, not a change in resource reserves.
`rolling_h2_balance_shortage_cost` (`policy/model_policy.py:149`)	`10000000.0`	Penalty on supported rolling H2 shortage variables. Nonzero shortage indicates relaxed service/balance requirements.
`rolling_industry_cap_shortage_cost` (`policy/model_policy.py:150`)	`10000000.0`	Penalty for supported rolling industry-capacity shortfalls. Report the slack rather than treating it as physical capacity.
`rolling_terminal_fuel_soc_tolerance` (`policy/model_policy.py:151`)	`0.0`	Allowed end-inventory target deviation in the applicable fuel-stock guidance rule.
`rolling_terminal_fuel_soc_slack_cost` (`policy/model_policy.py:152`)	`10000000.0`	Penalty for fuel terminal/end-stock slack. Inspect native stock and objective units before changing it.
`rolling_terminal_fuel_soc_lead_hours` (`policy/model_policy.py:153`)	`8760`	Lead time over which terminal fuel-stock targets begin influencing dispatch.
`rolling_terminal_fuel_soc_ramp_exponent` (`policy/model_policy.py:154`)	`1.0`	Shape exponent of the terminal-stock target ramp; does not change the underlying stock balance.
`storage_year_continuity_mode` (`policy/model_policy.py:155`)	`'cyclic'`	Select the supported annual storage boundary convention. The delivered cyclic setting should not be described as unrestricted interannual stock carryover.
`efuel_dh_heat_feedstock_aliases` (`policy/model_policy.py:156`)	`('heat', 'dh_heat', 'district_heat', 'district heating')`	Labels recognized as district-heat feedstock to e-fuel conversion. Such feedstock is served by the heat balance, not a free exogenous input.
`stage2_quantity_guidance_tolerance` (`policy/model_policy.py:157`)	`0.15`	Relative band around stage-1 quantity guidance in stage 2.
`stage2_quantity_guidance_abs_tolerance` (`policy/model_policy.py:158`)	`1e-06`	Absolute comparison allowance accompanying the relative guidance band.

Setting and source	Entered value / expression	Meaning and effect
`stage2_quantity_guidance_slack_cost` (`policy/model_policy.py:159`)	`1000000.0`	Penalty on allowed quantity-guidance deviations. Guidance compliance and physical feasibility are different checks.
`oil_refinery_product_generation_change_limit` (`policy/model_policy.py:160`)	`0.1`	Rate limit used when the extra refinery output-decline control is enabled.
`refinery_process_fuel_share` (`policy/model_policy.py:161`)	`0.08`	Refinery process-energy fraction, default 0.08. This is additional process demand, not product output efficiency by itself.
`refinery_process_fuel_co2_key` (`policy/model_policy.py:162`)	`'Oil'`	Fuel/emission-factor key used for refinery process combustion.
`transport_ice_hybrid_fossil_share_decline_scope` (`policy/model_policy.py:163`)	`'annual'`	Scope of the optional ICE/hybrid fossil-share decline rule; annual is the delivered setting.
`transport_ice_fuel_share_change_limit` (`policy/model_policy.py:165`)	`0.2`	Change allowance for ICE fuel composition when the corresponding main ramp is enabled.
`biomass_heating_fuel_aliases` (`policy/model_policy.py:166`)	`('biomass',)`	Fuel labels recognized by biomass-heating restrictions.
`h2_transmission_cost_aliases` (`policy/model_policy.py:169`)	`{'H2_pipe': 'H2CH'}`	Map standalone H2 pipeline technologies to supported cost keys. Does not add missing technical or geographic assets.
`transmission_transformer_units` (`policy/model_policy.py:170`)	`{'400kV_AC': 'transformer_AC', 'HVDC': 'transformer_HVDC'}`	Map electricity link technologies to converter/transformer cost and equipment keys, including HVDC.
`h2_pipeline_compressor_unit` (`policy/model_policy.py:174`)	`'comp_station'`	Technology name for standalone H2 compressor requirements.
`transmission_duplicate_policy` (`policy/model_policy.py:175`)	`'first'`	How duplicate link records are reconciled. Delivered first keeps the first applicable record; deterministic selection is not proof conflicting inputs are correct.
`active_zone_overrides` (`policy/model_policy.py:177`)	`None`	Optional explicit active-geography override. None retains normal workbook/configuration scoping; demands and links must still cover any new active zones.
`simple_industry_active_zones` (`policy/model_policy.py:180`)	`(('HR1', 'HR'), ('RS1', 'RS'), ('BA1', 'BA'))`	Active zone/System pairs when the separate product module is configured. The delivered HR1/RS1/BA1 example is different from an arbitrary NUTS3 case.
`simple_industry_grid_fallback_tco2_per_mwh` (`policy/model_policy.py:183`)	`0.6`	Fallback electricity intensity used by applicable product-industry reporting. Distinguish fallback reporting factors from physical modeled territorial generation emissions.
`electricity_cbam_future_intensity_tco2_per_mwh` (`policy/model_policy.py:186`)	`{}`	Optional exogenous future zone/year factors for electricity CBAM lookahead. Missing required realized history must not be silently replaced by zero.
`co2_storage_capex_input_unit` (`policy/model_policy.py:189`)	`'MEUR/t'`	Declared geological/buffer storage cost basis. MEUR/t is a stock-capacity cost, not MEUR per tonne/hour of injection.
`co2_amine_tank_capex_input_unit` (`policy/model_policy.py:190`)	`'MEUR/t'`	Declared amine inventory-capacity cost basis. Keep tonne-stock and hourly capture equipment costs separate.

Setting and source	Entered value / expression	Meaning and effect
`stat_grid_eff` (`policy/model_policy.py:191`)	`0.9`	Generic stationary electricity-storage/grid delivery-efficiency parameter in its reader/balance convention. Do not apply it twice to stock and delivered output.
`ceep_price` (`policy/model_policy.py:192`)	`1000`	Penalty for surplus electricity in the implemented objective. It can remain active when ceep_limit=False.
`noload_el_cost` (`policy/model_policy.py:195`)	`100000`	Penalty for unserved electricity. It is an entered scarcity parameter, not a measured national value of lost load.
`discount_rate` (`policy/model_policy.py:196`)	`0.05`	Economic discount rate for cross-year objective weighting. Different from capital financing and numerical objective scaling.
`discount_base_year` (`policy/model_policy.py:197`)	`2025`	Reference year for discount factors. Changing it changes the reported present-value basis.
`capital_finance_rate` (`policy/model_policy.py:198`)	`0.05`	Finance rate for annual capital-service payments over equipment life.
`fuel_storage_capex_aliases` (`policy/model_policy.py:204`)	`{'N2_stor': 'Gas_stor'}`	Explicit cost-key fallbacks for carrier stores; default N2_stor uses Gas_stor costs as an inherited assumption.
`fuel_storage_fixed_opex_fraction` (`policy/model_policy.py:205`)	`0.01`	Annual fixed-cost fraction for supported fuel storage equipment, relative to its configured capital basis.
`oil_gas_initial_soc_fraction` (`policy/model_policy.py:211`)	`{'Oil': 0.5, 'Gas': 0.5}`	Initial Oil/Gas inventory fractions under the supported supply boundary. Check zero-initial-stage-1 and rolling-stock rules before assuming the fraction is always applied.
`industry_electrification_full_applicability` (`policy/model_policy.py:215`)	`True`	Use the supported full-electrification applicability setting while retaining other process, transition, resource and capacity limits. It does not mandate 100% electrification.
`industry_heat_pumps_active` (`policy/model_policy.py:216`)	`True`	Enable eligible industrial HP technologies and their shared applicability/source budgets. Independent of building HP switches.
`industry_hp_profiles_dat` (`policy/model_policy.py:217`)	`'./data/industry_category_profiles.csv'`	Useful-service profile source used for industrial HP applicability/demand checks.
`complex_industry_sectors` (`policy/model_policy.py:223`)	`('Non_metallic_minerals',)`	Exact selected complex process branches. Only these replace corresponding legacy service demand; delivered default is Non_metallic_minerals.
`industry_external_steam_efficiency` (`policy/model_policy.py:226`)	`0.9`	Conversion assumption used when externally supplied industry steam/heat is attributed to input energy and emissions. Do not treat imported heat as automatically emission-free.

policy/runtime_policy.py

Setting and source	Entered value / expression	Meaning and effect
`dsm_availability_full_year` (`policy/runtime_policy.py:5`)	`2030`	Calendar year at which the enabled linear DSM rollout reaches its target; based on the original global base year.
`dsm_daily_energy_fraction` (`policy/runtime_policy.py:6`)	`0.07`	Daily-capable moved-energy pool including the weekly subset. It is a block-energy ceiling, not an hourly MW share.
`dsm_weekly_energy_fraction` (`policy/runtime_policy.py:7`)	`0.015`	Weekly-capable subset of the daily pool when both classes are enabled. The two values must not be added as independent full potentials.
`dsm_shift_fraction` (`policy/runtime_policy.py:8`)	`0.1`	Shared hourly absolute adjustment power cap as a fraction of baseline electricity load.
`dsm_capacity_cost_eur_per_mw_year` (`policy/runtime_policy.py:9`)	`10000.0`	Annual enabling resource cost per MW of available DSM adjustment, in EUR/MW/year.
`dsm_shift_cost_eur_per_mwh` (`policy/runtime_policy.py:10`)	`5.0`	Real activation/discomfort cost per MWh moved, counted once, in EUR/MWh.
`partial_foresight_representative_days` (`policy/runtime_policy.py:15`)	`4`	Number of future-year sampled blocks in ordinary partial foresight.
`partial_foresight_day_hours` (`policy/runtime_policy.py:16`)	`24`	Hours per partial-foresight representative block; delivered 24.
`partial_foresight_representative_day_start_hours` (`policy/runtime_policy.py:17`)	`(337, 2521, 4705, 6889)`	Explicit first-hour positions of future-year blocks. Keep them valid, nonoverlapping and compatible with the configured block count/length.
`partial_foresight_lookahead_years` (`policy/runtime_policy.py:20`)	`10`	Calendar-year reach of each ordinary partial-foresight solve. None exposes the remaining horizon; 10 gives current year plus decision points no later than current+10.
`representative_full_year_hours` (`policy/runtime_policy.py:21`)	`8760`	Full-year hours used in representative annualization. The delivered 8760 convention is not automatically leap-year data support.
`two_stage_stage1_export_timing` (`policy/runtime_policy.py:22`)	`'before_stage2'`	When stage-1 outputs are written. before_stage2 preserves capacity/history exports; off/skip/none can deprive later myopic chunks of expected investment CSVs.
`representative_weeks_count` (`policy/runtime_policy.py:23`)	`4`	Number of representative capacity-stage blocks. Despite its historical name, interpret with representative_week_hours.
`representative_week_hours` (`policy/runtime_policy.py:25`)	`24`	Length of each representative capacity block. Delivered 24, so four blocks are four days, not four weeks.
`representative_week_start_hours` (`policy/runtime_policy.py:26`)	`partial_foresight_representative_day_start_hours`	Start positions of representative capacity-stage blocks. Delivered aliases the four seasonal-day starts.
`long_storage_guidance_default_band` (`policy/runtime_policy.py:27`)	`0.0`	Default allowed band for applicable long-duration storage guidance; zero requests the corresponding tight target band.

Setting and source	Entered value / expression	Meaning and effect
`rolling_dispatch_granularity` (`policy/runtime_policy.py:28`)	`'day'`	day sets 48-hour lookahead/24-hour realization; week sets 336/168; custom uses the explicit window/step/realized settings.
`rolling_window_size` (`policy/runtime_policy.py:30`)	`48`	Hours optimized per rolling solve when granularity does not overwrite it.
`rolling_step_size` (`policy/runtime_policy.py:31`)	`24`	Hours by which the rolling origin advances; coordinate with the realized horizon.
`rolling_realized_horizon` (`policy/runtime_policy.py:32`)	`24`	Hours committed/exported from each rolling solve. Later lookahead hours are guidance, not committed full-year results.
`rolling_forecast_avg_dev_percent` (`policy/runtime_policy.py:33`)	`5.0`	Magnitude of the configured rolling forecast-deviation assumption; zero changes that forecast experiment, not perfect foresight automatically.
`stage1_capacity_reserve_margin` (`policy/runtime_policy.py:34`)	`0.0`	Extra capacity margin in the applicable stage-1 sizing constraints; delivered zero. It is not a full reliability/N-1 reserve model.
`rolling_unmet_industry_heat_cost` (`policy/runtime_policy.py:35`)	`1000000000.0`	Penalty for supported rolling industrial-heat shortage. Review physical MWh and actual slack alongside its cost.
`rolling_unmet_dh_heat_cost` (`policy/runtime_policy.py:36`)	`1000000000.0`	Penalty for supported rolling district-heat shortage.
`rolling_unmet_indv_heat_cost` (`policy/runtime_policy.py:37`)	`1000000000.0`	Penalty for supported rolling individual-heat shortage.
`data_read_cache_max_entries` (`policy/runtime_policy.py:38`)	`256`	Maximum retained local read-cache entries; separate from API/download caching and quota estimates.
`solver_stage1_method` (`policy/runtime_policy.py:41`)	`2`	Primary Gurobi algorithm for capacity planning; delivered 2 (barrier). Method selection does not change physical constraints.
`rolling_solver_method` (`policy/runtime_policy.py:42`)	`1`	Primary rolling dispatch algorithm; delivered 1 (dual simplex).
`solver_stage1_numeric_focus` (`policy/runtime_policy.py:45`)	`1`	Numerical-care setting for the ordinary stage-1 solve. Larger values can increase numerical effort and runtime.
`solver_diagnostic_method` (`policy/runtime_policy.py:49`)	`1`	Algorithm used to diagnose ambiguous/failing outcomes; separate from the normal method.
`solver_diagnostic_time_limit_s` (`policy/runtime_policy.py:53`)	`0`	Diagnostic solve time limit in seconds. Nonpositive is treated as unlimited by the applicable wrapper.
`solver_diagnostic_numeric_focus` (`policy/runtime_policy.py:54`)	`3`	Numerical focus for the independent diagnostic solve.
`solver_diagnostic_aggregate` (`policy/runtime_policy.py:55`)	`0`	Presolve aggregation setting in the diagnostic solve.
`solver_objective_scale` (`policy/runtime_policy.py:56`)	`1000.0`	Positive multiplier applied to the assembled MEUR objective before optimization. The delivered value 1000 changes its numerical scale; restore native economic units when interpreting solver objectives and duals.

Setting and source	Entered value / expression	Meaning and effect
`solver_stage1_scale_flag` (`policy/runtime_policy.py:57`)	`2`	Gurobi scaling choice for the primary solve, in addition to explicit model row/column scaling.
`solver_stage1_aggregate` (`policy/runtime_policy.py:61`)	`0`	Primary presolve aggregation setting; delivered zero to preserve difficult transport equalities.
`solver_stage1_annual_pf_retry_aggregate` (`policy/runtime_policy.py:64`)	`1`	Alternate presolve aggregation in the eligible annual retry; does not mean all future operations are modeled hourly.
`annual_accounting_row_scale` (`policy/runtime_policy.py:65`)	`1000.0`	Explicit scaling divisor for applicable annual accounting rows. Physical audits must restore the original units.
`fuel_storage_balance_row_scale` (`policy/runtime_policy.py:66`)	`1000.0`	Explicit scaling divisor for carrier-stock balances; not a physical storage efficiency.
`fuel_storage_capacity_row_scale` (`policy/runtime_policy.py:67`)	`1000.0`	Explicit scaling divisor for carrier-store capacity rows; not a capacity multiplier.
`co2_storage_cost_row_scale` (`policy/runtime_policy.py:68`)	`1000.0`	Explicit scaling for CO2 storage cost equations; not a change in the cost assumption.
`solver_stage1_bar_homogeneous` (`policy/runtime_policy.py:69`)	`1`	Homogeneous barrier selection in the applicable ordinary stage-1 branch.
`solver_stage1_annual_s_curve_bar_homogeneous` (`policy/runtime_policy.py:72`)	`0`	Annual S-curve branch override of the ordinary homogeneous barrier setting.
`solver_stage1_homogeneous_recovery` (`policy/runtime_policy.py:73`)	`True`	Permit the independent homogeneous-barrier recovery path when applicable.
`solver_stage1_bar_conv_tol` (`policy/runtime_policy.py:74`)	`1e-08`	Barrier convergence tolerance for the ordinary applicable branch; dimensionless numerical criterion, not a physical MW tolerance.
`solver_stage1_annual_cc_bar_conv_tol` (`policy/runtime_policy.py:77`)	`1e-06`	Barrier tolerance override in eligible annual carbon-capture solves.
`solver_stage1_crossover` (`policy/runtime_policy.py:78`)	`0`	Gurobi crossover choice after barrier; zero disables normal crossover in the delivered case.
`solver_stage1_crossover_basis` (`policy/runtime_policy.py:79`)	`-1`	Crossover basis-construction setting used when that path is active.
`solver_stage1_bundle_feasibility_tol` (`policy/runtime_policy.py:80`)	`1e-09`	Primal feasibility tolerance used in the applicable dispatch-bundle solve/export path.
`solver_stage1_time_limit_s` (`policy/runtime_policy.py:81`)	`0`	Stage-1 time limit in seconds. Nonpositive is treated as unlimited by the wrapper.
`solver_stage1_auto_extend_attempts` (`policy/runtime_policy.py:82`)	`0`	Number of optional automatic time-limit extensions. Zero disables extensions, not the primary solve.
`solver_stage1_auto_extend_s` (`policy/runtime_policy.py:83`)	`3600`	Seconds added per enabled time-limit extension.
`solver_stage1_retry_numeric_focus` (`policy/runtime_policy.py:84`)	`3`	Numerical focus in the optional generic stage-1 retry.

Setting and source	Entered value / expression	Meaning and effect
`solver_stage1_retry_scale_flag` (`policy/runtime_policy.py:85`)	`1`	Gurobi scaling selection for that retry.
`solver_stage1_retry_bar_homogeneous` (`policy/runtime_policy.py:86`)	`1`	Homogeneous barrier setting for that retry.
`solver_provisional_max_constr_vio` (`policy/runtime_policy.py:87`)	`0.2`	Maximum solver-scaled constraint violation considered by the provisional-acceptance path. Physical-unit audits are independent.
`physical_energy_balance_abs_tolerance` (`policy/runtime_policy.py:92`)	`0.1`	Absolute physical-energy residual tolerance for audited families after undoing row/column scaling; delivered 0.1 in their MW/MWh basis.
`solver_provisional_max_bound_vio` (`policy/runtime_policy.py:93`)	`0.0001`	Maximum solver bound violation considered for provisional acceptance.
`solver_provisional_max_dual_vio` (`policy/runtime_policy.py:94`)	`0.0001`	Maximum dual violation considered for provisional acceptance.
`solver_provisional_max_dual_residual` (`policy/runtime_policy.py:95`)	`3.0`	Dual-residual acceptance threshold in the provisional path.
`solver_provisional_max_barrier_gap` (`policy/runtime_policy.py:96`)	`0.03`	Relative barrier-gap threshold used by provisional acceptance; it is not proof of certified optimality.
`solver_provisional_max_returned_objective_deviation` (`policy/runtime_policy.py:97`)	`0.02`	Allowed objective deviation in the relevant provisional returned-point comparison.
`solver_require_annual_cc_quality` (`policy/runtime_policy.py:100`)	`True`	Require the additional annual-capture numerical/physical quality checks even for a nominally optimal point.
`amine_mass_balance_abs_tolerance_t` (`policy/runtime_policy.py:101`)	`0.001`	Absolute amine-associated CO2 mass residual tolerance in physical tonnes; delivered 0.001.
`solver_stage1_bar_iter_limit` (`policy/runtime_policy.py:105`)	`1000`	Maximum primary barrier iterations before applicable recovery logic; not an hourly simulation count.
`solver_stage1_threads` (`policy/runtime_policy.py:107`)	`32`	Requested primary solver threads. More threads do not guarantee faster solves or more accurate inputs.
`solver_stage1_bar_order` (`policy/runtime_policy.py:108`)	`-1`	Barrier fill-reducing ordering setting; -1 is automatic in the supplied configuration.
`solver_auto_iis_max_rows` (`policy/runtime_policy.py:109`)	`2000000`	Matrix-size threshold for automatic IIS computation. An IIS is a conflict diagnosis, not a feasible solution.
`solver_bar_stall_time_s` (`policy/runtime_policy.py:110`)	`0`	Configured barrier-stall timeout control; inspect the callback's zero/off handling rather than treating zero as an immediate abort.
`solver_bar_stall_min_runtime_s` (`policy/runtime_policy.py:111`)	`2400`	Minimum runtime before the applicable stall logic considers intervention.
`solver_bar_stall_rel_improve` (`policy/runtime_policy.py:112`)	`0.05`	Relative improvement criterion in barrier-stall detection.

Setting and source	Entered value / expression	Meaning and effect
`solver_bar_stall_accept_gap` (`policy/runtime_policy.py:113`)	`0.002`	Gap threshold for considering an otherwise stalled barrier point acceptable for the relevant recovery path.
`solver_bar_stall_accept_compl` (`policy/runtime_policy.py:114`)	`1e-05`	Complementarity threshold in stall-point acceptance.
`solver_bar_stall_accept_primal_inf` (`policy/runtime_policy.py:115`)	`0.001`	Primal infeasibility threshold in stall-point acceptance.
`solver_bar_stall_accept_dual_inf` (`policy/runtime_policy.py:116`)	`0.0001`	Dual infeasibility threshold in stall-point acceptance.
`solver_stalled_barrier_finalize_time_s` (`policy/runtime_policy.py:117`)	`3600`	Time budget for an enabled crossover/finalization attempt after an eligible stall.
`solver_fallback_numeric_focus` (`policy/runtime_policy.py:118`)	`3`	Numerical focus in the simplex fallback.
`solver_fallback_time_limit_s` (`policy/runtime_policy.py:119`)	`0`	Fallback time budget; nonpositive is treated as unlimited in the applicable wrapper.
`solver_fallback_threads` (`policy/runtime_policy.py:120`)	`32`	Requested fallback thread count.
`electricity_price_zero_threshold_eur_per_mwh` (`policy/runtime_policy.py:123`)	`1.0`	Small electricity-price sanitation threshold in EUR/MWh. It can affect near-zero trade incentives and must be disclosed.
`fuel_trade_price_zero_threshold_eur_per_mwh` (`policy/runtime_policy.py:124`)	`1.0`	Small fuel-trade price sanitation threshold in EUR/MWh; does not authorize free feedstock generally.
`transport_stock_variable_scale_by_sector` (`policy/runtime_policy.py:126`)	`{'Passenger_cars': 1000.0, 'commercial_vehicles': 1000.0, 'Motor_coaches_buses_and_trolley_buses': 10.0, 'Domestic_shipping': 1000.0, 'Domestic_aviation': 100.0}`	Physical vehicles per scaled solver stock unit by sector. Export and audit quantities must be converted back to vehicles.
`transport_service_variable_scale` (`policy/runtime_policy.py:135`)	`1000.0`	Default service-variable scaling; changes numerical coefficients, not demand in physical units.
`transport_service_variable_scale_by_sector` (`policy/runtime_policy.py:136`)	`{'Passenger_cars': 1.0, 'commercial_vehicles': 10.0, 'Motor_coaches_buses_and_trolley_buses': 10.0, 'Domestic_aviation': 1.0}`	Sector overrides of service-variable scaling.
`numeric_abs_min` (`policy/runtime_policy.py:142`)	`1e-09`	Generic lower numerical sanitation bound used by supported readers/helpers. It is not a universal physical minimum flow.
`numeric_abs_max` (`policy/runtime_policy.py:143`)	`100000000.0`	Generic upper numerical sanitation bound; inspect the field-specific reader before treating it as a physical capacity limit.
`price_abs_max` (`policy/runtime_policy.py:144`)	`10000.0`	Upper absolute price sanitation bound in the applicable reader conventions.
`var_cost_abs_max` (`policy/runtime_policy.py:145`)	`100000.0`	Upper absolute variable-cost sanitation bound in applicable readers.
`transmission_km_abs_max` (`policy/runtime_policy.py:146`)	`100000.0`	Upper link-length sanitation bound in kilometres.
`fuel_storage_eff_min` (`policy/runtime_policy.py:147`)	`0.0001`	Minimum accepted/sanitized carrier storage efficiency.

Setting and source	Entered value / expression	Meaning and effect
`fuel_storage_eff_max` (`policy/runtime_policy.py:148`)	`1.0`	Maximum accepted/sanitized carrier storage efficiency; default one.
`cap_inv_limit_active_max` (`policy/runtime_policy.py:149`)	`500000.0`	Reader threshold used in deciding whether large investment limits are treated as active bounds. It is not a national resource estimate.
`transport_coeff_eps` (`policy/runtime_policy.py:150`)	`1e-05`	Small coefficient threshold used in transport equations/data handling.
`transport_co2_block_hours` (`policy/runtime_policy.py:151`)	`168`	Hours per transport-carbon aggregation block; affects numerical formulation, not annual vehicle demand.
`transport_historical_stock_zero_epsilon` (`policy/runtime_policy.py:152`)	`1e-06`	Small historical-stock cleanup threshold in the supported physical/scaled history path.
`heating_stock_zero_epsilon_mw` (`policy/runtime_policy.py:154`)	`1e-06`	MW threshold for negligible reconstructed heating stock.
`heating_history_zero_epsilon_mwh` (`policy/runtime_policy.py:155`)	`0.001`	MWh threshold for negligible reconstructed heating history.
`rolling_storage_state_roundoff_tolerance` (`policy/runtime_policy.py:158`)	`1e-06`	Tolerance for small negative stock roundoff during rolling carryover. Materially negative inventory is not authorized.
`rolling_h2_reserve_margin_mwh` (`policy/runtime_policy.py:160`)	`0.001`	Small MWh numerical headroom in the applicable H2 inventory reserve envelope.
`transport_physical_bound_tolerance` (`policy/runtime_policy.py:162`)	`1e-06`	Physical vehicle bound-audit tolerance; distinguish it from solver-scaled column tolerances.
`levelized_cost_min_annual_output_mwh` (`policy/runtime_policy.py:164`)	`1.0`	Minimum annual output for meaningful reported levelized EUR/MWh costs. Below-threshold ratios are not interpreted as reliable costs.
`transport_inherited_stock_zero_epsilon` (`policy/runtime_policy.py:167`)	`0.1`	Physical vehicle threshold for negligible inherited net/retrofit stock; delivered 0.1 vehicle.
`INDUSTRY_SHARE_ROUNDING_TOLERANCE` (`policy/runtime_policy.py:168`)	`0.01`	Share-rounding allowance used by industry input checks; not a new annual adoption budget.
`COMPLEX_INDUSTRY_DEMAND_AUDIT_REL_TOL` (`policy/runtime_policy.py:169`)	`0.01`	Relative complex useful-service demand reconciliation tolerance.
`COMPLEX_INDUSTRY_DEMAND_AUDIT_ABS_TOL` (`policy/runtime_policy.py:170`)	`0.001`	Absolute complex-demand reconciliation tolerance, used alongside the relative allowance in native service units.
`transport_fixed_stock_abs_tolerance` (`policy/runtime_policy.py:172`)	`0.1`	Absolute physical-vehicle tolerance around fixed stage-2 stock pins.
`transport_stock_relative_tolerance` (`policy/runtime_policy.py:175`)	`0.001`	Relative band/headroom around supported fixed stocks and technology/share caps; delivered 0.001 means 0.1%.

policy/transport_pathways.py

Setting and source	Entered value / expression	Meaning and effect
`transport_marine_capex_multiplier` (`policy/transport_pathways.py:9`)	`{'methanol': 1.09, 'ammonia': 1.1}`	New marine methanol/ammonia equipment CAPEX multipliers relative to the supported baseline; not energy efficiencies.
`transport_marine_pilot_energy_share` (`policy/transport_pathways.py:10`)	`0.05`	Pilot-fuel energy fraction in supported marine conversion recipes. Include both pilot and main-fuel emissions/costs.
`transport_marine_retrofit_capex_fraction` (`policy/transport_pathways.py:13`)	`{'methanol': 0.16, 'ammonia': 0.24}`	Cost fractions for supported marine methanol/ammonia retrofits. Retrofitting consumes eligible incumbent stock.
`transport_marine_retrofit_remaining_years` (`policy/transport_pathways.py:15`)	`10`	Remaining service life assigned to supported marine retrofits.
`transport_physical_upgrade_capex_fraction` (`policy/transport_pathways.py:20`)	`{'bio_diesel': 0.03, 'ethanol': 0.05, 'methanol': 0.1}`	Baseline-CAPEX fractions for supported physical fuel upgrades by fuel family.
`transport_physical_upgrade_remaining_years_by_sector` (`policy/transport_pathways.py:23`)	`{'Passenger_cars': 5, 'commercial_vehicles': 5, 'Motor_coaches_buses_and_trolley_buses': 8, 'Rail_transport': 10, 'Rail_metro_and_tram': 10, 'Domestic_shipping': 10, 'international_shipping': 10}`	Remaining life of paid physical upgrades by transport sector.
`transport_physical_upgrade_capex_meur_per_vehicle` (`policy/transport_pathways.py:31`)	`{}`	Optional explicit per-vehicle upgrade CAPEX overrides. Empty uses the implemented fraction-based treatment.
`transport_upgraded_max_volume_share` (`policy/transport_pathways.py:32`)	`{'bio_diesel': 1.0, 'ethanol': 0.85, 'methanol': 1.0}`	Maximum supported upgraded-fuel blend volume share. Volume shares require the model's energy-conversion treatment before use in energy balances.
`transport_drop_in_fuels` (`policy/transport_pathways.py:34`)	`('e_diesel', 'e_petrol', 'e_kerosene', 'SAF')`	Fuel labels treated as compatible drop-in pathways in the supported equipment representation. This is a model compatibility assumption, not universal real-engine certification.
`transport_fame_max_volume_share` (`policy/transport_pathways.py:36`)	`0.07`	Unmodified-equipment FAME volume-blending ceiling, default 7%.
`transport_ethanol_max_volume_share` (`policy/transport_pathways.py:37`)	`0.1`	Unmodified-equipment ethanol volume-blending ceiling, default 10%.
`transport_methanol_max_volume_share` (`policy/transport_pathways.py:38`)	`0.03`	Unmodified-equipment methanol volume-blending ceiling, default 3%.
`transport_synthetic_jet_max_volume_share` (`policy/transport_pathways.py:41`)	`1.0`	Synthetic jet volume-blending ceiling in the supported scenario; default one. Treat certification and real fleet applicability separately.

Conditional input paths and table-level selections

The top-level assignment catalogue is supplemented by conditional paths such as `policy_dat_fuels` (e-fuel carrier price workbook), `demand_co2` (external CO2 demand in the non-e-fuel branch) and the selected ammonia-demand path. These are derived from module choices. They are not additional independent physical options.

Workbook controls are equally important: `setup` years/hours, `active_zones`, detailed-industry `include_in_model`/native inclusion fields, `CC_active`, technology/capacity/cost rows, terminal activation and allowed import/export directions, RPS/CO2 paths, transport targets, market mandatory flags and supplier/route eligibility. Exact capitalization and placement are file-specific; follow the reader and data reference.

Individual flags do not guarantee investment or dispatch. They make a technology, equation or policy available; the optimizer still chooses within costs, demand, build rates, resource limits and all other active constraints. Likewise, removing one policy can leave another policy binding.

Input, output and units reference

Applies to: the preparation repository and the documented H2RES local model, 12 November 2026. This chapter explains the core contracts; machine-readable inventories include every examined workbook sheet/header and native CSV header.

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1. Complete schema inventories

Inventory	Contents
workbook_schema.csv	All examined workbook/sheet names, stored dimensions and exact first-row headers; blank/duplicate headers preserved
model_workbooks.json	Sheet-name lists for all live model workbooks
csv_schema.json	Exact headers for each native CSV
model_data_files.csv	All examined data files, sizes and SHA-256 digests
preparation_cli_options.csv	Argument names, source defaults/choices/help and line positions for runner and sector CLIs
model_configuration.csv	Entered source configuration assignments
workbook_reference_urls.csv	Reference URL occurrences with workbook/sheet/cell context

Some workbooks omit stored worksheet dimensions; their row count is marked unknown in the inventory rather than invented. Header inventories describe the files examined, not an assurance that all retained historical variants are active. Rows/columns can differ in another release.

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2. Common keys and calendar

Field	Meaning and rule
`System`	Model system label; often ISO2 country, but can be a custom aggregate
`Zone` / `zone`	Model zone; capitalization is file-specific and significant
`country`	Source-country label in spatial mappings/provenance; it can differ from a merged System
`nuts3` / `NUTS3` / `nuts3_id`	Source NUTS3 geography; sector maps use documented aliases
`period`	One-hour model position, normally integer 1–8760
`year` / year-named column	Planning/observation year according to table; not interchangeable with a weather year
`unit_name`	Technology/asset identifier; must match profile, cost and compatibility mappings
`Industry`	Exact native industry identifier, not arbitrary display text
`category`	Service/temperature class, separate from Industry

Field	Meaning and rule
`branch` , `component`	Complex-industry product branch and useful-service component
`include_in_the_model` , `active` , `CC_active`	Table-specific inclusion flags; inclusion does not itself create missing demand or source data

Hourly records with long geography need unique `(System,zone,period)` keys. Wide power profiles need unique `period` and matching unit-name columns. Different files use different index orders internally; never infer an operational dictionary key order from a superficially similar variable. The native transport primary-fuel dictionary, for example, starts with physical fuel.

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3. Units and conversion discipline

Quantity	Normal basis	Frequent mistake
Electrical/thermal power	MW	Treating MW as annual MWh
One-hour flow	MWh per one-hour interval, often numeric MW	Multiplying by 8760 twice
Annual energy	MWh/year; sometimes source GWh/ktoe	Confusing 1000 MWh/GWh with 11630 MWh/ktoe
Storage power	MW	Using an energy-capacity cost
Storage energy	MWh, or native GWh converted explicitly	Confusing stock with throughput
Fuel input	Carrier MWh, usually lower-heating-value energy conventions	Applying useful-service efficiency a second time
CO2 flow/capture capacity	tCO2/hour	Treating as tonnes of stored inventory
CO2 stock	tCO2	Applying MW-based CAPEX
Product activity	Tonnes or explicit index units	Labelling an index as tonnes
Fleet	Continuous vehicle-equivalent counts	Assuming discrete integer purchases
Capital cost	MEUR per native capacity unit	Applying MEUR/MWh to MW or €/vehicle without conversion
Variable cost	EUR/MWh or declared physical input basis	Counting purchased feedstock at multiple conversion stages
Fixed OPEX	MEUR per capacity-year or configured fraction	Treating a source annual rate as upfront CAPEX
Terminal capacity	TWh/year in standalone terminal workbook	Treating as tank energy capacity

For JRC industry extraction, `1 ktoe = 11630 MWh` is the implemented conversion. `1 GWh = 1000 MWh` ; `1 TWh = 1000000 MWh` ; `1 MEUR = 1000000 EUR` . Currency labels do not by themselves identify a price year or inflation basis. Record those separately where source metadata provide them.

4. Native technology and policy workbook families

Mode	No e-fuels	E-fuels
Regular industry	`flex_tech.xlsx`	`flex_tech_efuel.xlsx`
Category industry	`flex_tech_categories.xlsx`	Conservative: `flex_tech_efuel_categories.xlsx`; technical: `variants/technical_2050/flex_tech_efuel_categories.xlsx`
Complex industry	`flex_tech_complex_industry.xlsx`	`flex_tech_efuel_complex_industry.xlsx`

Category/complex flags are mutually exclusive. Other retained filenames, including combined category/complex variants and copies, are archived alternatives rather than an instruction to activate them together. Regular and complex technology files can be paired with case-specific process workbooks as described below.

Policy uses `policy.xlsx` or `policy_efuel.xlsx`; e-fuel carrier prices use `policy_fuels.xlsx` where enabled. Read the active path printed/selected by the model, not the workbook whose filename happens to be open in Excel.

5. Electricity and hydro files

5.1 `generators.csv` / prepared `power/GENCO.csv`

Power assets use the source's native GENCO columns, including `unit_name,cap_mw,max_cap,fuel_type,decom_start_existing_cap,life_time,decom_start_new,final_life_cap,efficiency,cost_no_fuel,RampingCost,CO2Intensity,technology,RampUpRate,RampDownRate,CHPType,STOCapacityGWh,STOSelfDischarge,STOMaxChargingPower,STOChargingEfficiency,CHPPowerToHeat,CHPPowerLossFactor,CHPMaxHeat,System,Zone` in the examined native CSV.

`cap_mw` is installed power, `max_cap` a capacity ceiling where used, and storage energy is explicitly GWh in this native representation. A detailed intermediate can instead contain `STOCapacity` in MWh and extra coordinates/NUTS fields; use the exporter conversion rather than copying it directly into the native file. `CO2Intensity` /non-fuel costs must be interpreted on the reader's generation/CHP-equivalent basis, not assumed to be raw fuel factors.

5.2 Profiles

File	Shape	Meaning
`electricity_demand.csv`	`period,Power,System,Zone`	Baseline electric load in MW
`renewable_availability.csv`	`period` plus exact generator names	Fractions for VRE and run-of-river
`hydro_inflows.csv`	`period` plus exact HDAM/HPHS names	Normalized inflow convention expected by reader
`electricity_prices.csv`	Hourly/zone price columns in examined input	Selected trade/electricity price assumptions
`import_prices.csv`	Native import-price long format	Price input according to selected cost path
`outside_system_normalized_curves.csv`	Outside-source/period curve fields	External source availability patterns

File	Shape	Meaning
`outside_system_supply_shares.csv`	System/source/share fields	Mix for external availability calibration

Profiles must match the canonical inventory's unit naming and aggregation. Plant/source mapping and curve-quality CSVs are part of the preparation evidence. Do not rename plant rows without updating their curves and cost/investment keys.

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6. Shared technology workbook sheets

6.1 Geography and networks

`active_zones` identifies active `(zone,System)` combinations. `keys_links`, `keys_borders`, transmission and outside-link sheets establish which modeled and external endpoints exist. Input files may retain inactive countries and templates. Only valid active mappings should contribute to a run, but inherited network/policy assumptions still require review.

6.2 Generic flexible technologies

`flex_tech` contains assets with fields such as installed/maximum capacity, annual build limits, efficiency, electric auxiliary demand, lifetime/retirement, reserve contributions, input/output carriers, storage linkage, System/zone and activity flags. `Tech_type` and carrier fields determine how a row is interpreted; identical-looking numeric `cap_mw` fields in different asset classes should not be assumed to mean identical delivered services.

6.3 `tech_cost_years`

The first column identifies year; technology/component keys are columns. Values follow each asset's documented capital-capacity basis. Complex component costs use `component_unit_name` keys, with generic donors only when explicitly recorded. Interpolated/mapped data are not all direct quotes for that component.

For the active workbook, the optional `literature_fixed_opex.csv` links annual maintenance rates to workbook, technology, year, native capacity unit, conversion and source references. Its `source_cells`, `source_sheets` and `source_price_years` should accompany a cost audit. A match is specific to the chosen workbook/version; source references alone do not prove every row has been applied.

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7. Ordinary/category industry inputs

7.1 Equipment tables

`Industry_data` includes System/zone, Industry, optional category, `unit_name`, physical `fuel_type`, group, efficiency, inclusion, capacity, initial service share, maximum service share, lifetime/decommissioning, CO2 information and scalar/annual build limits. Category profiles must use exactly the same native sector/category identifiers.

`simple_industry` represents a pooled remainder rather than an additional copy of selected detailed sectors. It must be recalculated when the detailed inclusion selection changes. `industry_CC` is the dedicated capture authority where used. `industry_heat_stor` exists only in variants supporting category industrial thermal storage.

7.2 Demand profiles

`industry_heat.csv` contains `period,System,zone` plus sector service demands. Despite the historical filename, full native service demand includes relevant non-heat components. `industry_category_profiles.csv` uses names such as `Iron_steel[100_400C]` and includes `NON_HEAT` where generated. `industry_electricity.csv` is wide by model zone and records the baseline electricity associated with the chosen source convention.

The legacy `heat_demand_industry.csv` can be process-heat-only; do not use it as the full native file merely because its name sounds appropriate.

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8. Complex industry workbooks

Possible native names are `industry_complex.xlsx`, `industry_complex_efuel.xlsx` and their `_electricity_technical_2050.xlsx` variants.

Sheet	Key fields	Interpretation
`complex_industry`	System, zone, Industry, branch, component, unit_name	Process-equipment alternatives and observed/assumed fuel parameters
`Process_data`	System, zone, Industry, branch, component	Useful-service/product intensity and independent process emissions
`Product_demand`	System, zone, Industry, branch, year columns	Annual product activity in declared unit
`CC`	System, zone, Industry, branch	Branch-level capture/reboiler/storage assumptions
`JRC_source_cells`	Sector/branch/component/fuel/year	Source observations and exact useful/final/emission addresses
`industry_heat_pumps`	Unit/class	COP, sink temperature, lifetime and cost assumptions
`Methodology` / screening/provenance tables	Descriptive keys	Source/version/assumption explanations; runtime tables remain authoritative

8.1 Critical process fields

`activity_unit` and `product_unit` must agree within a branch. `energy_intensity_MWh_per_activity` and `process_CO2_t_per_activity` are authoritative generic intensities. Legacy `_per_t` fields are suitable only for physical product tonnes. `reference_activity` is the observation basis. `activity_factor` modifies activity-to-service conversion and must not be applied twice.

`historical_process_CO2_*` preserves observed history; `excluded_H2_reforming_CO2_*` records the guarded Croatian accounting exception; modeled `process_CO2_*` is the remaining additional process term. `combustion_CO2_t_per_MWh` is observed final-fuel combustion, distinct from the legacy pure-fuel `CO2Intensity` and from product process intensity.

`fossil_waste_final_energy_share` applies to the mixed biomass/waste final-energy stream. `source_waste_split` must distinguish exact single-class evidence from assumed sector blends. A positive observed fossil combustion factor cannot be erased because a row is named Biomass.

`hp_below60_share`, `hp_below80_share`, `hp_below125_share`, `hp_below150_share` and `hp_source_heat_share` are shared eligibility/source budgets. They are fractions of useful service, not individual independent technology potentials that can simply be added.

8.2 Industry identifiers

The source maps JRC ISI, NFM, CHI, NMM, PPA, FBT, TRE, MAE, TEL, WWP and OIS into the eleven exact model labels listed in the preparation manual. The spelling `Not_elsewhere_specified_(industry)` and legacy branch key `cemment` are intentional compatibility identifiers. Correcting display spelling inside keys without migrating dependent inputs breaks joins.

---

9. Transport tables and profiles

`transport_all` identifies sector, equipment, type, demand/load coefficients, primary efficiencies, stock counts, maximum stock/share, storage/charging, retirement/lifetime, grid efficiency and inclusion, plus year-based investment/share fields according to variant. The extended e-fuel workbook contains persisted `compat_*` and parent/conversion metadata.

Names such as `total_load`, `number`, `tot_num`, `efficiency_primary`, `max_share` and year columns must be read together with the selected native transport method. `total_load` is a service/demand coefficient, not a universal electrical-MW field. Annual activity normalization and stock-derived coefficients determine its physical interpretation. Source table and reader equations are the authority for road versus marine/rail proxy classes; a column name alone is insufficient for a cross-sector conversion.

`transport_ICE_prim` maps primary/secondary fuels and efficiencies/shares to the applicable engine/service rows. `compat_parent` permits inherited availability for converted equipment. Paid upgrades start with no independently observed additional stock; their conversion/cost recipes are template assumptions.

Shipping source aliases use the native route keys consistently across equipment, compatibility lookup and hourly profile headers: `ICE_MARINE_DOM` → `ICE_MARINE_SHORT` (`Domestic_shipping`) and `ICE_MARINE_INT` → `ICE_MARINE_LONG` (`international_shipping`). The source service quantities and initial stocks are retained. Both newly prepared files and cached legacy files passed through export use this normalization; simultaneous legacy/current profile columns are rejected as ambiguous.

File	Values
`transport_demand.csv`	`period,zone,System` plus service activity shares, normally normalized to annual total 1
`transport_profiles.csv`	`period,zone,System` plus equipment/parent plug-in availability fractions

Fleet, service shares, fuel volume shares and fuel energy shares are different. Use the documented density/LHV/efficiency conversion in the pathway method rather than interpreting a volume percentage as an energy percentage.

10. Heating, DH, solar and cooling

Prepared technology tables are `flex_tech_heat`, `flex_tech_DH`, `thermal_tech`, `thermal_tech_DH`, `solar_indv` and `solar_DH`; storage sheets exist according to variant. Electricity/capacity and delivered-heat units must follow the class-specific reader. Heat-pump COP is not a boiler fuel efficiency.

`heating_individual.csv` and `heating_district.csv` are wide by model zone with `period`. `heating_electricity.csv` is the associated baseline electric demand. `solar_thermal_availability.csv` has zone/technology series according to the native reader. Cooling has separate individual/district files; their presence in the template does not imply they were freshly generated or enabled in a run.

11. Hydrogen, e-fuels, primary supply and capture

The selected flex workbook contains enabled e-fuel conversion, feedstock, primary-energy, trade, storage and network sheets. The exact sheet set is in the inventory because retained variants differ. Technology input/output recipes use carrier-specific energy or material units. Source notes and cost aliases are part of the contract.

Standalone terminals use `h2_terminals.xlsx`:

Terminal asset field	Unit/meaning
`cap_TWh`, `max_cap_TWh`	Annual terminal throughput capacity, TWh/year
`max_new_TWh_per_year`	New annual-throughput capacity allowed per calendar year
`loss`	Fractional handling loss
`el_MWh_per_MWh`	Handling electricity per H2 throughput energy
`var_EUR_per_MWh`	Handling cost per throughput energy
`life_years`, `existing_remaining_life_years`	New/existing lifetime inputs
`fixed_opex_pct`	Asset-specific annual maintenance fraction where present

Shipping-route fields include origin/destination, `delay_hours`, `cargo_MWh`, departure interval/offset and ships en route. A terminal row alone does not define an internal corridor. The workbook is an aggregate H2 representation and does not declare a separate ammonia vessel/cracker model.

Capture fields include `CC_cap`, capture electricity, capture rate, reboiler capacity/stripper heat and optional tank capacity. Their annual build limits have different units and cannot all be scaled as MW. Carbon-source tagging and allowed geological storage follow the chosen model switches, not merely the existence of a CC row.

12. Annual investment limits

Asset / field prefix	New capacity allowed per calendar year
Industry `max_inv_period_YYYY`	MW useful-service equipment capacity
Industry heat-storage `max_inv_period_YYYY`	MWh storage energy
`CC_max_inv_period_YYYY`	tCO2/hour capture capacity

Asset / field prefix	New capacity allowed per calendar year
`Reboiler_max_inv_period_YYYY`	MW reboiler capacity
`Tank_stor_max_inv_period_YYYY`	tCO2 intermediate inventory capacity
Generic asset or generation tables	Their declared native capacity basis
Transport/year fields	Native vehicle/service investment convention; inspect sheet and active flag

The year-specific cell takes precedence. Blank/missing annual cells fall back to the matching scalar. If that is also blank/missing, the limit is unrestricted. Zero is a binding prohibition on additions. Limits are not interpolated merely because future years are sparse.

The model sums calendar-year allowances over the global decision interval; the initial decision uses its own year. Installed stock and an annual build rate are not the same number. A total capacity limit and an annual build limit can both bind. Capture table precedence and sharing across categories prevent multiplying one physical branch capture budget by category count.

---

13. Policy tables and scenario JSON

`setup` holds the modeled years and first-row controls (`chunk_size,step_size,first_hour,last_hour`). Other policy sheets can hold RPS, CO2 limits, fuel/import/carbon prices, transport technology floors, biomass supply limits, generation calibration, hydro caps and outside trade targets. Exact names/headers are in the schema inventory. Many policy rows are System-scoped, not country-independent constants.

The separate preparation adapter's scenario JSON supports `preset,scope,years,start_year,end_year,year_step,target_year,start_share,investment_multiplier,investment_overrides` . Native `main.py` does not automatically load this JSON. Preparation CLI choices override corresponding JSON fields. Years are increasing/equally spaced and must be supported by costs. A whole-energy RES preset requires the e-fuel representation and a compatible model adapter; this documentation did not certify the adapter against all 12 November extensions.

Investment overrides select supported sheets, a native column, value, optional row conditions and years. Unsupported sheet names are errors. A reference to `Industry_data` is not automatically an override of the separate `complex_industry` workbook. The supported complex multiplier can scale its build limits; use documented direct editing for unsupported per-row complex changes.

---

14. Manifests and evidence files

File	What it establishes
`model_reference/manifest.json`	Template origin, captured file/source-reader hashes
`model_inputs/bundle_manifest.json`	Prepared sectors, countries, chosen variants, replaced sheets, template-retained inputs and reported repairs
`model_inputs/validation.json`	Implemented key/schema/hour/profile/zone coverage checks
`model_inputs/scenario.json`	Requested future policy and integration assumptions

File	What it establishes
`complex_generation.json`	Process sources, allocation, case, cost donors, profile basis and limitations
`source_reconciliation.csv`	Source-to-generated energy/emission closure for selected observation bases
`reference_calibration_diff.csv`	Matching HR1 process-equipment differences for the defined comparison fields
`scenario_reader_validation.json`	Separate actual-model input-reader check, if explicitly performed
`validation/*.json`	Bounded development verification; read its scope and timestamp

---

15. Minimum review before handing inputs to H2RES

Check requested geography against active/generated keys, unique hourly periods, matching unit/profile names, full sector/year coverage, useful/final-energy distinctions, scenario investment units, non-energy feedstocks, process/combustion carbon separation, finite efficiencies and stock/cost bases. Inspect warnings about fallback allocations and retained reference rows. Preserve the raw data and exact hashes so the handoff can be reproduced.

The model will also require consistent flag selection and any separately implemented whole-energy policy equations. Structural input compatibility is an interface property; historical accuracy and scenario feasibility require additional review.

---

16. New product-industry input contract

`data/simple_industry.xlsx` is read only when `model_simple_industry=True`. It is unrelated to the pooled-remainder worksheet of that name inside a flex-tech workbook. The product mode disables overlapping legacy industry/transport, e-fuel, hydrogen and industrial capture/storage blocks. Active example zones are configured separately in model policy.

Worksheet	Principal quantities / purpose
`plants`	Fixed product/route/geography, t/h capacity, MWh/t electricity, MW connection, operating years, flexibility and capital
`inputs_per_tonne`	Non-electric GJ/t recipes, emissions tCO2/GJ; avoid duplicate coke/process carbon
`economics`, `fuel_prices`	Plant/year economics and fuel/year EUR/GJ
`markets`, `market_years`	Eligibility and mandatory/optional markets; annual tonnes, sale/delivery costs and hourly ceilings
`carbon_policy`	Plant/year industrial price, benchmark, allocation and credits
`import_suppliers`, `import_years`	Outside origin/eligibility and year-specific supply limits, prices, embedded footprints and credits
`trade_routes`	Optional enabled supplier/market pairs with delivery costs and hourly limits
`profiles`	Named full-year profiles; `flat` is built in

Worksheet	Principal quantities / purpose
`baseline_load`	Explicit year-specific MWh to remove from basic electricity before adding product-industry consumption
`metadata`	Example/calibration status, boundaries and provenance

The example contains illustrative BF-BOF/EAF plants, mandatory domestic/Europe product demand and bounded outside offers. Zero baseline removals mean added electricity demand. Every modeled plant/year, market/year and required fuel/year needs its corresponding input row. Read the supplied module notes and exact headers in the schema inventory before changing these inputs.

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17. DSM parameters and additional output contracts

Basic-electricity DSM uses `main.py` switches and `policy/runtime_policy.py` parameters; it does not require a new DSM input workbook. Defaults are documented in model manual section 19 and the supplied DSM notes. The energy fractions, hourly power cap, rollout and costs are explicit scenario assumptions.

DSM results include `dsm_hourly_YEAR.csv`, `dsm_classes_YEAR.csv`, `dsm_summary_YEAR.csv` and `dsm_assumptions.json`. Product-industry results include hourly production, fuels, sales, markets, summary, imports, demand, fiscal and border-carbon tables. Geography, year, physical/embedded carbon and annualization must be read with their recorded boundaries. These result names describe enabled module outputs, not files created while writing this documentation.

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18. Formula caches and workbook handling

The screenshots deliberately show stored formulas. The inventory's cached-value read does not recalculate Excel formulas and is not a formula-validity audit. Inspect both formulas and numeric values when editing `Industry_data`, `_remainder_calc`, `simple_industry`, annual limits or dependent policy tables. Preserve the workbook, sheet names, header rows and native stacked-table layout. For missing or stale cached results, recalculate a working copy in a spreadsheet application and verify its saved values with the model reader.

The 12 November model also supplies read-only checking instructions in `data/README.md`, including `python -m scripts.repair_flex_workbooks data/flex_tech_efuel_categories.xlsx --check`. Read that helper's options before using a repair mode. No repair or recalculation was performed as part of this manual update.

Assumptions, limitations and evidence register

This register distinguishes observations from processing rules and scenario choices. It is part of both manuals. An internet source can contain analytically reconstructed data, and a referenced technology report does not validate a specific national parameter transferred from a template.

---

1. Classification

Class	Definition	Example
Source observation	Quantity taken from the identified source version/cell/response	JRC annual useful/final energy; ENTSO-E hourly load
Derived quantity	Reproducible transformation with a stated equation and units	Useful/final efficiency; capacity-weighted profile
Spatial/temporal proxy	Allocation using related evidence rather than direct local measurement	Raster shares, inherited hourly shapes, flat fallback
Inherited template assumption	Value retained from the supplied model/cost workbook	Lifetime, annual build limit, some conversion/cost rows
Explicit scenario assumption	Deliberately selected future or technical policy	2050 renewable target, investment multiplier, compatible replacement envelope
Unverified inherited provenance	A value/comment/reference whose original calibration cannot be established here	Unattributed technology defaults or missing source price-year details

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2. Parameter and method register

ID	Item	Classification / implication	Required study disclosure
A01	Country selection	Processing/output filter; bulk source geography can be wider	Selected codes, observed output System/zone, cache/source scope
A02	Geographic allocation	Source NUTS3 data plus proxy/custom mapping	Boundary vintage, raster/facility evidence, equal/custom weights, cross-border assumptions
A03	Power technical defaults	Mixed observed asset attributes and inherited technology defaults	Source snapshot, operating-status audit, donor parameters and remaining lives
A04	Weather/renewable profiles	Reanalysis/proxy conversion, optional national scaling	Weather year/backend, technology assumptions, IRENA extract and scaling
A05	Hydro profiles	Bundled runoff shapes plus scaling/matching assumptions	Runoff source/year, mapping distances, exclusions, natural/closed-loop distinction
A06	Category industry	Older annual/source profile vintages plus inherited temperature allocation	Reference years and what has/has not been recalibrated for the selected country

ID	Item	Classification / implication	Required study disclosure
A07	Complex industry initial MW	Annual useful energy / 8760	Average-utilization assumption; not observed nameplate plant census
A08	Process product activity	Observed activity to source end; constant thereafter	Future growth/structural-change scenario, fallback observation years
A09	Process electrical/HP applicability	Service screening and inherited equipment assumptions	Conservative/technical case, partial-furnace coefficient, COP/lifetime, cumulative/source budgets
A10	Biomass/waste split	Exact single-class evidence or assumed common sector blend	Strict/blend choice, exact source evidence, future mixed-stream composition assumption
A11	Material H2 convention	Non-energy gas/oil services represented as H2-fed demand	Chemical-route justification and possible process-emissions overlap
A12	Croatian ammonia exception	Guarded inherited accounting interpretation	Exact country/branch/year/amount; independent evidence required elsewhere
A13	Biomass/lignin substitution	Baseline biomass ceiling and bounded local lignin option	Share versus absolute annual MWh distinction; sustainability/resource scenario
A14	Transport stock/profiles	Bundled national data, service coefficients and allocation proxies	Source vintage, fleet/service units, availability, fractional stock and OSM fallback
A15	Paid conversions/blends	Technology-screening assumptions, including marine transfers	Engine approvals, pilot fuel, conversion cost/remaining life, actual fleet calibration
A16	Heating allocation	Annual observations plus population/HDD/density proxies	DH boundary, fuels, weather/population vintage and baseline electricity convention
A17	Costs/lifetimes/build limits	Inherited/reference or explicit scenario	Source workbook/cells, capacity basis, price year, finance/lifetime, total versus annual ceilings
A18	External energy supply	Network/capacity/price/availability assumptions	Import origin, hourly limits, fuel-specific trade prices and active endpoints
A19	Non-energy materials/CO2	Separate recipe and carbon classifications	Energy/material units, capture/reuse/storage distinction and imported stream tagging
A20	Future chronology	Full hourly or sampled/rolling mode	Period sets/weights, boundary conditions, what has actually been operationally validated
A21	Renewable preset	Explicit energy-origin constraints, not a forecast or net-zero definition	Target scope, adapter version, imports and cleared stocks, process/lifecycle exclusions
A22	Numerical acceptance	Solver tolerances and physical-unit quality gates	Status, primary/recovery, scaled/unscaled violations, slack and accepted history

3. Current boundary conditions worth checking

The live model's category/e-fuel selection does not automatically activate the process compiler's all-sector selection. The process policy defaults to non-metallic minerals until changed. Conversely, a workbook retaining data for other countries does not activate them if the active-zone scope is correct.

Several local entered values are Croatian plant/technology calibrations. Changing `--countries` can replace historical demand/inventory without supplying a new transmission network, biomass potential, outside-trade target, CAPEX forecast or heating/transport policy for that country. Complete native coverage and country-specific scientific calibration are separate review tasks.

Ordinary/category and complex industry have distinct vintages and definitions. Their totals may differ even when both are internally correct. Do not force equality by adding a remainder that duplicates replaced services. The model's process-replacement audit preserves ordinary services separately.

The documented conservative whole-energy preset blocks unverified imported energy and clears generic uncertified boundary stocks. That is stricter than permitting certified renewable imports. It can make a target infeasible even when another origin-tracing formulation would allow it. Stating "100% renewable" without the trade/storage boundary definition is insufficient.

4. Evidence levels

Level	Typical evidence	Does not establish
Dependencies/startup	Import checks, hidden interface construction	Source availability or correct national calibration
File structure/coverage	Validation JSON, unique keys, 8760 periods	A feasible optimization scenario
Source calibration	Cell reconciliation and baseline comparisons	Future technology realism or complete lifecycle emissions
Model reader	Actual native input reader completes	All constraints/objective built or a solve accepted
Small adapter/model test	Sample periods, bounded equation/LP tests	Full national chronology and all modes
Full solve	Exact input/version, solver logs and quality certificates	Empirical truth outside the modeled assumptions
Independent study validation	Historical reproduction and external plausibility/sensitivity checks	Universal validity for other countries/scenarios

The preparation verification reports record passing bounded tests. The supplied critical-rebuild folder also contains synthetic fixtures, unpromoted candidate checks and user-reported remote results. These evidence classes must not be merged into a claim of an independently reproduced full transition in this documentation session.

5. Source and reference completeness

The bibliography contains verified public data/method references and retained identifiable references from the industry handoff and cost-source records. Exact endpoint/cell inventories support further audit. Some publisher pages block automated retrieval or large PDFs exceed the browser tool's limit; those cases are labelled rather than represented as fully re-read.

Inherited parameters without a verified primary source remain marked as inherited/unattributed. A catalogue URL, generic literature citation or software repository is not enough to reconstruct an individual numeric price, efficiency or lifetime. A publication-ready parameter table needs the exact edition, page/table/sheet/cell, physical units, price year and all conversion steps.

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6. Recommended sensitivity and handoff record

Archive at least: command/configuration; selected countries/sectors/mapping; raw data/version/hashes; future years/demand; technology costs/lifetimes; annual/total investment limits; resource/biomass assumptions; industrial route and HP eligibility; fuel/engine compatibility; external prices/origins; storage initial/final policies; chronology/sampling; solver settings; accepted residuals/slack; resume history and exported results.

Useful sensitivity dimensions include high/low demand growth, renewable potential, grid/terminal expansion, weather years, profile calibration, industrial electrification scope, HP source limits, biomass ceilings, transport turnover/retrofit cost, renewable imports, financing and different chronological representations. Report which constraints bind and which inputs dominate the result.

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7. Additional 12 November model assumptions

The new DSM energy pools, hourly power cap, rollout and enabling/activation costs are scenario assumptions. They are not independently measured values for all active countries. The supplied simple-industry plants, world/European offers, product demand and carbon footprints are illustrative and require calibration. Product trade can move emissions outside the modeled territory; falling territorial emissions alone do not establish falling embedded emissions or domestic technology replacement.

The external workbook `simple_industry.xlsx` and the flex-tech worksheet `simple_industry` have different meanings. Enabling the product model disables overlapping legacy blocks. Baseline electricity removal must be calibrated when a plant's load is already included in the basic profile.

This version enables separated process/other carbon balances. Its default solver settings also permit provisional outcomes and warning-only postoptimal checks. Output files alone are not an optimization-quality certificate. This manual is a static inspection with source/input hashes, not a new national solve or a certification of the preparation adapter against every 12 November extension.

References and source register

Documentation/source check date: **2026-11-12**.

R entries are the curated scientific/data/software references. H entries preserve additional identifiable sources from the supplied industry handoff. T entries preserve the live fixed-OPEX workbook sources. Multiple URLs can describe the same work; entries are not a count of independent validation studies.

The verification note records whether a primary page/PDF was actually inspected or metadata were retained from supplied files. No missing source, price year or parameter calibration has been invented. A reference to a technology catalogue is not proof that every local default was transferred correctly.

Citation practice

For a study, cite the H2RES optimization formulation (R02), the actual dataset releases, and the technology sources used. Also identify the local model hash, preparation version and scenario assumptions. Historical H2RES simulation and later optimization versions must not be treated as the same code.

Record download/request dates and local hashes for mutable URLs, mirrored archives and manually prepared statistics extracts. The source's licence and the web-service usage terms are separate matters; check both before redistribution. OSM attribution follows its provider's licence terms. This register documents source conditions, not a legal opinion.

Curated references

R01 — Increasing renewable energy sources in island energy supply: case study Porto Santo

Duić, Neven and da Graça Carvalho, Maria. (2004). *Increasing renewable energy sources in island energy supply: case study Porto Santo*. article.

Primary/preserved source. DOI: `10.1016/j.rser.2003.11.004`.

Use: Historical H2RES island-energy lineage; not a specification of the current Python optimization implementation.

Verification: Author-hosted paper metadata/content checked.

Additional checked page.

R02 — A long-term capacity investment and operational energy planning model with power-to-X and flexibility technologies

Feijoo, Felipe and Pfeifer, Antun and Herc, Luka and Groppi, Daniele and Duić, Neven. (2022). *A long-term capacity investment and operational energy planning model with power-to-X and flexibility technologies*. article.

Primary/preserved source. DOI: `10.1016/j.rser.2022.112781`.

Use: Published optimization formulation and model lineage; later local additions are documented from code.

Verification: Publisher search record/article text and H2RES project publication listing checked; DOI redirect sometimes blocked.

R03 — The management of an energy system in the realm of rapid energy transition and degasification as a consequence of energy crisis, examination in H2RES energy model

Herc, Luka and Feijoo, Felipe and Kodba, Ana and Dorotić, Hrvoje and Stunjek, Goran and Beljan, Doris and Pukšec, Tomislav and Krajačić, Goran and Pfeifer, Antun and Duić, Neven. (2024). *The management of an energy system in the realm of rapid energy transition and degasification as a consequence of energy crisis, examination in H2RES energy model*. article.

Primary/preserved source. DOI: `10.1016/j.enconman.2024.118782`.

Use: H2RES application background; not an independent validation of this source snapshot.

Verification: Croatian research information system publication record and project bibliography checked.

R04 — Optimal decarbonisation pathways for the Italian energy system: Modelling a long-term energy transition to achieve zero emission by 2050

Pastore, Lorenzo Mario and Groppi, Daniele and Feijoo, Felipe and Lo Basso, Gianluigi and Astiaso Garcia, Davide and de Santoli, Livio. (2024). *Optimal decarbonisation pathways for the Italian energy system: Modelling a long-term energy transition to achieve zero emission by 2050*. article.

Primary/preserved source. DOI: `10.1016/j.apenergy.2024.123358`.

Use: Italian H2RES application background. This preparation package does not automatically reproduce its input scenario.

Verification: Institutional repository PDF opened; first page confirms full author list, journal/article number, publication date and DOI. Direct DOI fetch was blocked.

Additional checked page.

R05 — H2RES Integrated Energy System Model: project website

H2RES project. (n.d.). *H2RES Integrated Energy System Model: project website*. online.

Primary/preserved source.

Use: Model lineage, publication resources and version guidance; the local model remains version-specific.

Verification: Project website opened.

R06 — JRC-IDEES-2021

Rózsai, Máté and Jaxa-Rozen, Marc and Salvucci, Raffaele and Sikora, Przemyslaw and Tattini, Jacopo and Neuwahl, Frederik. (2024). *JRC-IDEES-2021*. dataset.

Primary/preserved source. DOI: `10.5281/zenodo.15895830`.

Use: Annual source database used by category industry and buildings; local endpoint is a mirrored bulk archive.

Verification: Zenodo release metadata and its primary JRC catalogue pointer checked. Publication date 20 May 2024; mirror record created later.

Source conditions: Check the JRC source licence and the exact mirror/version; do not infer all upstream dataset rights from Zenodo hosting.

R07 — JRC-IDEES-2023

Rózsai, Máté and Jaxa-Rozen, Marc and Salvucci, Raffaele and Sikora, Przemyslaw and Gea Bermudez, Juan and Neuwahl, Frederik. (2026). *JRC-IDEES-2023*. dataset.

Primary/preserved source. DOI: `10.2905/JRC.JPXYRT8`.

Use: Country Industry, EnergyBalance and EmissionBalance source workbooks for complex industry.

Verification: Official JRC overview, technical-report metadata, country download endpoint and inherited dataset DOI checked.

Additional checked page.

Source conditions: JRC overview identifies CC BY 4.0; archive the actual release.

R08 — SECURES-Energy: Hourly electricity demand and supply profiles for historical climate and climate change projections in Europe until 2100

Schöniger, Franziska and Resch, Gustav and Suna, Demet and Widhalm, Peter and Totschnig, Gerhard and Pardo-Garcia, Nicolas and Hasengst, Florian and Formayer, Herbert and Maier, Philipp and Leidinger, David. (2024). *SECURES-Energy: Hourly electricity demand and supply profiles for historical climate and climate change projections in Europe until 2100*. dataset.

Primary/preserved source. DOI: `10.5281/zenodo.14615500`.

Use: National hourly profiles; the category pipeline downloads the historical ERA5-Land 1981–2010 archive, not every future climate scenario.

Verification: Zenodo v2 metadata, files, coverage notes and project report link checked.

R09 — powerplantmatching: tools and combined power plant data

PyPSA powerplantmatching contributors. (n.d.). *powerplantmatching: tools and combined power plant data*. software.

Primary/preserved source.

Use: Non-hydro inventory source; actual code downloads the repository master powerplants.csv.

Verification: Maintainer repository and code endpoint identified; actual source hash/retrieval date is needed because master is mutable.

R10 — JRC Hydro-power database

European Commission, Joint Research Centre. (2026). *JRC Hydro-power database*. dataset.

Primary/preserved source. DOI: `10.2905/JRC.JBF7S58`.

Use: Hydro inventory and annual plant metadata. Bundled hourly runoff profiles are an additional local resource.

Verification: Official catalogue citation and release link checked. Catalogue citation year differs from earlier literature references to the original database.

Source conditions: Official catalogue identifies CC BY 4.0.

R11 — Energy and Industry Geography Lab and demand raster distribution

European Commission, Joint Research Centre. (n.d.). *Energy and Industry Geography Lab and demand raster distribution*. dataset.

Primary/preserved source.

Use: 2019 electricity/fuel spatial-allocation rasters; spatial reference year is independent of demand/statistical year.

Verification: Official overview and directory listing checked; layer copyright file is separately listed by provider.

Additional checked page.

R12 — Transparency Platform REST API implementation guide and token management

ENTSO-E. (n.d.). *Transparency Platform REST API implementation guide and token management*. documentation.

Primary/preserved source.

Use: Hourly actual total load requests and credential/code-list requirements.

Verification: Official API guide/token document search records checked. Legacy Guide.html URL could not be opened; current account workflow should be checked with provider.

Additional checked page.

R13 — OpenStreetMap data and Geofabrik country extracts

OpenStreetMap contributors and Geofabrik GmbH. (n.d.). *OpenStreetMap data and Geofabrik country extracts*. dataset.

Primary/preserved source.

Use: Selected-country PBF transport allocation indicators; the Geofabrik index itself is multi-country.

Verification: Provider download page and OpenStreetMap copyright/licence page opened.

Additional checked page.

Source conditions: OSM data are ODbL; use required contributor attribution and consult provider terms for redistribution.

R14 — heat_tot_curr_density: current heat-density raster

Hotmaps project. (n.d.). *heat_tot_curr_density: current heat-density raster*. dataset.

Primary/preserved source.

Use: Heat-density spatial proxy used by heating/DH allocation.

Verification: Project metadata opened; dataset describes 2015 heat density. Source release and licence must be preserved.

R15 — Historical Weather API documentation

Open-Meteo. (n.d.). *Historical Weather API documentation*. documentation.

Primary/preserved source.

Use: Default site weather/temperature source for VRE and heating preparation.

Verification: Provider documentation opened; service/data usage conditions are separate from locally derived profile assumptions.

R16 — The ERA5 global reanalysis

Hersbach, Hans and others. (2020). *The ERA5 global reanalysis*. article.

Primary/preserved source. DOI: `10.1002/qj.3803`.

Use: Scientific description of ERA5 reanalysis; does not validate the local power-curve conversion.

Verification: Publisher article metadata and primary publication record checked.

R17 — ERA5-Land: a state-of-the-art global reanalysis dataset for land applications

Muñoz-Sabater, Joaquín and others. (2021). *ERA5-Land: a state-of-the-art global reanalysis dataset for land applications*. article.

Primary/preserved source. DOI: `10.5194/essd-13-4349-2021`.

Use: Scientific description of ERA5-Land underlying weather/runoff sources.

Verification: Publisher article and PDF identified.

R18 — Renewables.ninja documentation and model service

Renewables.ninja. (n.d.). *Renewables.ninja documentation and model service*. documentation.

Primary/preserved source.

Use: Optional token-authenticated PV/wind simulation backend.

Verification: Provider documentation page opened; authenticated documentation may require login. The generic /api page returned 404 and is not used as a user guide.

R19 — Renewable capacity statistics 2025 and renewable electricity statistics

International Renewable Energy Agency. (2025). *Renewable capacity statistics 2025 and renewable electricity statistics*. dataset.

Primary/preserved source.

Use: Context for annual capacity/generation calibration; actual local extract and its source year must be recorded separately.

Verification: Official statistics/PDF search records checked; /Data direct fetch was blocked. No claim that the local extract equals the latest online release.

Additional checked page.

R20 — Direct electrification of industrial process heat: An assessment of technologies, potentials and future prospects for the EU

Rehfeldt, Matthias and Bußmann, Simon and Fleiter, Tobias and Rissman, Jeffrey. (2024). *Direct electrification of industrial process heat: An assessment of technologies, potentials and future prospects for the EU*. report.

Primary/preserved source.

Use: Industrial electrical-service applicability background; national screening coefficients remain explicit local assumptions.

Verification: Publisher page lists authors, version 1.1, publication 5 June 2024 and revision 6 June 2024.

R21 — Renewables for Industry: Electrification of low-temperature heat and steam

International Energy Agency. (2025). *Renewables for Industry: Electrification of low-temperature heat and steam*. report.

Primary/preserved source.

Use: Industrial low-temperature heat/steam electrification background.

Verification: Official report page, publication date 23 December 2025 and suggested citation checked.

R22 — Technology Data for Industrial Process Heat

Danish Energy Agency. (2026). *Technology Data for Industrial Process Heat*. report.

Primary/preserved source.

Use: Technology catalogue/background and inherited industrial heat cost assumptions.

Verification: Official catalogue page checked; it lists April 2026 catalogue and January 2026 datasheet updates. Individual numeric transfers require sheet/cell provenance.

R23 — Croatian National Inventory Report 2025

Croatian national greenhouse-gas inventory authorities. (2025). *Croatian National Inventory Report 2025*. report.

Primary/preserved source.

Use: Inherited evidence for the guarded HR ammonia-reforming accounting exception, table 4.3-1 as identified in supplied method.

Verification: Original publisher PDF endpoint identified; browser retrieval exceeded its size limit. The full inventory/table was not independently reread in this documentation session.

R24 — Croatian National Inventory Report 2020

Croatian national greenhouse-gas inventory authorities. (2020). *Croatian National Inventory Report 2020*. report.

Primary/preserved source.

Use: Inherited ammonia steam-reforming process description, section 4.3.1 as identified in supplied method.

Verification: Publisher PDF endpoint identified; full file exceeded browser size limit. Country/accounting interpretation retained as inherited provenance.

R25 — Preparing Container Vessels for Conversion to Green Fuels

Mærsk Mc-Kinney Møller Center for Zero Carbon Shipping. (2022). *Preparing Container Vessels for Conversion to Green Fuels*. report.

Primary/preserved source.

Use: Context for inherited marine conversion assumptions. Transfer to national model proxy vessels is a scenario assumption.

Verification: Provider PDF identified; publication label October 2022 checked. No claim that every local conversion fraction was independently recalibrated.

R26 — GeoPandas installation documentation

GeoPandas developers. (n.d.). *GeoPandas installation documentation*. documentation.

Primary/preserved source.

Use: Python spatial dependency installation and platform guidance.

Verification: Maintainer documentation checked; locally installed versions are recorded separately.

R27 — Climate Data Store: CDSAPI setup

Copernicus Climate Change Service and ECMWF. (n.d.). *Climate Data Store: CDSAPI setup*. documentation.

Primary/preserved source.

Use: Optional CDS client/token setup and dataset terms acceptance.

Verification: Official instructions opened; no credentials copied into documentation.

R28 — Gurobi Optimizer reference manual: optimization status codes

Gurobi Optimization. (n.d.). *Gurobi Optimizer reference manual: optimization status codes*. documentation.

Primary/preserved source.

Use: Meaning of solver termination statuses; local acceptance is additionally governed by configured quality gates.

Verification: Official manual opened; current online manual can be newer than the installed solver.

R29 — Guidelines for Numerical Issues

Gurobi Optimization. (n.d.). *Guidelines for Numerical Issues*. documentation.

Primary/preserved source.

Use: General numerical conditioning and tolerance interpretation.

Verification: Official manual opened; does not certify the local model's physical residuals.

R30 — Extraction-Condensing CHP example

PyPSA developers. (n.d.). *Extraction-Condensing CHP example*. documentation.

Primary/preserved source.

Use: Background reference explicitly cited in the local CHP-equivalent energy-accounting helper.

Verification: Code-cited primary documentation URL opened and its CHP feasible-region example checked; exact local equation is documented from energy_accounting.py.

R31 — NUTS statistical regions: GISCO distribution

Eurostat GISCO. (2021). *NUTS statistical regions: GISCO distribution*. dataset.

Primary/preserved source.

Use: 2021 NUTS boundaries and hierarchy used for shared geography.

Verification: Official distribution page opened; exact scale/CRS/file/hash should be archived with each run.

R32 — Population grids and GEOSTAT

Eurostat GISCO. (2021). *Population grids and GEOSTAT*. dataset.

Primary/preserved source.

Use: Population-allocation background; package uses the supplied 2021 TIFF or its configured mirror.

Verification: Official GEOSTAT page redirects users to population grids. Exact local raster and its mirror/licence remain part of source provenance.

R33 — JRC-IDEES-2023: the Integrated Database of the European Energy System — Data update and technical documentation

Rózsai, Máté and Jaxa-Rozen, Marc and Salvucci, Raffaele and Sikora, Przemyslaw and Gea Bermudez, Juan and Neuwahl, Frederik. (2026). *JRC-IDEES-2023: the Integrated Database of the European Energy System — Data update and technical documentation*. report.

Primary/preserved source. DOI: `10.2760/6945021`.

Use: Authoritative documentation of the JRC analytical database release.

Verification: Official record lists authors, report JRC144707 and date 14 January 2026; source content describes the underlying 2025 release.

R34 — H2RES-core public source repository

H2RES-model contributors. (n.d.). *H2RES-core public source repository*. software.

Primary/preserved source.

Use: Public project code context; it is not asserted to be byte-identical to the local documented copy.

Verification: Maintainer repository opened.

R35 — Assessment of the regionalised demand response potential in Germany using an open source tool and dataset

Heitkoetter, Wilko and Schyska, Bruno U. and Schmidt, Danielle and Medjroubi, Wided and Vogt, Thomas and Agert, Carsten. (2020). *Assessment of the regionalised demand response potential in Germany using an open source tool and dataset*. article.

Primary/preserved source. DOI: `10.48550/arXiv.2009.05122`.

Use: Regional demand-response methods and cost-potential background. German results do not calibrate Croatian residual basic-load DSM or the local 7%/1.5% assumptions.

Verification: Primary author preprint landing page, author list and abstract checked 12 November 2026; no numerical transfer or full-text replication claimed.

R36 — Economic potential for future demand response in Germany – Modeling approach and case study

Gils, Hans Christian. (2016). *Economic potential for future demand response in Germany – Modeling approach and case study*. article.

Primary/preserved source. DOI: `10.1016/j.apenergy.2015.10.083`.

Use: Demand-response modeling and sensitivity background, Applied Energy 162, 401–415. Not a country-specific calibration of local DSM costs.

Verification: Primary DLR repository metadata and abstract checked 12 November 2026. Repository lists 2015 publication/deposit metadata; journal volume is conventionally cited as 2016 in the supplied model notes.

R37 — Objective Function — DIETERpy documentation

DIETERpy project. (n.d.). *Objective Function — DIETERpy documentation*. online.

Primary/preserved source.

Use: Primary model documentation illustrating annualized DSM enabling/fixed and activation-cost terms. H2RES uses its own moved-energy counting convention.

Verification: Primary documentation equations and DSM sections checked 12 November 2026; no assertion that the two models have identical formulations.

Additional industry handoff references

H01 — JRC-IDEES-2023: the Integrated Database of the European Energy System - Data update and technical documentation

Rózsai, M. and Jaxa-Rozen, M. and Salvucci, R. and Sikora, P. and Gea Bermudez, J. and Neuwahl, F.. (2026). *JRC-IDEES-2023: the Integrated Database of the European Energy System - Data update and technical documentation*. inherited reference.

Primary/preserved source. DOI: `10.2760/6945021`.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `jrc_idees_documentation`.

H02 — Renewables for Industry: Electrification of low-temperature heat and steam

{International Energy Agency}. (2025). *Renewables for Industry: Electrification of low-temperature heat and steam*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `iea_renewables_industry`.

H03 — Direct electrification of industrial process heat: An assessment of technologies, potentials and future prospects for the EU

{Fraunhofer ISI}. (2024). *Direct electrification of industrial process heat: An assessment of technologies, potentials and future prospects for the EU*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `fraunhofer_agora_electrification`.

H04 — CO2-neutral process heat using electrification and hydrogen: Technologies, barriers and required action

Fleiter, T. and Rehfeldt, M. and Neusel, L. and Hirzel, S. and Neuwirth, M. and Schwotzer, C. and Kaiser, F. and Gondorf, C.. (2024). *CO2-neutral process heat using electrification and hydrogen: Technologies, barriers and required action*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `fleiter_process_heat`.

H05 — High-Temperature Heat Pumps, Task 3: Applications and Transition

{IEA HPT TCP Annex 58}. (2024). *High-Temperature Heat Pumps, Task 3: Applications and Transition*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `hpt_annex58_task3`.

H06 — Innovation Outlook: Renewable Methanol

{IRENA} and {Methanol Institute}. (2021). *Innovation Outlook: Renewable Methanol*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `irena_methanol`.

H07 — Green hydrogen for industry: A guide to policy making

{IRENA}. (2022). *Green hydrogen for industry: A guide to policy making*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `irena_hydrogen_industry`.

H08 — Biomass boilers for industrial process heat

{Energy Efficiency and Conservation Authority}. (n.d.). *Biomass boilers for industrial process heat*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `eeca_biomass`.

H09 — Decarbonisation technologies

{Ceramics UK}. (n.d.). *Decarbonisation technologies*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `ceramics_uk`.

H10 — VOLTA: One year demonstrating hybrid melting at industrial scale

{Saint-Gobain}. (2026). *VOLTA: One year demonstrating hybrid melting at industrial scale*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `saint_gobain_volta`.

H11 — Arvedi ESP: Endless casting and rolling

{Primetals Technologies}. (n.d.). *Arvedi ESP: Endless casting and rolling*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `primetals_esp`.

H12 — Ammonia Technology Roadmap

{International Energy Agency}. (2021). *Ammonia Technology Roadmap*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `iea_ammonia`.

H13 — World's First 1MW Ammonia Boiler Successfully Demonstrated by UK Consortium via Amburn Project

{Amburn project consortium}. (2026). *World's First 1MW Ammonia Boiler Successfully Demonstrated by UK Consortium via Amburn Project*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `amburn_demo_2026`.

H14 — Technical Verifications for Practical Application Completed on Ammonia Single Firing Burner for Thermal Power Plant Boilers

{IHI Corporation}. (2026). *Technical Verifications for Practical Application Completed on Ammonia Single Firing Burner for Thermal Power Plant Boilers*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `ihi_ammonia_burner_2026`.

H15 — Methanol Industrial Boilers

{Methanol Institute}. (n.d.). *Methanol Industrial Boilers*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `methanol_boilers_2020`.

H16 — DME: Production, distribution and storage

{IEA Advanced Motor Fuels TCP}. (n.d.). *DME: Production, distribution and storage*. inherited reference.

Primary/preserved source.

Use: Identifiable reference retained from the supplied industry methodology/bibliography. Some URLs duplicate a curated work using a different endpoint.

Verification: Bibliographic metadata retained from supplied handoff; not every document or numeric transfer independently re-read in this documentation session.

Supplied bibliography: `industry\handoff\model\analysis\industry_fuel_compatibility_2026_10_31\references.csv`, key `iea_amf_dme`.

Fixed-OPEX source workbooks

T01 — Technology-data workbook identified as DEA_CCS in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_CCS in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 416 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T02 — Technology-data workbook identified as DEA_FUELS in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_FUELS in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 1456 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T03 — Technology-data workbook identified as DEA_HEAT in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_HEAT in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 1144 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T04 — Technology-data workbook identified as DEA_NETWORK in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_NETWORK in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 416 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T05 — Technology-data workbook identified as DEA_POWER in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_POWER in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 41288 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T06 — Technology-data workbook identified as DEA_PROCESS in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_PROCESS in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 2288 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

T07 — Technology-data workbook identified as DEA_STORAGE in the supplied fixed-OPEX source table

Danish Energy Agency. (n.d.). *Technology-data workbook identified as DEA_STORAGE in the supplied fixed-OPEX source table*. technology workbook.

Primary/preserved source.

Use: Source endpoint retained for 208 technology/year/workbook rate records; see fixed_opex_sources.csv for an example sheet/cell trace.

Verification: Source URL and extraction provenance read from the live model CSV. Full downloaded catalogue and every numeric conversion were not independently re-audited here.

Provenance appendices

- URLs extracted from executable source constants: includes APIs, XML namespaces and plotting resources; not every URL is a dataset/reference.
- Workbook source/reference URL occurrences: exact workbook/sheet/cell context.
- Fixed-OPEX source groups: retained source endpoints and example extraction cells.
- Documented file hashes and input hashes.

Unresolved provenance

Generic technology defaults, some future demand/build/policy inputs, bundled profile/national-table assumptions and source price-year gaps remain inherited assumptions unless their own metadata establish a specific primary source. The manuals identify these rather than assigning a misleading scientific citation. Complete parameter-level

re-auditing would require the original extraction workbooks, editions, page/table/cell references and conversion calculations.